Rigid Analytic Geometry

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1 Introduction - Spectral Spaces and Spaces of Valuation Spectra

1.1 Spectral Spaces

Definition 1.1.1. Let X be a topological space. A non-empty subset Y is irreducible if it satisfies one of the following equivalent conditions.

i. If $Y \subseteq Y_1 \cup Y_2$ for some closed sets then $Y \subseteq Y_1$ or $Y \subseteq Y_2$.

ii. For any open sets U, V of X such that $U \cap Y, V \cap Y \neq \emptyset$ then $U \cap V \cap Y \neq \emptyset$.

Proposition 1.1.1. Let X be a topological space. For a point $x \in X$,

$$\overline{\{x\}} = \{y \in X \mid \text{every neighborhood of } y \text{ in } X \text{ contains } x\}.$$

Definition 1.1.2. A point $x \in X$ is a generic point if $\overline{\{x\}} = X$, (i.e. $\{x\}$ is dense in X).

Notation 1.1.1. For a topological space X, we denote the collection of all open sets to be $\mathscr{O}_X$.

Definition 1.1.3. Let X be a topological space and $x, y \in X$. We say x, y are topologically distinguishable if there is $U \in \mathscr{O}_X$ such that either $x \in U$ and $y \notin U$ or $x \notin U$ and $y \in U$. They are otherwise indistinguishable.

Definition 1.1.4. A topological space X is T_0 or Kolmogorov if for any pair of points are distinguishable.

Definition 1.1.5. A topological space X is sober if it satisfies one of the following equivalent conditions.

1. Every irreducible closed subset of X has precisely one generic point.

2. X is T_0 and every irreducible closed subset has at least one generic point.

Remark 1.1.1. It is easy to see that these are local conditions.

Definition 1.1.6. A topological space is T_1 if it satisfies one of the following equivalent conditions.

1. For any $x, y \in X$, $x \neq y \Rightarrow$ there is $U \in \mathscr{O}_X$ such that $x \in U$ and $y \notin U$.

2. Every point of X is closed.

It is T_2 or Hausdorff if it satisfies one of the following equivalent conditions.

1. Any pair of points $x, y \in X$ has disjoint neighborhoods U, V such that $x \in U$ and $y \in V$.

2. The diagonal set $\{(x, x) \mid x \in X\}$ is closed in $X \times X$.

Definition 1.1.7. X is quasi-compact if every open covering admits a finite subcover. A subset M of X is quasi-compact if it is a quasi-compact as a topological space with the subset topology from X.

Proposition 1.1.2. If X can be covered by finitely many quasi-compact open neighborhoods then it is quasi-compact.

Proposition 1.1.3. Every closed subset of a quasi-compact space is again quasi-compact.

Proposition 1.1.4. Let $X \xrightarrow{f} Y$ be a continuous with X being quasi-compact and Y being Hausdorff, then f is closed (i.e. every closed set A of X has a closed image $f(A)$ in Y).

Definition 1.1.8. A subset $\mathcal{B} \subseteq \mathscr{O}_X$ is a topology basis for X if for any $U \in \mathscr{O}_X$, we have $U = \bigcup_{V \in \mathcal{B}_U} V$ where $\mathcal{B}_U = \{V \in \mathcal{B} \mid V \subseteq U\}$.

Remark 1.1.2. The subsets of the form

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}$$

form a topology basis for $\operatorname{Spec} A$.

Proposition 1.1.5. For a topological space X, the following conditions are equivalent.

i. $\mathfrak{Q}_C(X) := \{\Omega \in \mathscr{O}_X \mid \Omega \text{ is quasi-compact}\}$ is a topology basis closed under finite intersection (including the empty intersection $\bigcap_\emptyset = X$).

ii. There is a topology base $\mathcal{B} \subseteq \mathfrak{Q}_C(X)$ such that $\mathcal{B}$ consists of quasi-compact and it is closed under the finite intersection in X.

Definition 1.1.9. A topological space X is spectral if it is sober and satisfies one of the conditions in Proposition 1.1.5.

Definition 1.1.10. A spectral map is a map $X \xrightarrow{f} Y$ between spectral spaces such that for any quasi-compact open subset Ω in Y, $f^{-1}(\Omega)$ is quasi-compact subset of X.

Remark 1.1.3. A spectral map is automatically continuous.

Proposition 1.1.6. If $\mathcal{B} \subseteq \mathscr{O}_Y$ is a topology base as in the second condition of Proposition 1.1.5 for Y, then f is spectral if and only if $\forall \Omega \in \mathcal{B}, f^{-1}(\Omega) \in \mathfrak{Q}_C(X)$.

Example 1.1.1. $\operatorname{Spec} A$ is spectral and every ring homomorphism defines a spectral map and $\{D(f) \mid f \in A\}$ are such base as in Proposition 1.1.5 for $\operatorname{Spec} A$.

Example 1.1.2. The underlying space of qcqs (i.e. quasi-compact and quasi-separated) scheme is spectral.

Remark 1.1.4. Every closed and every quasi-compact open sets of a spectral space is spectral.

Remark 1.1.5. A Noetherian space X (i.e. $\mathfrak{Q}_C(X) = \mathscr{O}_X$) is spectral if and only if it is sober.

Proposition 1.1.7. Every non-empty closed subset of a spectral space X contains a closed point.

Proof. By Remark 1.1.4, it suffices to show that every spectral space $X \neq \emptyset$ contains a closed point as non-empty closed subsets are again spectral. Set,

$$\mathcal{M} := \{\emptyset \neq Z \subseteq X \mid Z \text{ is closed}\}.$$

Equip $\mathcal{M}$ with a partial order by $\subseteq$. Recall that as $\bigcap_\emptyset = X \in \mathfrak{Q}_C(X)$, X itself is quasi-compact. Let $(Z_i)_{i\in I}$ be a descending chain in $\mathcal{M}$. Then $(X\backslash Z_i)_{i\in I}$ is an ascending chain of open sets. If the intersection $\bigcap_{i\in I} Z_i$ is empty, then

$$\emptyset = (X)^C = \bigcap_{i\in I} Z_n = \left(\bigcup_{i\in I} X\backslash Z_n\right)^C = \left(\bigcup_{i=1}^{k} X\backslash Z_i\right)^C \neq \emptyset.$$

This is a contradiction, hence it has a minimal element with respect to inclusion. Let Z be such minimal element in $\mathcal{M}$. If Z has two different element x, y, then by T_0 condition, we have $U \in \mathscr{O}_X$ such that $x \notin U$ and $y \in U$. Then $Z' = Z\backslash U$ is in $\mathcal{M}$. As $x \in Z'$, this contradicts to the minimality of Z as $y \in Z\backslash Z'$. □

Definition 1.1.11. A filter base of a set M is a non-empty subset $\mathcal{B}$ of $\mathfrak{P}(M)$ such that

i. $\emptyset \notin \mathcal{B}$,

ii. for $N, O \in \mathcal{B}$ there is $P \in \mathcal{B}$ with $P \subseteq N \cap O$.

A filter is a filter base $\mathcal{F}$ such that for $N \in \mathcal{F}$, and $N \subseteq O \subseteq M$ implies $O \in \mathcal{F}$.

A filter generated by a filter base $\mathcal{B}$ is such that

$$\mathcal{F} := \{N \mid N \subseteq M \text{ and there is } O \in \mathcal{B} \text{ such that } O \subseteq N\}.$$

A direct image under a map $M \xrightarrow{f} X$ of a filter $\mathcal{F}$ is such that

$$f_*\mathcal{F} := \{Y \mid Y \subseteq X \text{ and } f^{-1}(Y) \in \mathcal{F}\}.$$

Lemma 1.1.1. A filter is closed under finite intersection.

Proof. Let $N, O \in \mathcal{F}$. Then there is $P \in \mathcal{F}$ such that $P \subseteq N \cap O$. Thus $N \cap O \in \mathcal{F}$. □

Example 1.1.3. A neighborhood filter $\mathcal{N}_x$ of $x \in X$ is such that

$$\mathcal{N}_x := \{N \subseteq X \mid \exists U \in \mathscr{O}_X \text{ s.t. } x \in U \subseteq N\}.$$

Example 1.1.4. A principal filter $\mathcal{P}_x$ of $x \in X$ is such that

$$\mathcal{P}_x = \{N \subseteq X \mid x \in N\}.$$

Remark 1.1.6. A neighborhood base for $x \in X$ is a filter base $\mathcal{B}$ such that it generates the neighborhood filter of x.

Proposition 1.1.8. For a filter $\mathcal{F}$ of X, the following statements are equivalent.

1. $\mathcal{F}$ is a $\subseteq$-maximal element of the set of filters on X.

2. If $M, N \subseteq X$ and $M \cup N \in \mathcal{F}$ then $M \in \mathcal{F}$ or $N \in \mathcal{F}$.

3. For $M \subseteq X$, precisely one of M and $X \backslash M$ is in $\mathcal{F}$.

Definition 1.1.12. Such filter $\mathcal{F}$ satisfying the conditions in Proposition 1.1.8 is called an ultrafilter.

Theorem 1.1.1 (Tarski)**.** Every filter on X is contained in some ultrafilter.

Proof. Follows from Zorn's lemma. □

Proposition 1.1.9. Let $\mathcal{U}$ be an ultrafilter of X then all direct images of $\mathcal{U}$ are ultrafilters. If $\iota : A \to X$ is the inclusion of a subset then this induces,

$$\{\text{ultrafilters on } A\} \xrightarrow{\iota_*} \{\text{ultrafilters on } X \text{ containing } A\},$$

which is a bijection.

Definition 1.1.13.

$$\operatorname{Lim} \mathcal{F} = \{x \in X \mid \mathcal{N}_x \subseteq \mathcal{F}\}.$$

is the set of limits of $\mathcal{F}$. If it contains precisely one $x \in X$, we denote it by

$$\operatorname{Lim} \mathcal{F} = x.$$

Definition 1.1.14. Let $\mathcal{F}$ be a filter on X. We say that $x \in X$ is a point of accumulation of $\mathcal{F}$ (sometimes called point of condensation) if for every neighborhood $V \in \mathcal{N}_x$ and every $M \in \mathcal{F}$, we have $V \cap M \neq \emptyset$.

Proposition 1.1.10. Let $x \in X$ then

$$\overline{\{x\}} = \operatorname{Lim} \mathcal{P}_x = \text{point of accumulation of the principal filter of } x = \operatorname{Lim} \mathcal{N}_x.$$

Proposition 1.1.11. Let $\mathcal{F}, \mathcal{G}$ be filters such that $\mathcal{F} \subseteq \mathcal{G}$ and $\mathcal{U}$ be a ultrafilter.

1. Every limit of $\mathcal{F}$ is a point of accumulation of $\mathcal{F}$.
2. Every limit of $\mathcal{F}$ is a limit of $\mathcal{G}$.
3. Every point of accumulation of $\mathcal{G}$ is a point of accumulation of $\mathcal{F}$.
4. A point $x \in X$ is a limit of $\mathcal{U}$ if and only if it is a point of accumulation of $\mathcal{U}$.
5. The sets of its limits and its point of accumulations are closed for $\mathcal{F}$.
6. Let $\iota : A \to X$ be a canonical inclusion where A has an induced topology from X. Then for an ultrafilter $\mathcal{F}$ on A, $\operatorname{Lim}_A \mathcal{F} = A \cap \operatorname{Lim} \iota_* \mathcal{F}$.

Proof. **1**. let $x \in \operatorname{Lim} \mathcal{F}$. For all $V \in \mathcal{N}_x, V \in \mathcal{F}$. Since $\mathcal{F}$ is a filter thus does not contain $\emptyset$ and closed under finite intersection, derived the statement. **2** is trivial as $\mathcal{N}_x \subseteq \mathcal{F} \subseteq \mathcal{G}$, similarly for **3**.

4. If a is a point of accumulation of $\mathcal{U}$ and $V \in \mathcal{N}_x$, then $X \backslash V \notin \mathcal{U}$ as x is a point of accumulation. Hence $V \in \mathcal{U}$ by Proposition 1.1.8.

5. let $x \notin \operatorname{Lim} \mathcal{F}$, (respectively x not a point of accumulation of $\mathcal{F}$) then there is $U \in \mathcal{N}_x$ such that $U \notin \mathcal{F}$ (respectively, such that there is $M \in \mathcal{F}$ such that $U \cap M = \emptyset$). Then no element of U is a limit (respectively, point of accumulation) of $\mathcal{F}$.

6. Suppose $x \in \operatorname{Lim}_A \mathcal{F}$, then $\mathcal{N}_x^A \subseteq \mathcal{F}$ by definition. By definition of direct image and that the topology is induced from X, we see $\iota_* \mathcal{N}_x^A \subseteq \mathcal{N}_x^X$. Hence $\subseteq$ is shown. For the other inclusion, again follows from the induced topology and $x \in A$. □

Proposition 1.1.12. For a subset M of a topological space X, the following subsets all coincide.

a. $\{$limits of ultrafilters $\mathcal{U}$ of X such that $M \in \mathcal{U}\}$,

b. $\bigcup_{\substack{\mathcal{F} \text{ filter} \\ M \in \mathcal{F}}} \operatorname{Lim} \mathcal{F}$.

c. $\{x \in X \mid \exists \mathcal{F}$ filter s.t. $M \in \mathcal{F}$ and x is a point of accumulation of $\mathcal{F}\}$.

d. $\bigcap_{\substack{M \subseteq Z \subseteq X \\ Z \text{ closed}}} Z$.

Proof. Denote those sets by M_a, M_b, M_c, M_d. Then,

$$M_a \underbrace{\subseteq}_{\text{trivial}} M_b \qquad \underbrace{\subseteq}_{\text{Proposition 1.1.11}} \qquad M_c.$$

For $M_c \subseteq M_d$, note that $x \notin M_d$ implies that there is a neighborhood U of x such that $U \cap M = \emptyset$. Therefore x is not an accumulation point of any filter containing M.

For the reverse direction, note that $M_d = \bigcap_{\substack{U \in \mathscr{O}_X, \\ U \cap M = \emptyset}} (X \backslash U)$. If $x \in M_d$, then

$$\{U \cap M \mid U \in \mathcal{N}_x\}$$

is a filter base containing M as $\mathcal{N}_x$ is a neighborhood base of x and $M \cap U = \emptyset \Rightarrow x \notin U$. The filter generated by this contains $\mathcal{N}_x$. By Theorem 1.1.1, it is contained in some ultrafilter $\mathcal{U}$ and by definition $x \in \operatorname{Lim} \mathcal{U}$. □

Definition 1.1.15. The set of above is called the closure of M in X.

Remark 1.1.7. The closure of a point is a special case of this.

Corollary 1.1.1. For a subset A of a topological space X, the following conditions are equivalent.

i. If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a limit of $\mathcal{F}$ then $x \in A$.

ii. If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a point of accumulation of $\mathcal{F}$ then $x \in A$.

iii. If $\mathcal{U}$ is a ultrafilter on X such that $A \in \mathcal{U}$, every limit of $\mathcal{U}$ in X is in A.

iv. A is closed.

Proof. **i**⇒**ii** follows from that every limit of a filter is a point of accumulation Proposition 1.1.11.1. **ii**⇒**iii** follows from that limits and points of accumulation coincide for ultrafilters Proposition 1.1.11.4. **iii**⇒**iv** follows $M_a = M_d$ and M_d being closed Proposition 1.1.12. Similarly **iv**⇒**i** follows from $M_d = M_b$. □

Proposition 1.1.13. For a map $f : X \to Y$ of topological spaces. The following conditions are equivalent.

i. f is continuous.

ii. If $\mathcal{F}$ is a filter on X and $x \in \operatorname{Lim} \mathcal{F}$ then $f(x) \in \operatorname{Lim} f_* \mathcal{F}$.

iii. If $\mathcal{F}$ is a filter on X and $x \in X$ is a point of accumulation of $\mathcal{F}$, then $f(x)$ is a point of accumulation of $f_* \mathcal{F}$.

iv. If $\mathcal{U}$ is a ultrafilter on X and $x \in \operatorname{Lim} \mathcal{U}$, then $f(x) \in \operatorname{Lim} f_* \mathcal{U}$.

Proof.

i ⇒**ii**. This follows from that $f_* \mathcal{N}_x \supseteq \mathcal{N}_{f(x)}$ and f_* preserves inclusion.

ii ⇒ **iii**. By Proposition 1.1.11, every limit is a point of accumulation.

iii⇒**iv**. By Proposition 1.1.11, every point of accumulation of a ultrafilter is a limit of the filter and f_* maps ultrafilters to ultrafilters.

iv⇒**i**. If $A \subseteq Y$ is closed, we want to show that $f^{-1}(A)$ is closed. Let $\mathcal{U}$ be an ultrafilter in X such that $f^{-1}(A) \in \mathcal{U}$. Then $f_* \mathcal{U}$ is a ultrafilter on Y containing A and by Corollary 1.1.1, any limit of $f_* \mathcal{U}$ is in A. Since limits of $\mathcal{U}$ are mapped to limits of $f_* \mathcal{U}$, $f^{-1}(A)$ contains limits of $\mathcal{U}$. Again by Corollary 1.1.1 **iii**⇒**iv**, $f^{-1}(A)$ is closed. □

Remark 1.1.8. Definition 1.1.14 can be stated in terms of closures. Let A be the set of points of accumulation of $\mathcal{F}$. Then by Proposition 1.1.12,

$$A \subseteq \bigcap_{M \in \mathcal{F}} \overline{M} \subseteq \bigcap_{\substack{Z \in \mathcal{F} \\ \text{closed}}} Z = A'.$$

Now if $x \notin A'$, there is $X \backslash Z = U$ such that $x \in U$. Then $U \in \mathcal{N}_x$ but $U \cap Z = \emptyset$, therefore x is not a point of accumulation. Therefore, we obtained that above three sets all coincide.

Proposition 1.1.14. Let X be a topological space. The following conditions are equivalent.

i. X is Hausdorff.

ii. Every filter has at most one limit.

iii. Every ultrafilter on X has at most one limit.

Proof.

i⇒ii. If $x \neq y$, they have disjoint neighborhoods $x \in U$ and $y \in V$. If $U \in \mathcal{F}$ and $V \in \mathcal{F}$, then $\emptyset = U \cap V \in \mathcal{F}$, which is not possible. Thus precisely one of x and y is in $\operatorname{Lim} \mathcal{F}$.

ii⇒iii is trivial.

iii⇒i. Assume $x \neq y$ and consider the set

$$\{U \cap V \mid U \in \mathcal{N}_x, V \in \mathcal{N}_y\}.$$

This is a filter unless it contains $\emptyset$. In that case, it is contained in some ultrafilter $\mathcal{U}$. By definition of limits, x, y are both limits of $\mathcal{U}$ thus $x = y$ which is a contradiction. □

Proposition 1.1.15. Let X be a topological space. X is quasi-compact if and only if for arbitrary collection of closed sets $(Z_i)_{i \in I}$,

$$\forall J \subseteq I, \left[|J| < \infty \Rightarrow \bigcap_{j \in J} Z_j \neq \emptyset \right] \Rightarrow \bigcap_{i \in I} Z_i \neq \emptyset.$$

Proposition 1.1.16. Let X be a topological space. The following conditions are equivalent.

i'. X is quasi-compact.

ii'. Every filter has at least one point of accumulation.

iii'. Every ultrafilter has at least one limit.

Proof.

i'⇒ii' By Remark 1.1.8,

$$\{x \mid x \text{ is a point of accumulation of } \mathcal{F}\} = \bigcap_{M \in \mathcal{F}} \overline{M}.$$

By quasi-compactness X and applying it to the complement, this is a finite intersection of elements in $\mathcal{F}$. By definition, any filter is closed under finite intersection and does not contain the emptyset, therefore this is not empty.

ii'⇒iii'. Every filter is contained in some ultrafilter and limits and points of accumulation are the same in ultrafilter by Proposition 1.1.11.

iii'⇒i'. Let $(A_\lambda)_\Lambda$ be a family of closed subsets such that for any finite subset $F \subseteq \Lambda$, we have,

$$\bigcap_{\lambda \in F} A_\lambda \neq \emptyset.$$

Then the collection,

$$\left\{ \bigcap_{\lambda \in F} A_\lambda \;\middle|\; F \subseteq \Lambda, |F| < \infty \right\}$$

is a filter base, thus it is contained in some ultrafilter $\mathcal{U}$. Let $x \in \operatorname{Lim}\mathcal{U}$ then as A_λ is closed for each $\lambda \in \Lambda$, using **iv⇒iii** of Corollary 1.1.1, $A_\lambda \in \mathcal{U}$ hence $x \in A_\lambda$. Therefore, $x \in \bigcap_{\lambda \in \Lambda} A_\lambda$ which is not empty. Using the equivalent condition for the quasi-compactness, we have shown the statement. □

Proposition 1.1.17. Let X be a quasi-compact topological space such that the set of quasi-compact open subsets forms a topology basis (i.e. as in Proposition 1.1.5), then for every ultrafilter $\mathcal{U}$ on X, the closed subset $\operatorname{Lim}\mathcal{U} \subseteq X$ is irreducible.

Proof. First $\operatorname{Lim}\mathcal{U} \neq \emptyset$ as X is quasi-compact by Proposition 1.1.16. Let $\Omega, \Theta \in \mathfrak{Q}_C(X)$ such that $\Omega \cap \operatorname{Lim}\mathcal{U} \neq \emptyset$ and $\Theta \cap \operatorname{Lim}\mathcal{U} \neq \emptyset$. That is $\Omega \in \mathcal{N}_x, \Theta \in \mathcal{N}_y$ for some $x, y \in \operatorname{Lim}\mathcal{U}$. Hence, $\Omega, \Theta \in \mathcal{U}$ and thus $\Omega \cap \Theta \in \mathcal{U}$. As $\Omega \cap \Theta$ is quasi-compact $\mathcal{U} \cap \mathscr{O}_{\Omega \cap \Theta}$ has a limit $x \in \Omega \cap \Theta$ by Proposition 1.1.16. By Proposition 1.1.11, $x \in \operatorname{Lim}\mathcal{U}$.

Let U and V be open subsets such that $U \cap \operatorname{Lim}\mathcal{U}$ and $V \cap \operatorname{Lim}\mathcal{U}$ are not empty. By the assumption, there are $\Omega \subseteq U$ and $\Theta \subseteq V$, hence $U \cap V \cap \operatorname{Lim}\mathcal{U}$ is not empty. □

Definition 1.1.16. We say that $x \in X$ is a generic limit of a ultrafilter $\mathcal{U}$ if $\operatorname{Lim}\mathcal{U} = \overline{\{x\}}$.

Definition 1.1.17. Let X be a topological space in which every ultrafilter $\mathcal{U}$ has a unique generic limit. Then we denote such limit by

$$\operatorname{glim}\mathcal{U} = \operatorname{glim}_X \mathcal{U}.$$

We say that a subset $M \subseteq X$ is closed under generic limit in X if for every ultrafilter $\mathcal{U}$, $M \in \mathcal{U}$ implies $\operatorname{glim}\mathcal{U} \in M$.

Proposition 1.1.18.

1. Suppose ultrafilters of X admit unique generic limits. Then X is T_0 if and only if every ultrafilter on X has at most one generic limit.

2. If every ultrafilter on X has some generic points then every irreducible closed subset $Z \subseteq X$ has some generic points.

Proof. For **1**. the sufficiency of being T_0 follows from that X is Hausdorff by Proposition 1.1.14 and by Definition 1.1.5. If x and y are topologically indistinguishable, then $\overline{\{x\}} = \overline{\{y\}} = Z$ for some Z. Furthermore, the principal filter $\mathcal{P}_x$ of x which has $\operatorname{Lim} \mathcal{P}_x = Z$ from Example 1.1.4 has both x and y as generic limits, therefore $x = y$.

2. Irreducibility of Z implies that $\mathscr{O}_Z \backslash \{\emptyset\}$ is a filter base in X. Hence it is contained in some ultrafilter $\mathcal{U}$. By Definition 1.1.15 and the closedness of Z implies $Z = \operatorname{lim} \mathcal{U}$. Hence every $\operatorname{glim} \mathcal{U}$ is a generic point of Z. □

Proposition 1.1.19. Assume every ultrafilter $\mathcal{U}$ on X has a unique generic limit $\operatorname{glim} \mathcal{U}$. Then the following statements hold.

1. The set of subsets of X which are closed under glim is closed under arbitrary intersection and finite unions.

2. An open subset $V \subseteq X$ is closed under glim if and only if V is quasi-compact.

Proof. For **1**, let $(M_i)_{i \in I}$ be the collection of subsets closed under glim. Then, if $M_I = \bigcap_{i \in I} M_i$, $M \in \mathcal{U}$, we have,

$$\forall i \in I, M_I \subseteq M_i \Rightarrow M_i \in \mathcal{U}.$$

Therefore for each $i \in I$, $\operatorname{glim} \mathcal{U} \in M_i$. Thus $\operatorname{glim} \mathcal{U} \in M_I$.

If $M \cup N \in \mathcal{U}$ then $M \in \mathcal{U}$ or $N \in \mathcal{U}$ by Proposition 1.1.8. Hence $\operatorname{glim} \mathcal{U} \in M \subseteq M \cup N$ or $\operatorname{glim} \mathcal{U} \in N \subseteq M \cup N$.

Let $\mathcal{V}$ be an ultrafilter on V and $\mathcal{U}$ be the image of $\mathcal{V}$ under the embedding $V \hookrightarrow X$. If V is closed under glim then $\operatorname{glim} \mathcal{V} \in V$ is a limit of $\mathcal{U}$ as

$$\operatorname{Lim} \mathcal{V} = V \cap \operatorname{Lim} \mathcal{U}$$

by Proposition 1.1.11. Under the bijection of Proposition 1.1.9, every ultrafilter of V admits a limit and by Proposition 1.1.14, V is quasi-compact. Conversely, if V is quasi-compact, $\mathcal{V}$ has a limit $x \in V$ and $x \in \operatorname{Lim} \mathcal{V} = V \cap \operatorname{Lim} \mathcal{U}$ by Proposition 1.1.11. Let $y \in X$ be such that $x \in \overline{\{y\}}$ hence $y \in V$. Therefore V is closed under glim. □

Definition 1.1.18. A spectral space X is quasi-separated if for any ultrafilter $\mathcal{U}$ on X, if $\operatorname{Lim} \mathcal{U}$ is non-empty implies that $\operatorname{Lim} \mathcal{U}$ is an irreducible closed subset.

Proposition 1.1.20.

1. A topological space X is spectral if and only if every ultrafilter on X has a unique generic limit and the set

$$\mathfrak{O} = \{\Omega \in \mathscr{O}_X \mid \Omega \text{ is closed under glim}\}$$

is a topology base.

2. An open subset of a spectral space X is quasi-compact if and only if it is closed under glim.

3. A map $f : X \to Y$ is spectral if and only if it is continuous and $\mathrm{glim}_Y f_*\mathcal{U} = f(\mathrm{glim}_X \mathcal{U})$ for every ultrafilter $\mathcal{U}$ on X. (This is proved by the above assertion).

Proof. For **1**, if X is spectral, then by Proposition 1.1.17, $\mathrm{Lim}\,\mathcal{U}$ is always a closed and irreducible subset, hence $\mathrm{glim}\,\mathcal{U}$ exists uniquely by the sobriety of X.

By Proposition 1.1.19, one has $\mathfrak{O} = \mathfrak{Q}_C(X)$ and this is a topology base by the spectrality of X. If any ultrafilter has a unique generic limit, then X is T_0 and every closed, in particular, irreducible subset contains a generic point by Proposition 1.1.18. By Proposition 1.1.19 $\mathfrak{O} = \mathfrak{Q}_C(X)$ and it is closed under finite intersection in X. Therefore X is spectral. **2** follows from **1** and Proposition 1.1.19.

For **3**, $\Rightarrow$ immediately follows from Proposition 1.1.19. Let $\Omega \subseteq Y$ be open and $\mathcal{U}$ be an ultrafilter on X.

$$\mathrm{glim}_Y f_*\mathcal{U} \in \Omega \Leftrightarrow \Omega \in f_*\mathcal{U} \Leftrightarrow f^{-1}(\Omega) \in \mathcal{U} \Leftrightarrow \mathrm{glim}_X \mathcal{U} \in f^{-1}(\Omega) \Leftrightarrow f(\mathrm{glim}_X \mathcal{U}) \in \Omega.$$

Therefore $\mathrm{glim}_Y f_*\mathcal{U}$ and $f(\mathrm{glim}_X \mathcal{U})$ are topologically indistinguishable. As X is T_0, they must coincide. □

Lemma 1.1.2. Let $\mathcal{B}$ be a topology base of the topological space X.

a. Let $\mathcal{U}$ be an ultrafilter on X and $g \in X$ such that for all $\Omega \in \mathcal{B}$, $g \in \Omega \Leftrightarrow \Omega \in \mathcal{U}$. Then g is a generic limit of $\mathcal{U}$.

b. If X is T_0 and for every ultrafilter $\mathcal{U}$ on X, there is $g \in X$ as in a, then X is spectral and the elements of $\mathcal{B}$ are quasi-compact.

Proof. Let $U \in \mathcal{N}_g$. Then there is $\Omega \in \mathcal{B}$ such that $g \in \Omega \subseteq U$. By our assumption, $\Omega \in \mathcal{U}$ hence $U \in \mathcal{U}$ thus g is a limit of $\mathcal{U}$. Let $x \notin \overline{\{g\}}$ then there is $U \in \mathcal{N}_x$ such that $g \notin U$. As $\mathcal{B}$ is a topology base there is $\Omega \in \mathcal{B}$ such that $x \in \Omega \subseteq U$. Then $g \notin \Omega$, hence $\Omega \notin \mathcal{U}$. Since Ω is a neighborhood of x, this shows $x \notin \mathrm{Lim}\,\mathcal{U}$. Thus g is a generic limit. This proves a.

For b), since X is T_0, by Proposition 1.1.18, we see that g in assumption is a unique generic limit of an ultrafilter. As elements of $\mathcal{B}$ are closed under such generic limits, the claim follows from Proposition 1.1.20 1. □

Theorem 1.1.2 (M.Hochster 1969)**.** A topological space X is spectral if and only if there is a ring A such that X is homeomorphic to $\mathrm{Spec}\,A$.

Proof. We omit the hard part which is $\Rightarrow$ direction (the existence of such ring A for every spectral space X). That $\operatorname{Spec} A$ is spectral can be seen as follows. Let $\mathcal{U}$ be an ultrafilter on $\operatorname{Spec} A$ and set

$$\mathfrak{g}_{\mathcal{U}} := \{a \in A \mid V(a) \in \mathcal{U}\} = \{a \in A \mid \{\mathfrak{p} \in \operatorname{Spec} A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

We show $\mathfrak{g}_{\mathcal{U}} \in \operatorname{Spec} A$. Let $a_1, a_2 \in \mathfrak{g}_{\mathcal{U}}$. Then $V(a_1), V(a_2) \in \mathcal{U}$. As $V(a_1, a_2) \subseteq V(a_1 + a_2)$ and $\mathcal{U}$ is closed under finite intersections, $V(a_1 + a_2) \in \mathcal{U}$. In particular $a_1 + a_2 \in \mathfrak{g}_{\mathcal{U}}$ hence it is an ideal. Now suppose $a_1 a_2 \in \mathfrak{g}_{\mathcal{U}}$ but $a_2 \notin \mathfrak{g}_{\mathcal{U}}$. Then $V(a_2) \notin \mathfrak{g}_{\mathcal{U}}$ but $V(a_1 a_2) = V(a_1) \cup V(a_2) \in \mathcal{U}$. Thus $V(a_1) \in \mathcal{U}$ that is $a_1 \in \mathfrak{g}_{\mathcal{U}}$ i.e. it is prime.

Recall that principal open subsets form a topology basis of $\operatorname{Spec} A$. Since $X = V(f) \cup D(f)$, and that

$$\mathfrak{g}_{\mathcal{U}} \in D(f) \Leftrightarrow f \notin \mathfrak{g}_{\mathcal{U}} \Leftrightarrow V(f) \notin \mathcal{U} \Leftrightarrow D(f) \in \mathcal{U}.$$

By Lemma 1.1.2, $\mathfrak{g}_{\mathcal{U}} = \operatorname{glim} \mathcal{U}$ and all $D(f)$ are glim-closed. □

Corollary 1.1.2. The subset $D(a) \subseteq \operatorname{Spec} A, a \in A$ are quasi-compact.

Proof. Lemma 1.1.2 and that the set of principal opens is a topology basis. □

Definition 1.1.19. Let X be a topological space. A subset U which is open and closed is called clopen.

Corollary 1.1.3. For a compact (thus Hausdorff) X, the following conditions are equivalent.

a. X is spectral.

b. The clopen subsets of X form a topology base.

In this case, the open subsets U is quasi-compact if and only if it is clopen and for a spectral space Y, a map $Y \to X$ is spectral if it is continuous.

Proof. For a compact space X, very ultrafilter $\mathcal{U}$ of X admits a unique limit by Proposition 1.1.14 and Proposition 1.1.16, such limits are trivially generic limits. By Corollary 1.1.1, the open subsets which are invariant under taking limits are closed.

From the observation above, the second statement is equivalent to saying that $\mathfrak{Q}_C(X)$ is a topology base for X. We further see that if $Y \xrightarrow{f} X$ is continuous and $\Omega \in \mathfrak{Q}_C(X)$, then Ω is clopen hence $f^{-1}(\Omega)$ is clopen in Y, hence $f^{-1}(\Omega) \in \mathfrak{Q}_C(Y)$. □

Definition 1.1.20. A set $\mathcal{A}$ of subsets of a set X is a boolean algebra if the following conditions hold.

1. $\emptyset \in \mathcal{A}$.
2. $M \in \mathcal{A} \Rightarrow X \backslash M \in \mathcal{A}$.

3. $\mathcal{A}$ is closed under taking finite unions.

Remark 1.1.9. From the basic axioms of boolean algebras. Clearly we have $X \in \mathcal{A}$ and $M_1, M_2 \in \mathcal{A} \Rightarrow M_1 \cap M_2 \in \mathcal{A}$.

Lemma 1.1.3. Let X be a spectral space and $M \subseteq X$ be a subset. If M is closed under glim then so is $X \backslash M$.

Proof. Omitted. □

Proposition 1.1.21. Let X be a spectral space. The following boolean algebras of subsets of X coincide.

1. The boolean algebra generated by $\mathfrak{Q}_C(X)$.
2. $\mathfrak{M} = \{M \subseteq X \mid M \text{ is closed under generic limits of ultrafilters}\}$.

Let $\mathcal{K}_X$ be this boolean algebra of subsets of X. The topology on X obtained by taking $\mathcal{K}_X$ as a topology base coincides with the one for which taking the subsets $M \subseteq X$ which are closed under glim of ultrafilters as closed subsets.

Finally let X_{con} be X equipped with this topology. Then X_{con} is a spectral hausdorff space (i.e. X_{con} as in Corollary 1.1.2), and $X_{\text{con}} \overset{\text{id}_X}{\to} X$ is a spectral map.

For every ultrafilter $\mathcal{U}$ of X, $\lim_{X_{\text{con}}} \mathcal{U} = \text{glim}_X \mathcal{U}$.

Proof:(Caution : This is not required for the exam). Let $\mathcal{K}_1$ be th boolean algebra generated by $\mathfrak{Q}_C(X)$ and $\mathcal{K}_2$ be the boolean algebra generated by $\mathfrak{M}$. By Lemma 1.1.3, we see,

$$\mathcal{K}_2 := \{M \mid M \text{ and } X \backslash M \text{ closed under glim}\}.$$

By Proposition 1.1.19, $\mathcal{K}_2$ is a Boolean algebra (note that $\emptyset$ is closed under glim as no ultrafilter contains it). By Proposition 1.1.20 b), and Corollary 1.1.1, $\mathfrak{Q}_C(X) \subseteq \mathcal{K}_2$, hence we have an inclusion $\mathcal{K}_1 \subseteq \mathcal{K}_2$. Let X_1 be the topological space with underlying space X and the topology generated by $\mathcal{K}_1$. Let X_2 be the topological space again with the same underlying set such that for a subset $A \subseteq X$ is closed if and only if it is closed under glim of ultrafilters. By Proposition 1.1.19, X_2 is indeed a topological space. Since $\mathcal{K}_x \subseteq \mathcal{K}_2$, X_2 is finer or equal to X_1.

We first show that the space X_1 is Hausdorff. As X is T_0, if $x \neq y$ then without loss of generality exchanging x and y if necessary, there is $U \in \mathscr{O}_X$ such that $x \in U$ and $y \notin U$. As $\mathfrak{Q}_C(X)$ is a topology base, hence without loss of $U \in \mathfrak{Q}_C(X)$. Then $U \in \mathcal{K}_X$ and $X \backslash U \in \mathcal{K}_1$ and they are distinguished neighborhoods of x and y in X_1. Moreover, we can show that X_2 is quasi-compact. Indeed if we consider $\iota_2 : X_2 \to X$ a canonical bijection, then ι_2 is continuous. Let $\mathcal{U}$ be an ultrafilter of X_2, then $\iota_2 * \mathcal{U}$ is also an ultrafilter of X thus admits a limit. Therefore, $\mathcal{U}$ admits a limit and by Proposition 1.1.16, X_2 is quasi-compact.

Now we can use Proposition 1.1.4 and the fact that a bijective closed map is a homeomorphism. Consider the map $\mathrm{id}_X : X_2 \to X_1$. As $X_2 = X_1$ and X_2 has finer topology than X_1, this is continuous. As X_2 quasi-compact and X_1 Hausdorff, by Proposition 1.1.4 this is closed. Therefore topologies coincide.

Let $X_{\mathrm{con}} = X_1 = X_2$. Note that X_{con} is compact thus every ultrafilter admits only a single limit. Let $\Omega \in \mathcal{K}_2$ then by Corollary 1.1.1, Ω is clopen in X_{con}. Since $\mathcal{K}_1$ is a topology base, Ω is a union of elements of $\mathcal{K}_1$ and $\Omega \in \mathfrak{Q}_C(X_{\mathrm{con}})$, the union can be chosen to be finite. Therefore, $\Omega \in \mathcal{K}_1$ as $\mathcal{K}_1$ is a Boolean algebra.

We will now set, $\mathcal{K}_X = \mathcal{K}_1 = \mathcal{K}_2$. Since the elements of $\mathcal{K}_X$ are clopen and $\mathcal{K}_X$ is a topology base for X_{con}, and X_{con} compact, X_{con} is spectral by Corollary 1.1.3. Finally, by the definition of topology on X_{con}, any quasi-compact set in X is again quasi-compact in X_{con} thus $\mathrm{id} : X_{\mathrm{con}} \to X$ is spectral. □

Definition 1.1.21. X_{con} is called the constructible topology on X and the elements of $\mathcal{K}_X$ are called constructible subsets.

Corollary 1.1.4. A subset of X is an element of $\mathcal{K}_X$ if and only if it is clopen in X_{con}.

Corollary 1.1.5. A map $X \xrightarrow{f} Y$ between spectral spaces is spectral if and only if it is continuous and $X_{\mathrm{con}} \xrightarrow{f} Y_{\mathrm{con}}$ is also continuous.

Corollary 1.1.6. For a spectral space X, $\mathfrak{Q}_C(X) = \mathscr{O}_X \cap \mathcal{K}_X$.

Definition 1.1.22. Let X be a spectral. We say that $x \in X$ is a specialization of $y \in X$ and conversely y is a generalization of x if $x \in \overline{\{y\}}$. We denote such relation by $x \preceq y$.

Proposition 1.1.22.

1. A subset $Z \subseteq X$ is closed in X if and only if it is closed in X_{con} and closed under specialization. That is $x \preceq y, y \in Z \Rightarrow x \in Z$.

2. A subset $U \subseteq X$ is open in X if and only if it is open in X_{con} and closed under generalization.

3. Let Z be closed in X_{con} and $\overline{Z}$ be its closure in X. Then $\overline{Z}$ is the set of specialization of elements of Z.

Proof. By taking complement, we see 1⇔2 and 3⇒1. Thus enough to prove the third one. If $z \in Z$ and $x \preceq z$, then $x \in \overline{\{z\}} \subseteq \overline{Z}$. If $x \in \overline{Z}$ by Proposition 1.1.12, there is an ultrafilter $\mathcal{U}$ on X such that $Z \in \mathcal{U}$ and $x \in \mathrm{Lim}\mathcal{U}$. As Z is closed in X_{con}, $g = \mathrm{glim}\mathcal{U} \in Z$ thus $x \preceq g$. □

Corollary 1.1.7. For a spectral space X, the following statements are equivalent.

1. X is T_1.

2. $X = X_{\text{con}}$ as topological space.

3. Every quasi-compact subset of X is closed.

4. X is Hausdorff.

5. X is T_4.

Corollary 1.1.8. For a map $X \xrightarrow{f} Y$ between spectral spaces, the following statements are equivalent.

1. $\text{glim}_Y f_*\mathcal{U} = f(\text{glim}_X \mathcal{U})$ for all ultrafilters $\mathcal{U}$ on X, and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.

2. $X_{\text{con}} \xrightarrow{f} Y_{\text{con}}$ is continuous and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.

3. f is spectral.

Corollary 1.1.9. For a subset $Z \subseteq X$, the following statements are equivalent.

a. Z is spectral with induced topology and $Z \to X$ is spectral.

b. Z is closed in X_{con}.

In this case, we denote Z_{con} to be the topology on Z induced from topology X_{con}

1.2 The Valuation Spectrum of a Rings

Definition 1.2.1. An ordered abelian group is an abelian group Γ (under multiplication) with a linear ordering $\leq$ such that

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

Example 1.2.1. $\{1\}$ and $(0,\infty)^{\times}_{\mathbb{R}}$ are ordered abelian groups.

Remark 1.2.1. The inequality,

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

is strict if one of the inequalities in the left hand side is strict. In particular, $g > 1$ implies $g^k > 1$ and $g < 1$ implies $g^k < 1$ for all $k \in \mathbb{N}$. Thus Γ has no torsion.

Definition 1.2.2. A (general) valuation of a ring R is a map $R \xrightarrow{v} \Gamma \cup \{0\}$ where Γ is an ordered abelian group.

Definition 1.2.3. A valuation of a ring R is a map $R \xrightarrow{v} \Gamma \cup \{0\}$ where Γ is an ordered abelian group with multiplication such that the following hold.

i. $\forall x \in \Gamma \cup \{0\}, 0 \cdot x = x \cdot 0 = 0, 0 \leq x$.

ii. $v(xy) = v(x)v(y)$

iii. $v(1) = 1$

iv. $v(0) = 0$

v. $v(x + y) \leq \max\{v(x), v(y)\}$.

Remark 1.2.2.

$$v(-1) = 1, \quad v(y) < v(x) \Rightarrow v(x + y) = v(x).$$

Definition 1.2.4. Let Γ_v be the subgroup of Γ generated by $\{v(r) \,|\, r \in R\}\backslash\{0\}$. We say that valuations v and w are equivalent if there is an order preserving isomorphism $\Gamma_v \overset{i,\sim}{\to} \Gamma_w$ such that

$$w(r) = i(v(r)),$$

where we put $i(0) = 0$.

Proposition 1.2.1. The valuations v and w are equivalent if and only if

$$\{(r, s) \in R^2 \mid v(r) \leq v(s)\} = \{(r, s) \in R^2 \mid w(r) \leq w(s)\}.$$

Definition 1.2.5.

$$\operatorname{supp} v = \{r \in R \mid v(r) = 0\}.$$

Remark 1.2.3. $\operatorname{supp} v$ is a prime ideal. It stable under multiplication by any element since,

$$\forall s \in R, v(r) = 0 \Rightarrow v(rs) = v(r)v(s) = 0.$$

It is closed under addition since,

$$r, s \in \operatorname{supp} v \to v(r + s) \leq \max\{v(r), v(s)\} = 0.$$

Therefore it is an ideal. Clearly $1 \notin \operatorname{supp} v$. If $v(rs) = 0$, then $v(r)v(s) = 0$. Thus $v(r) = 0$ or $v(s) = 0$.

Remark 1.2.4. The support only depends on the equivalence classes.

Example 1.2.2. If $\mathfrak{p}$ is a prime ideal and

$$v(r) = \begin{cases} 1, & r \notin \mathfrak{p}, \\ 0, & r \in \mathfrak{p}. \end{cases}$$

the trivial valuation belonging to $\mathfrak{p}$.

Definition 1.2.6. A valuation is trivial if it is equivalent to this kind of valuation. That is $\Gamma_v = \{1\}$.

Example 1.2.3. If $R = K$ is a field, then $v(s) = 1$ where $s^k = 1$ for some $k \in \mathbb{N}$ as Γ has no torsion. In particular any finite field only admits trivial valuations.

Example 1.2.4. The p-adic valuation of $\mathbb{Q}$ or $\mathbb{Q}_p$ is a valuation.

Definition 1.2.7. Let $\mathrm{Spv}(R)$ be the set of equivalent classes of valuation of R. For $x \in \mathrm{Spv}(R)$, we choose a representative $|\cdot|_x$. Let Γ_x be the group generated by those $|r|_x$ which are non-zero. For $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in R^{m+1}$, let

$$\begin{aligned}\mathcal{R}(f_0|f_1, \cdots, f_m) &= \mathcal{R}_{\mathrm{Spv}(R)}(f_0|f_1, \cdots, f_m), \\ &:= \{x \in \mathrm{Spv}(R) \mid 0 \neq |f_0|_x \geq |f_i|_x, 1 \leq i \leq m\}.\end{aligned}$$

These are called rational open subsets. We equip $\mathrm{Spv}(R)$ with the topology for which the set of rational open subsets is a topology base.

If $A \xrightarrow{\alpha} B$ is a ring homomorphism then this induces

$$\mathrm{Spv}(B) \overset{\mathrm{Spv}(\alpha)}{\to} \mathrm{Spv}(A)$$

sends $x \in \mathrm{Spv}(B)$ to the equivalence class of the valuation,

$$a \mapsto |\alpha(a)|_x$$

of A. We set,

$$V(a) = \mathrm{Spv}(A) \backslash \mathcal{R}(a|).$$

Remark 1.2.5. The topology is therefore the coarsest for which

$$\mathrm{Spv}(A) \overset{x \mapsto \mathrm{supp}(x)}{\to} \mathrm{Spec}(A)$$

is continuous and all

$$\{x \in \mathrm{Spv}(A) \backslash V(a) \mid |\alpha|_x \leq |a|_x\}$$

for $(a, \alpha) \in A^2$ are open.

Remark 1.2.6. The set of rational open subset is closed under finite intersections in $\mathrm{Spv}(R)$. Indeed,

$$\begin{aligned}\mathcal{R}(1|) &= \mathrm{Spv}(R), \\ \mathcal{R}(r_0|r_1, \cdots, r_m) \cap \mathcal{R}(s_0|s_1, \cdots, s_n) &= \mathcal{R}(r_0 s_0 | r_i s_j, 0 \leq i \leq n, 0 \leq j \leq n).\end{aligned}$$

If you assume $m \leq 1$ or even $m = 1$, you only get a topology subbase.

One way of constructing generic limits of ultrafilters in $\mathrm{Spv}(R)$ would be to use Proposition 1.2.1 and

$$\{(r, s) \in R^2 \mid |r|_g \leq |s|_g\} := \{(r, s) \in R^2 \mid \{x \in \mathrm{Spv}(R) \mid |r|_x \leq |s|_x\} \in \mathcal{U}\}.$$

This would work and was shown by Huber. This approach will not be taken in this course.

Definition 1.2.8. Let $\mathcal{U}$ be an ultrafilter on Λ and $(X_\lambda)_{\lambda\in\Lambda}$ be a family of sets indexed by Λ. We call the elements $(x_\lambda)_{\lambda\in\Lambda}$ and $(\xi_\lambda)_\Lambda$ of $\prod_{\lambda\in\Lambda} X_\lambda, \mathcal{U}$-equivalent if

$$\{\lambda \mid x_\lambda = \xi_\lambda\} \in \mathcal{U}.$$

The ultraproduct $\prod_{\mathcal{U}} X_\lambda$ is the set of $\mathcal{U}$-equivalence classes of elements of $\prod_{\lambda\in\Lambda} X_\lambda$. If each X_λ-s has a group structure, $(x\xi) \in \prod_{\mathcal{U}} X_\lambda$ is given as the $\mathcal{U}$-equivalence classes of $(x_\lambda\xi_\lambda)_\Lambda$. If each X_λ is equipped with linear ordering, put

$$(x_\lambda)_{\lambda\in\Lambda}/\sim_{\mathcal{U}} \leq (\xi_\lambda)_{\lambda\in\Lambda}/\sim_{\mathcal{U}}$$

if $\{\lambda \in \Lambda \mid x_\lambda \leq \xi_\lambda\} \in \mathcal{U}$.

Proposition 1.2.2. Suppose $X_\lambda \neq \emptyset$ for all $\lambda \in \Lambda$.

1. If $\mathcal{U} = \{\Theta \subseteq \Lambda \mid \lambda_0 \in \Theta\}$ is the principal ultrafilter belonging to λ_0, then $\prod_{\mathcal{U}} X_\lambda = X_{\lambda_0}$.
2. If F is a finite set and all $\lambda \in \Lambda, X_\lambda = F$. Then $\prod_{\mathcal{U}} F = F$ as for an arbitrary $x \in \prod_{\mathcal{U}} F$, $\Lambda = \bigcup_{f\in F}\{\lambda \mid x_\lambda = f\}$ (i.e. disjoint finite union) hence there is precisely one $f \in F$ such that $\{\lambda \mid x_\lambda = f\} \in \mathcal{U}$.
3. If $X_\lambda = X'_\lambda \cup X''_\lambda$ (disjoint unions, $\forall\lambda \in \Lambda, X'_\lambda, X''_\lambda \neq \emptyset$) Then $\prod_{\mathcal{U}}(X'_\lambda \cup X''_\lambda) = (\prod_{\mathcal{U}} X'_\lambda) \cup (\prod_{\mathcal{U}} X''_\lambda)$ which is also a disjoint union as $\Lambda = \{\lambda \in \Lambda \mid x_\lambda \in X'_\lambda\} \cup \{\lambda \in \Lambda \mid x_\lambda \in X''_\lambda\}$.
4. Then $\prod_{\mathcal{U}}(\Gamma_\lambda \cup \{0\}) = (\prod_{\mathcal{U}} \Gamma_\lambda) \cup \{0\}$ and $\prod_{\mathcal{U}} \Gamma_\lambda$ is an ordered abelian group where all Γ_λ are ordered abelian groups.

Remark 1.2.7. When some of the X_λ may be empty, it is better to define $\prod_{\mathcal{U}} X_\lambda$ as the set of equivalent classes of pairs (M, ξ) where $M \in \mathcal{U}$ is a subset and ξ a function defined on M such that $\xi(\lambda) \in X_\lambda$ where $\lambda \in M$. Two pairs $(M, \xi), (N, \nu)$ are equivalent if

$$\{\lambda \in M \cap N \mid \xi(\lambda) = \nu(\lambda)\} \in \mathcal{U}.$$

With this, we define,

$$\prod_{\mathcal{U}} \left(\bigcup_{i=1}^{m} X_\lambda^{(i)} \right) \cong \bigcup_{i=1}^{m} \prod_{\mathcal{U}} X_\lambda^{(i)}.$$

Remark 1.2.8. $\prod_{\mathcal{U}}$(ordered abelian group) is again an ordered abelian group and

$$\prod_{\mathcal{U}} (\Gamma_\lambda \cup \{0\}) \cong \left(\prod_{\mathcal{U}} \Gamma_\lambda \right) \cdot \{0\}.$$

Proposition 1.2.3. For every ring R, $\mathrm{Spv}(R)$ is a spectral space and a subset of the form $\mathcal{R}(f_0|f_1, \cdots, f_m)$ with $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in R^{m+1}$ is quasi-compact.

The generic limit of the ultrafilter $\mathcal{U}$ on $\operatorname{Spv} R$ is given by the equivalence class of the valuation,

$$R \overset{r\mapsto \binom{\text{the } \mathcal{U}\text{-equivalence class of}}{(|r|_v)_{v\in \operatorname{Spv} R}}}{\longrightarrow} \left(\left(\prod_{v\in \operatorname{Spv} R}\right)_{\mathcal{U}} \Gamma_v\right) \cup \{0\} = \prod_{\mathcal{U}} (\Gamma_v \cup \{0\}). \quad (1.1)$$

Moreover, the map,

$$\operatorname{Spv}(R) \overset{v\mapsto \operatorname{supp}|\cdot|_v}{\longrightarrow} \operatorname{Spec} R$$

is spectral and for every ring homomorphism $R \overset{\rho}{\to} S$, the map

$$\operatorname{Spv}(S) \overset{\operatorname{Spv}(\rho)}{\to} \operatorname{Spv}(R)$$

is also spectral.

Proof. It is obvious from the definition and the properties of ultrafilters that (1.1) is indeed a valuation. To show that it defines a generic limit of $\mathcal{U}$ and that the rational open subsets are quasi-compact, we use Lemma 1.1.2. Set

$$\mathcal{B} = \{\mathcal{R}(f_0|f_1,\cdots,f_m) \mid m\in\mathbb{N}, (f_i)_{i=0}^m \in R^{m+1}\}.$$

Let $X = \operatorname{Spv} R$ and g be the equivalence class of (1.1),

$$\begin{aligned}
[g\in\mathcal{R}(f_0|f_1,\cdots,f_m)] &\Leftrightarrow |f_0|_g \neq 0 \text{ and for all } i=1,\cdots,m,\ |f_i|_g \le |f_0|_g,\\
&\overset{\text{By (1.1)}}{\Leftrightarrow} (|f_0|_x)_{x\in X}/\sim_{\mathcal{U}} \neq 0 \text{ and } (|f_i|_x)_{x\in X}/\sim_{\mathcal{U}} \le (|f_0|_x)_{x\in X}/\sim_{\mathcal{U}},\\
&\overset{\text{Def. } \mathcal{U}\text{-equivalence}}{\Leftrightarrow} \{x\in X \mid |f_0|_x \neq 0\} \in \mathcal{U} \text{ and}\\
&\qquad \forall i=1,\cdots,m, \{x\in X \mid |f_i(x)| \le |f_0(x)|\} \in \mathcal{U},\\
&\overset{\text{Def. filters}}{\Leftrightarrow} \{x\in X \mid |f_0|_x \neq 0\} \cap \bigcap_{i=1}^m \{x\in X \mid |f_i(x)| \le |f_0(x)|\} \in \mathcal{U},\\
&\overset{\text{Def. rational open}}{\Leftrightarrow} \mathcal{R}(f_0 \mid f_1,\cdots,f_m) \in \mathcal{U}.
\end{aligned}$$

Therefore, Lemma 1.1.2 can indeed be applied. That the induced map $\operatorname{Spv}(R) \overset{\text{supp}}{\to} \operatorname{Spec}(R)$ is spectral follows from

$$\operatorname{supp}(D(f)) = \mathcal{R}(f|).$$

By Proposition 1.1.6, that $\operatorname{Spv}(\rho)$ is spectral follows immediately from

$$\operatorname{Spv}(\rho)^{-1}\mathcal{R}(f_0|f_1,\cdots,f_m) = \mathcal{R}(\rho(f_0)|\rho(f_1),\cdots,\rho(f_m)).$$

□

Remark 1.2.9. For every ideal $I \subseteq R$, if $R \xrightarrow{\pi} R/I$ denotes the canonical projection, then the induced map $\mathrm{Spv}(R/I) \xrightarrow{\pi} \mathrm{Spv}(R)$ is an embedding. Explicitly, we have,

$$\mathrm{Spv}(R/I) = V_{\mathrm{Spv}(R)}(I) := \{x \in \mathrm{Spv}(R) \mid \mathrm{supp}\, x \supseteq I\}, \tag{1.2}$$

$$= \mathrm{Spv}(R) \setminus \bigcup_{f \in I} \mathcal{R}(f). \tag{1.3}$$

If I is finitely generated (i.e. $I = \langle g_1, \cdots, g_m \rangle_R$), then the right hand side of Equation (1.3) can be written as,

$$\mathrm{Spv}(R/I) = \bigcup_{i=1}^{m} \mathcal{R}(g_i)$$

which is the embedded subset. Alternatively, we have,

$$V_{\mathrm{Spv}(R)}(I) = \{x \in \mathrm{Spv}(R) \mid \forall f \in I, |f|_x = 0\}.$$

Furthermore, we can identify this with a homeomorphism,

$$\mathrm{Spv}(R/I) \overset{\mathrm{Spv}(\pi)}{\underset{\simeq}{\to}} V(I).$$

Let S be a multiplicative subset of S and $\iota : R \to R_S$ be a canonical inclusion then,

$$\mathrm{Spv}(R_S) \underset{\mathrm{Spv}(\iota)}{\overset{\simeq}{\to}} \bigcap_{s \in S} (\mathrm{Spv}(R) \setminus V(s)) = \bigcap_{s \in S} \mathcal{R}(s|).$$

Proposition 1.2.4. Let s be a specialization of $\eta \in \mathrm{Spv}(R)$, then

$$\mathrm{supp}\, \eta \subseteq \mathrm{supp}\, s.$$

Proof.

$$r \notin \mathrm{supp}\, s \iff s \in \mathcal{R}(r|) \Rightarrow \eta \in \mathcal{R}(r|) \iff r \notin \mathrm{supp}\, \eta.$$

□

Definition 1.2.9. Let $\gamma \in \Gamma \cup \{0\}$ and $R \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation. We then define,

$$K_{v \leq \gamma} := \{r \in R \mid v(r) \leq \gamma\}.$$

For an equivalence class $x \in \mathrm{Spv}(R)$ where $v \in x$, we may denote,

$$K_{x \leq \gamma} = K_{v \leq \gamma}.$$

For $\rho \in R$, we denote,

$$K_{v \leq \rho} := K_{v \leq v(\rho)}, K_{x \leq \rho} := K_{x \leq |\rho|_x}.$$

Similarly,

$$K_{v < \gamma} = K_{x < \gamma} := \{r \in R \mid v(r) < \gamma\}$$
$$K_{v < \rho} = K_{x < \rho} := K_{v < v(\rho)}.$$

Proposition 1.2.5. $K_{v\leq 1}$ is an integrally closed subring of R, and $K_{v<1}$ is a prime ideal in $K_{v\leq 1}$.

Proof. These being a subring and an ideal can be checked easily. Let $r \notin K_{r\leq 1}$, then if m is a positive integer and $(p_i)_{i=1,\cdots,m} \in K_{v\leq 1}^m$, we have,

$$\begin{aligned} v(r^m) &= v(r)^m, \\ &> v(r)^{m-1} \geq \max_{1\leq i\leq m-1} v(p_i)v(r)^i, \\ &= \max_{1\leq i\leq m-1} v(p_i r^i), \\ &\geq v\left(\sum_{i=1}^{m-1} p_i r^i\right). \end{aligned}$$

Hence,

$$r^m \neq \sum_{i=1}^{m-1} p_i r^i.$$

Thus r is not integral over $K_{v\leq 1}$, in particular, $K_{v\leq 1}$ is integrally closed.

For $K_{v<1}$ being prime, obviously $1 \notin K_{v<1}$. Let $a, b \in K_{v\leq 1}\backslash K_{v<1}$. Then $v(a) = v(b) = 1$. Therefore $v(ab) = 1$. Showing $K_{v<1}$ is a prime ideal. □

Proposition 1.2.6. Let s be a specialization of $\eta \in \mathrm{Spv}(R)$. We have,

$$a \in R\backslash \operatorname{supp} s \Rightarrow K_{\eta<a} \subseteq K_{s<a}.$$

Applying this to $a = 1$, (recall that $\operatorname{supp} s$ is a prime ideal), we get,

$$K_{\eta<1} \subseteq K_{s<1}.$$

Proof. By the assumption and Proposition 1.2.4, we know that $|a|_\eta \neq 0$. Therefore,

$$b \notin K_{\eta<a} \iff \eta \notin \mathcal{R}(b|a) \Rightarrow s \notin \mathcal{R}(b|a) \iff b \notin K_{s<a}.$$

□

Proposition 1.2.7. Let s be a specialization of $\eta \in \mathrm{Spv}(R)$.

$$a \in R\backslash \operatorname{supp}(s) \Rightarrow K_{s\leq a} \subseteq K_{\eta\leq a}.$$

In particular for $a = 1$, we get,

$$K_{s\leq 1} \subseteq K_{\eta\leq 1}.$$

Proof. By Proposition 1.2.4, $|a|_\eta \neq 0$. Hence,

$$b \in K_{s\leq a} \iff s \in \mathcal{R}(a|b) \Rightarrow \eta \in \mathcal{R}(a|b) \iff b \in K_{\eta\leq a}.$$

□

Definition 1.2.10. Let X be a linearly ordered set. For $a, b \in X$ with $a \leq b$, we define the line segment between a and b to be

$$[a, b]_X := \{x \in X \mid a \leq x \leq b\}.$$

A subset $C \subseteq X$ is convex if for any two elements $c, d \in C, c \leq d$, we have $[c, d]_X \subseteq C$.

Definition 1.2.11. A convex subgroup Θ is a subgroup of a linearly ordered group Γ which is convex.

Proposition 1.2.8. A subgroup Θ of a linearly ordered group Γ is convex if and only if for any $\theta \in \Theta$ with $\theta < 1$, $[\theta, \theta^{-1}]_\Gamma \subseteq \Theta$.

Recall that for a valuation $v : R \to \Gamma \cup \{0\}$, Γ_v is the subgroup of Γ generated by $v(R)\backslash\{0\}$.

Definition 1.2.12. $c\Gamma_v$ is the convex subgroup of Γ_v generated by $\{v(r) \mid r \in R, v(r) \geq 1\}$.

Remark 1.2.10. Equivalently, we can define $c\Gamma_v$ to be a subgroup generated by $\{v(r) \mid r \in R, v(r) > 1\}$.

Proposition 1.2.9.

$$c\Gamma_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \min\{\gamma, \gamma^{-1}\}v(r) \geq 1\}$$

Proof. Let $c\Gamma'_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \min\{\gamma, \gamma^{-1}\}v(r) \geq 1\}$. We will show that $c\Gamma_v = c\Gamma'_v$. If $r \in R$ is as in the construction of $c\Gamma'_v$, then $v(r) \geq 1$ thus $v(r) \in c\Gamma_v$. If $\gamma \in c\Gamma'_v$, then

$$v(r)^{-1} \leq \gamma \leq v(r).$$

Hence $\gamma \in c\Gamma_v$. In particular $c\Gamma'_v \subseteq c\Gamma_v$. On the other hand $c\Gamma'_v$ is a convex subgroup of Γ_v containing all $v(r)$ such that $v(r) \geq 1$. Thus by the minimality of $c\Gamma_v$, we have shown the statement. □

Remark 1.2.11. Even if H is a convex subgroup of Γ_v, it does not imply that $c\Gamma_v \subseteq H$.

Remark 1.2.12. If R is a field then $c\Gamma_v = \Gamma_v$.

Definition 1.2.13. Let $A \xrightarrow{v} \Gamma$ be a valuation of A and $c\Gamma \subseteq H \subseteq \Gamma_v$ a convex subgroup. Let

$$v|_H(r) := \begin{cases} v(r) & (v(r) \in H) \\ 0 & (\text{otherwise}). \end{cases}$$

Proposition 1.2.10. Under the assumption of Definition 1.2.10, $v|_H$ is indeed a valuation of A and $v|_H$ is a specialization of v in $\mathrm{Spv}(A)$.

Proof. Let M be the convex hull of $H \cup \{0\}$ in $\Gamma_v \cup \{0\}$. For $m \in M$, set,

$$\varphi(m) = \begin{cases} m, & (m \in H), \\ 0, & (\text{otherwise}). \end{cases}$$

Note that $0, 1 \in M$, therefore $[0,1]_{\Gamma \cup \{0\}} \subseteq M$ and since $c\Gamma_v \subseteq H \subseteq M$, we have $v(A) \subseteq M$. Set $w := v|_H = \varphi v$ which is well-defined by the previous observation. As $\varphi(0) = 0$, and $\varphi(1) = 1$, the axioms

$$w(0) = 0, w(1) = 1, w(a+b) \leq \max\{w(a), w(b)\}$$

follows from their counterparts for v. To show the remaining axioms for valuation, let $m_1, m_2 \in M$, we will show that

$$\varphi(m_1 m_2) = \varphi(m_1)\varphi(m_2).$$

This is obvious when $m_1, m_2 \in H$. Suppose $\varphi(m_1 m_2) = 0$ then we can assume without loss of generality that $m_1 \notin H$. Then we will show $m_1 < 1$ and $hm_1 < 1$ for all $h \in H$. Indeed if $m_1 \geq 1$, then there is some $h \in H$ such that $1 \leq m_1 \leq h$. Then $m_1 \in H$, which is a contradiction. Similarly if $hm_1 \geq 1$ for some h then $hm_1 \in H$ thus $m_1 \in H$ which is again a contradiction. This further shows that for $\mu \in M$ as we have $0 \leq \mu \leq h$ for some $h \in H$, $m_1 \mu < 1$ holds. In particular $m_1 m_2 < 1$. Thus w is a valuation.

To see w is a specialization of r, note that when $v(f_i) > v(f_0) \in H$ then either $v(f_i) \geq 1$ or $v(f_i) < 1$. For the former case, $v(f_i) \in H$, otherwise $v(f_0) < v(f_i) < 1$ which implies $v(f_i) \in H$ by the convexity of H. Hence $v(f_i)$ bounded from above by some elements of H. Hence $w(f_i) = v(f_i) > v(f_0) = w(f_0)$ if $v(f_i) > v(f_0)$. Finally, observe that,

$$\begin{aligned} w \in \mathcal{R}(f_0|f_1, \cdots, f_m) &\iff [w(f_0) \neq 0 \text{ and } w(f_0) \geq w(f_i), i = 1, \cdots, m], \\ &\iff [v(f_0) \in H \text{ and } v(f_0) \geq w(f_i), i = 1, \cdots, m], \\ &\iff [v(f_0) \in H \text{ and } v(f_0) \geq v(f_i), i = 1, \cdots, m], \\ &\iff [v(f_0) \neq 0 \text{ and } v(f_0) \geq v(f_i), i = 1, \cdots, m] \\ &\iff (v/\sim) \in \mathcal{R}(f_0|f_1, \cdots, f_m). \end{aligned}$$

□

Definition 1.2.14. In particular, $s \in \mathrm{Spv}(R)$ is called a horizontal specialization of $\eta \in \mathrm{Spv}(R)$ if there is $v \in \eta$ and a convex subgroup H such that $c\Gamma_v \subseteq H \subseteq \Gamma_v$ such that

$$v|_H \in s.$$

Corollary 1.2.1. If x is a closed point of $\mathrm{Spv}(A)$, then $c\Gamma_x = \Gamma_x$.

Proof. If this is not the case, $x|_{c\Gamma_x}$ is a proper specialization of x. □

Proposition 1.2.11. If $s \in \mathrm{Spv}(A)$ is a horizontal specialization of $\eta \in \mathrm{Spv}(A)$ and $\Omega = \mathcal{R}(f_0|f_1, \cdots, f_m)$ then $s \in \Omega \iff (\eta \in \Omega \text{ and } |f_0|_s \neq 0)$.

Proof. Indeed, the construction of the end of the proof of Proposition 1.2.10 imply

$$(w/\sim) \in \Omega \iff [(v/\sim) \in \Omega \text{ and } w(f_0) \neq 0].$$

□

Definition 1.2.15. Let $A \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation and a convex subgroup $H \subseteq \Gamma$ or $H \subseteq \Gamma_v$. The vertical specialization of v with respect to H is

$$(v/H)(a) = \begin{cases} v(a)H, & (v(a) \neq 0), \\ 0, & (\text{ otherwise }), \end{cases}$$

where $v(a)H$ denotes the coset represented by $v(a)$.

Remark 1.2.13. s is a vertical specialization of $\eta \in \mathrm{Spv}(A)$ if and only if s is a specialization of η and $\operatorname{supp} s = \operatorname{supp} \eta$.

Remark 1.2.14. In Huber's literature, he refers horizontal specialization as primary specialization and the vertical specializations as the secondary specializations.

Remark 1.2.15. Note that $v|_H$ is a specialization of v while v is a specialization of v/H.

1.3 Valuation Rings

Definition 1.3.1. Let K be a field. A subring $R \subseteq K$ is a valuation ring of K if for all $x \in K^\times, x \in R$ or $x^{-1} \in R$.

Example 1.3.1. Every discrete valuation rings is a valuation ring of its field of fraction. More concretely, $\mathbb{Z}_p$ is a valuation ring of $\mathbb{Q}_p$.

Example 1.3.2. If S is a valuation ring of L and $K \subseteq L$ is a subfield then $S \cap K$ is a valuation ring of K.

Example 1.3.3. If $\mathcal{L}$ is a set of valuation rings which is linearly ordered by $\subseteq$. Then $\bigcap_{R \in \mathcal{L}} R$ is a valuation ring of K (if $\mathcal{L} = \emptyset$ then the intersection would be K itself which is trivially a valuation ring) and if $\mathcal{L} \neq \emptyset$ then $\bigcup_{R \in \mathcal{L}} R$ is also a valuation ring of K.

Example 1.3.4. If R is a valuation ring of K and $R \subseteq S \subseteq K$ then S is a valuation ring of K.

Proposition 1.3.1. Let R be a valuation ring of K. We have the following.

a. The field of fractions of R is K.

b. For $a, b \in R$, either $a \mid b$ or $b \mid a$.

c. R is a Bézout domain. That is every finitely generated ideals are principal.

d. R is integrally closed in K.

e. The following conditions are equivalent for R

   i. R is Noetherian.

   ii. R is a discrete valuation ring.

   iii. The unique maximal ideal $\mathfrak{m}_R$ is principal.

   iv. R is a principal ideal domain.

f. The set of ideals of R is linearly ordered by the inclusion.

g. $\mathrm{Spv}(R)$ is linearly ordered by specialization.

h. An ideal $I \subseteq R$ is prime if and only if $I = \sqrt{I}$ and $1 \notin I$.

i. We have an order reversing bijection such that

$$\mathrm{Spec}(R) \underset{R\cap\mathfrak{m}_S \hookleftarrow S}{\overset{\mathfrak{p}\mapsto R_{\mathfrak{p}}}{\rightleftarrows}} \{S \mid R \subseteq S \subseteq K \text{ and } S \text{ is a subring of } K\}$$

j. The set

$$\{S \mid S \text{ is a subring of } K \text{ such that } R \subseteq S\}$$

is linearly ordered by inclusion.

Proposition 1.3.2. If R and S are valuation ring of K then $R \subseteq S$ is equivalent to $\mathfrak{m}_S \subseteq \mathfrak{m}_R$.

Proof.

$$\mathfrak{m}_R = \{0\} \cup \{x \in K^\times \mid x \notin R\}.$$

□

Proposition 1.3.3. We have following bijections,

$$\mathrm{Spv}(K) \underset{v_R/\sim \hookleftarrow R}{\overset{x\mapsto\{k\in K \mid |k|_x\leq 1\}}{\rightleftarrows}} \{\text{valuation rings } R \text{ of } K\}$$

where $v_R : K \to \Gamma_R$ is given by

$$v_R(x) = \begin{cases} x \cdot R^\times, & (x \neq 0), \\ 0, & (x = 0). \end{cases},$$

$$\Gamma_R = K^\times / R^\times.$$

The ordering on Γ_R is given as follows,

$$xR^\times \leq yR^\times \iff x \in yR.$$

Remark 1.3.1. Under the bijection in Proposition 1.3.3, if the embedding of K into a field extension L is denoted i, then $\mathrm{Spv}(i)(x_R) = x_{K\cap R}$ where R is a valuation ring of L, $x_R \in \mathrm{Spv}(L)$. Moreover, $x_R \in \mathcal{R}(f_0|f_1, \cdots, f_m)$ if and only if $f_0 \neq 0$ and $f_i \in f_0 R$.

Proposition 1.3.4. *x_R is a specialization of x_S if and only if $R \subseteq S$.*

Proof.
If $R \subseteq S$, then by Remark 1.3.1 and

$$\begin{aligned} x_R \in \mathcal{R}(f_0|f_1, \cdots, f_m) &\iff [f_0 \neq 0 \text{ and } \forall 1 \leq i \leq m, f_i \in f_0 R] \\ &\Rightarrow [f_0 \neq 0 \text{ and } \forall f_i \in f_0 S], \\ &\iff [x_S \in \mathcal{R}(f_0|f_1, \cdots, f_m)]. \end{aligned}$$

If x_R is a specialization of x_S, then $R = K_{x_R \leq 1} \subseteq K_{x_S \leq 1} = S$ by Proposition 1.2.7. □

Lemma 1.3.1. Let B be an A-algebra and $M \subseteq B$ be a finitely generated A-submodule such that $\forall b \in B, bM = 0 \Rightarrow b = 0$. Then if $b \in B$ satisfies $bM \subseteq M$, b is integral over A.

Proof. Let $(m_i)_{i=1}^l$ be generators of M as an A-module.

$$bm_i = \sum_{i=1}^{l} \alpha_{ij} m_j.$$

Set $p \in A[T]$ be the characteristic polynomial of (α_{ij}). Then $p(b)M = 0$ thus $p(b) = 0$, showing b is integral over A. □

Proposition 1.3.5. Let A be a subring of the field K. Then the integral closure of A in K is equal to the intersection of the valuation rings of K containing A.

Proof. Let A' be the intersection of all valuation rings containing A. Then A' contains the integral closure of A as all valuation rings are integrally closed. Thus it is sufficient to consider an element $x \in K$ which is not integral over A and show there is a valuation ring R of K which doesn't contain x. To this end, set

$$\mathcal{M} := \{A \subseteq B \subseteq K \mid B \text{ is an intermediate ring with } x \text{ is not integral over } B\}.$$

Since $A \in \mathcal{M}$, $\mathcal{M} \neq \emptyset$. Let $\mathcal{L} \subseteq \mathcal{M}$ be a non-empty set which is linearly ordered with respect to inclusion. Then $C = \bigcup_{B\in\mathcal{L}} B$ is in $\mathcal{M}$. That is $\mathcal{L}$ is bounded in $\mathcal{M}$. By Zorn's lemma, there is a maximal element R. It remains to show that R is a valuation ring.

Let $k \in K \backslash R$ such that $k^{-1} \notin R$. Then x is integral over $R[k]$ and $R[k^{-1}]$ by the maximality of R, hence there are polynomials

$$(p_j^+)_{j=0}^{d_+-1} \in R[T]^{d_+-1}, (p_j^-)_{j=0}^{d_--1} \in R[T]^{d_--1}$$

such that

$$x^{d_+} = \sum_{j=0}^{d_+-1} x^j p_j^+(k), x^{d_-} = \sum_{j=0}^{d_--1} x^j p_j^-(k^-).$$

Let $d = \max\{d_+, d_-\}$ and

$$e := \max\{\deg p_0^+, \cdots, \deg p_{d_+-1}^+, \deg p_0^-, \cdots, \deg p_{d_--1}^-\}.$$

Let $N \subseteq K$ be the R-submodule such that

$$N = \langle x^i k^j \mid 0 \leq i < d, 0 \leq j < 2e \rangle_R.$$

Then N is a finitely generated R-module and $xN \subseteq N$. Indeed, consider a generator $y = x^i k^j$ of N. First, we assume that $j < e$. If $i < d_+ - 1$ then xy is in the set generators of N. Otherwise,

$$xy = x^{i+1-d_+} \sum_{l=0}^{d_+-1} x^l p_l^+(k) k^j.$$

Note that

$$\deg_k p_l^+(k) k^j < 2e,$$
$$\deg_x x^{i+1-d_+} x^l \leq i + 1 - d_+ + d_+ - 1 = i.$$

Therefore $xy \in N$. Now assume $j \geq e$. If $i < d_- - 1$, then xy is in the set of generators of N. Otherwise,

$$xy = x^{i+1-d_-} \sum_{l=0}^{d_--1} x^l k^j p_l^-(k^{-1}) \in N.$$

Similarly as before $xy \in N$. By Lemma 1.3.1, x is integral over R. This contradicts that $R \in \mathcal{M}$. □

2 G_+-topological span and their sheaf cohomology

2.1 Grothendieck topologies and G_+-topological span

Let $\mathscr{C}$ be a category. In the application we are interested in, it will be given by a poset $(P, \preceq)$. Objects of $\mathscr{C}$ are the all of P and a morphism $x \xrightarrow{\mu} y$ exists if and only if $x \preceq y$, in which case μ is unique. In fact $P = \mathscr{O}_X$ (all open subsets of X) or P is some topology base of X, and $\preceq$ is containment $\subseteq$. However, the case where there may be more than one morphism between objects of $\mathscr{C}$ are very important in other applications like étale or crystalline cohomology.

Definition 2.1.1. A sieve on (or over) an object X of $\mathscr{C}$ or an X-sieve is a class $\mathscr{S}$ of morphisms $U \to X$ such that $[U \xrightarrow{\nu} X] \in \mathscr{S}$ then $\nu\omega \in \mathscr{S}$ where $V \xrightarrow{\omega} U$ is any morphism in $\mathscr{C}$.

Definition 2.1.2. If $\mathscr{S}$ is a X-sieve and $Y \xrightarrow{\xi} X$ then $\xi^*\mathscr{S}$ is the Y-sieve of all morphism $U \xrightarrow{\nu} Y$ such that $\xi\nu \in \mathscr{S}$.

Definition 2.1.3. A Grothendieck topology on $\mathscr{C}$ is a collection $(\mathcal{T}_X)_{X\in \mathrm{Ob}(\mathscr{C})}$ of X-sieves (where $\mathcal{T}_X$ being called the class of covering sieves of X) satisfying the following conditions.

I. (GT-Triv) The all-sieve $\mathscr{S}$ on X (i.e. the class of all morphisms with its target to X) is in $\mathcal{T}_X$.

II. (GT-Trans) If $\mathscr{S} \in \mathcal{T}_X$, then for any morphism $Y \xrightarrow{\nu} X$ in $\mathscr{C}$, we have $\nu^*\mathscr{S} \in \mathcal{T}_Y$.

III. (GT-Loc) Let $\mathscr{T}$ be a X-sieve. If there is $\mathscr{S} \in \mathcal{T}_X$ such that $\nu^*\mathscr{T} \in \mathcal{T}_U$ for any morphism $(U \xrightarrow{\nu} X) \in \mathscr{S}$, then $\mathscr{T} \in \mathcal{T}_X$.

Proposition 2.1.1.

1. If $\mathscr{S} \subseteq \mathscr{T}$ and $\mathscr{S} \in \mathcal{T}_X$, then $\mathscr{T} \in \mathcal{T}_X$.
2. If $\mathscr{S}_1, \mathscr{S}_2 \in \mathcal{T}_X$, then $\mathscr{S}_1 \cap \mathscr{S}_2 \in \mathcal{T}_X$.

Proof. If $[U \xrightarrow{\nu} X] \in \mathscr{S}$,

$$\nu^*\mathscr{S} = \{\sigma : V \to U \mid \nu\sigma : V \to X \in \mathscr{S}\}.$$

Thus for any $\sigma : V \to U \in \mathscr{C}$, $\sigma \in \nu^*\mathscr{S}$ by using the definition of sieves. Thus $\nu^*\mathscr{S}$ is the all-sieve on U. Observing that $\nu^*\mathscr{T} \supseteq \nu^*\mathscr{S}$, hence $\nu^*\mathscr{T} =$ (all sieve on U) $\in \mathcal{T}_U$ by GT-Triv. By GT-Loc, $\mathscr{T} \in \mathcal{T}_X$.

Let $\mathscr{T} = \mathscr{S}_1 \cap \mathscr{S}_2$. If $[U \xrightarrow{\nu} X] \in \mathscr{S}_1$, as in the previous argument $\nu^*\mathscr{S}_1$ is the all-sieve, thus $\nu^*\mathscr{T} = \nu^*\mathscr{S}_2 \in \mathcal{T}_U$. By GT-Trans and $\nu^*\mathscr{S}_2 = \mathscr{T} \in \mathcal{T}_U$. By GT-Loc $\mathscr{T} \in \mathcal{T}_X$. □

From now on, $\mathscr{C}$ will be given by some topology base $\mathcal{B}$ in a topology space X turned into a poset by the inclusion. We then give a more specific definition for sieves in $\mathscr{C}$.

Definition 2.1.4. Let $\mathscr{C}$ be a category whose objects are open subsets of X and morphisms are inclusions.

1. A sieve $\mathscr{S}$ on $U \in \mathscr{C}$ is a set of open subsets of U such that for any $V \in \mathscr{S}$, every open subsets W of V is also in $\mathscr{S}$.
2. Let $V \subseteq U$ and $\mathscr{S}$ be a sieve on U, then $\mathscr{S}|_V = \mathscr{S} \cap \{\text{open subsets of } V\}$.

Definition 2.1.5. Let X be a topological space and $\mathcal{B}_X$ be the topology base of X. Let $(U_i)_{i\in I}$ be a collection of elements of $\mathcal{B}_X$. Then,

$$[U_i|i \in I] := \bigcap_{\substack{X\text{-sieve } \mathscr{S} \\ \text{s.t. } \forall i\in I, U_i\in\mathscr{S}}} \mathscr{S},$$
$$= \{U \in \mathscr{O}_X \mid \exists i \in I \text{ s.t. } U \subseteq U_i\}.$$

Remark 2.1.1. In order to see the two definitions given above note that the latter is a sieve and contains all U_i and indeed a sieve on X which is minimal among ones contains all $\{U_i\}_{i\in I}$. The case I is finite follows from Proposition 2.1.1.

Definition 2.1.6. A W-sieve is finitely generated if it has the form $[\Omega_1, \cdots, \Omega_n]$ where all $\Omega \in \mathcal{B}_W$. If several topology bases are under consideration, we will use $[\Omega_1, \cdots, \Omega_n]_{\mathcal{B}}$ for disambiguation.

Let $U \in \mathscr{O}_X$. U-sieve $\mathscr{S}$ is admissible (denoted by $\mathscr{S} \vDash U$) by the Grothendieck topology $(\mathcal{T}_U)_{U\in\mathscr{O}_X}$ if $\mathscr{S} \in \mathcal{T}_U$. Also for an arbitrary sieve $\mathscr{S}$, we write $\mathscr{S}//U$ if $\mathscr{S}|_U \vDash U$.

Proposition 2.1.2. Let X be a topological space and $\mathcal{B}$ be a topology base for it. Then we have a bijection between the following two sets.

1. Grothendieck topologies $\mathcal{T}$ on $\mathcal{B}$.
2. Grothendieck topologies $\mathcal{T}$ on $\mathscr{O}_X$ with the additional property that $[\mathcal{B}_U] \in \mathcal{T}_U$ holds for all $U \in \mathscr{O}_X$.

The bijection is explicitly given by the following way.

a. If a Grothendieck topology $\mathcal{T}^{(\mathcal{B})}$ on $\mathcal{B}$ is given, $\mathscr{S} \in \mathcal{T}_U$ if and only if $\mathscr{S}\cap\mathcal{B}_\Omega \in \mathcal{T}_\Omega^{(\mathcal{B})}$ holds for all $\Omega \in \mathcal{B}_U$.

b. If a Grothendieck topology $\mathcal{T}$ on $\mathscr{O}_X$ is given, $\mathscr{S} \in \mathcal{T}_\Omega^{(\mathcal{B})}$ if and only if $[\mathscr{S}] \in \mathcal{T}_\Omega$, (where $[\mathscr{S}] = [\Theta \mid \Theta \in \mathscr{S}]$ is the sieve on $\mathscr{O}_\Omega$ generated by $\mathscr{S}$).

Proof.

We first show that the map in a is well-defined. GT-Triv is obvious as $\mathscr{O}_X \cap \mathcal{B} = \mathcal{B} \vdash \mathcal{T}^{(\mathcal{B})}$. For GT-Trans, let $\xi : V \hookrightarrow U$ and $\mathscr{S} \in \mathcal{T}$, then □

Definition 2.1.7. A G_+-topological space is a T_0-space X with a Grothendieck topology on $\mathscr{O}_X$ such that

1. $\emptyset \vDash \emptyset$ (implies $\mathcal{G}(\emptyset) = \{0\}$ for sheaves $\mathcal{G}$ of abelian groups).
2. If $\mathscr{S} \vDash U$ then $U = \bigcup_{V\in\mathscr{S}} V$.

Definition 2.1.8. If X is a G_+-space then a G_+-base for X is a subset $\mathcal{B} \subseteq \mathscr{O}_X$ such that $\mathcal{B}$ is a topology base in the ordinary sense and $[\mathcal{B}_U] \vDash U$ for all $U \in \mathscr{O}_U$. We say that $\mathcal{B}$ is G_{++} if in addition, we have,

a. $\Omega, \Theta \in \mathcal{B} \Rightarrow \Omega \cap \Theta \in \mathcal{B}$.

b. Membership in $\mathcal{B}$ is local in the sense that $\Omega \in \mathcal{B}, U \in \mathscr{O}_\Omega$ such that $U \subseteq \Omega$ and $[\Theta \in \mathcal{B}_\Omega \mid U \cap \Theta \in \mathcal{B}] \vDash \Omega$ then $U \in \mathcal{B}$.

Definition 2.1.9. If X is a G_+-space then $V \in \mathscr{O}_X$ is called G_+-quasi-compact (abbreviated as (G_{+qc})) if for all $\mathscr{S} \in \mathcal{T}_U$, there are finitely many $(U_i)_{i=1,\cdots,n} \in \mathscr{S}^n$ such that

$$[U_1, \cdots, U_n] \vDash U.$$

(i.e. every covering sieve of U has a finitely generated covering subsieve).

Example 2.1.1. If X is a T_0-space it gets the structure of a G_+-space by

$$\mathcal{T}_U = \left\{ \text{sieves } \mathscr{S} \text{ in } U \text{ such that } U = \bigcup_{V \in \mathscr{S}} V \right\}.$$

Proposition 2.1.3. Let X be a T_0-space, $\mathcal{B}$ a topology base for it such that $\Omega, \Theta \in \mathcal{B}$ implies $\Omega \cap \Theta \in \mathcal{B}$. For each $U \in \mathscr{O}_X$, put

$$\mathcal{T}_U = \left\{ \text{sieves } \mathscr{S} \text{ on } U \;\middle|\; \forall \Omega \in \mathcal{B}_U, \exists (\Theta_i)_{i=1}^n \in (\mathscr{S} \cap \mathcal{B}_\Omega)^n \text{ such that } \Omega = \bigcup_{i=1}^n \Theta_i \right\}$$

Then X becomes a G_+-topological space and $\mathcal{B}$ is a G_+-base and the elements of $\mathcal{B}$ are G_{+qc}.

Proof. GT-Triv follows since for any open set U, U itself is a subset of U. Let $\mathscr{S}$ be an admissible U-sieve and $\nu : V \hookrightarrow U$. Then as $\mathcal{B}_V \subseteq \mathcal{B}_U$, we have GT-Trans. Let $\mathscr{S}, \mathscr{T}$ be U-sieves such that $\mathscr{S}$ is admissible. Suppose for any $\nu : V \hookrightarrow U \in \mathscr{S}$, $\nu^* \mathscr{T} \in \mathcal{T}_V$ holds. For $\Omega \in \mathcal{B}_U$, take $\{\Theta_1, \cdots, \Theta_n\}$ as in the construction. Set $\nu_i : \Theta_i \to U \in \mathscr{S}$ for each $i = 1, \cdots, n$. Then for each $i = 1, \cdots, n$,

$$(\nu_i)_* \mathscr{T} \vDash \Theta_i \Rightarrow \exists (\Xi_j^i)_{j=1,\cdots,r_i} \in (\mathscr{T} \cap \mathcal{B}_{\Theta_i})^{r_i} \;\text{ s.t. }\; \Theta_i = \bigcup_{j=1}^{r_i} \Xi_j^i.$$

In particular, Ω is covered by finitely many elements of $\mathscr{T} \cap \mathcal{B}_\Omega$. Thus $\mathscr{T} \in \mathcal{T}_U$.

To show that it is a G_+-topological space. First, $\emptyset \vDash \emptyset$ is clear from the definition. The other parts of G_+ for X is also clear. Also for $U \in \mathscr{O}_X$, $[\mathcal{B}_U] \vDash U$ as for any $\Omega \in \mathcal{B}_U$, $\Omega \in [\mathcal{B}_\mathcal{U}]$ hence $n = 1, \Theta_1 = \Omega$ can be taken.

Finally let $\mathscr{S} \vDash \Omega$ for $\Omega \in \mathcal{B}$. If $\Theta_1, \cdots, \Theta_n$ are as in the formulation of the Proposition. Let $\mathscr{T} = [\Theta_1, \cdots, \Theta_n]$ then $\mathscr{T} \vDash \Omega$ as when $\Xi \in \mathcal{B}_\Omega$ and $\Xi_i = \Xi \cap \Theta_i \in \mathcal{B} \cap \mathscr{T}$ by the definition of topology bases and $\Xi = \bigcup_{i=1}^n \Xi_i$. □

Remark 2.1.2. The assertion of the Proposition 2.1.3 will still hold when we relax the condition as follows.

$$\Omega, \Theta \in \mathcal{B} \Rightarrow \exists n \in \mathbb{N}, \exists (\Xi_i)_{i=1,\cdots,n} \in \mathcal{B}^n \;\text{ s.t. }\; \Omega \cap \Theta = \bigcup_{i=1}^n \Xi_i.$$

Definition 2.1.10. The above Grothendieck topology on X is called the G_+-topology on X enforcing the quasi-compactness of the elements of $\mathcal{B}$.

Definition 2.1.11. A sieve in $\mathscr{O}_X$ is called a prime sieve if $n \in \mathbb{N}$ and $\bigcap_{i=1}^n U_i \in \mathscr{S}$, (intersection in X and all $U_i \in \mathscr{O}_U$) implies that there is $1 \leq i \leq n$ such that $U_i \in \mathscr{S}$.

Remark 2.1.3. It is sufficient to require this when $n = 0, (X \notin \mathscr{S})$ and $n = 2, (U \cap V \in \mathscr{S} \Rightarrow U \in \mathscr{S}$ or $V \in \mathscr{S})$.

Proposition 2.1.4. Let X be a G_+-space. For a subset $\xi \subseteq \mathscr{O}_X$ the following conditions are equivalent.

a. ξ satisfies

   i. $U \in \xi, V \in \mathscr{O}_X, U \subseteq V \Rightarrow V \in \xi$.

   ii. $U, V \in \xi \Rightarrow U \cap V \in \xi$ and moreover $X \in \xi$.

   iii. If $U \in \xi$ and $\mathscr{S} \in \mathcal{T}_U$ then $\mathscr{S} \cap \xi \neq \emptyset$.

b. $\mathscr{T} = \mathscr{O}_X \backslash \xi$ is a prime sieve containing every open subset covered by it ($U \in \mathscr{O}_X$ and $\mathscr{T}//U$ implies $U \in \mathscr{T}$).

Proof.
a⇒b). $\mathscr{T}$ is indeed a sieve on X. Let $U \hookrightarrow X \in \mathscr{T}$. Then for any $V \hookrightarrow U$, if $V \notin \mathscr{T}$ then $U \notin \mathscr{T}$ by i). Thus $V \in \mathscr{T}$. By ii), $\mathscr{T}$ is prime. Finally, if $\mathscr{T}//U$ and $U \in \xi$, then $\mathscr{T} \cap \xi \neq \emptyset$ which is impossible.
Taking contrapositives of these we see b)⇒a. □

Definition 2.1.12. A collection ξ of open subsets of a G_+-space X satisfying above equivalent conditions is called a van der Put point.

Definition 2.1.13. The set of vander Put points of X is denoted by X^*.

Proposition 2.1.5. Let X be a G_+-space and $U \in \mathscr{O}_U$ be an open subset. Equip U with the G_+-structure $\mathcal{T}^{(U)} = \mathcal{T}|_{\mathscr{O}_U}$, we then have a bijection

$$\{\text{van der Put points } \xi_U \text{ of } U\} \underset{\xi_U = \xi \cap \mathscr{O}_U \hookleftarrow \xi}{\overset{\xi_U \mapsto \xi = \{V \in \mathscr{O}_X \mid V \cap U \in \xi_U\}}{\longrightarrow}} U^* = \{\xi \in X^* \mid U \in \xi\}$$

We have $(U \cap V)^* = U^* \cap V^*$.

Proof. $\xi_U \mapsto \xi$ is justified by the counterparts for ξ_U by using that $\cdot \cap U$ preserves inclusion and $\cap$ is associative. □

Definition 2.1.14. We equip X^* with the topology for which $\{U^* \mid U \in \mathscr{O}_X\}$ is a topology base.

Proposition 2.1.6.

a. If $\mathscr{S} \vDash U$ then $U^* = \bigcup_{V \in \mathscr{S}} V^*$.

b. If $\mathcal{B}$ is a G_+-base then $\mathcal{B}^* = \{\Omega^* \mid \Omega \in \mathcal{B}\}$ is a topology base for X^*.

c. $\xi \in X^*$ is a specialization (i.e. $\xi \in \overline{\{u\}}$ of $u \in X^*$) if and only if $\xi \subseteq u$.

d. If $\mathscr{S}$ is a X-sieve which is maximal with respect to inclusion among all non-admissible X-sieves. Then $\xi = \mathscr{O}_U \backslash \mathscr{S}$ is a closed point of X^*.

Proof.

For a, $\xi \in U^*$ then $U \in \xi$ by the assumption. BY Proposition 2.1.4, for an admissible U-sieve $\mathscr{S}$, we have $\mathscr{S} \cap \xi \neq \emptyset$. Therefore $\xi \in V^*$ for some $V \in \mathscr{S}$.

By a and $[\mathcal{B}_U] \vDash U$, $\mathcal{B}^*$ covers the space. By Proposition 2.1.5,

$$\Xi \subseteq \Omega \cap \Theta \overset{\text{Prop. 2.1.4 a.i)}}{\Rightarrow} \Xi^* \subseteq \Omega^* \cap \Theta^*.$$

Therefore it is indeed a topology basis.

ξ is a specialization of u if and only if for any $\Omega^* \in \mathcal{B}^*$, $\xi \in \Omega^* \Rightarrow u \in \Omega^*$ if and only if $\Omega \in \xi \Rightarrow \Omega \in u$. This extends to the set of all open subsets by Proposition 2.1.4 a.i).

Finally for d), if $X \in \mathscr{S}$ then $\mathscr{S}$ is the all-sieve on X, therefore $X \notin \mathscr{S}$ by GT-Triv. Let $U_1, U_2 \notin \mathscr{S}$ and set $U = U_1 \cap U_2$. We now show that $\mathscr{S}$ is prime. By maximality of $\mathscr{S}$, $\mathscr{S}_i = \mathscr{S} \cup \mathscr{O}_{U_i} \in \mathcal{T}_X$. If $V \in \mathscr{S}$ then But $\mathscr{S}|_V = \mathscr{S}_2|_V$ is immediate as these are all sieves on V. Suppose $V \in \mathscr{O}_{U_1} \backslash \mathscr{S}$, we have $\mathscr{S}|_V \subseteq \mathscr{S}_2|_V$. Let $W \in \mathscr{O}_{U_2}|_V$. Then $W \hookrightarrow V \hookrightarrow U$. Since $U \in \mathscr{S}$ and by the definition of sieves, we see $W \in \mathscr{S}|_V$. But then $\mathscr{S} \in \mathcal{T}_X$ by GT-Loc, a contradiction. Thus $\mathscr{S}$ is a prime sieve. If $\mathscr{S}//V$ and $\mathscr{S} \cup \mathscr{O}_V \in \mathcal{T}_X$ then

$$W \in \mathscr{S} \Rightarrow \mathscr{S}|_W = \mathscr{O}_W \vDash W,$$
$$W \subseteq V \Rightarrow \mathscr{S}|_W = (\mathscr{S}|_V)_W \overset{\mathscr{S}|_V \vDash V, \text{GT-Trans}}{\vDash} W.$$

Thus by GT-Loc, $\mathscr{S} \vDash X$ which is a contradiction. Thus $\mathscr{S} \cup \mathscr{O}_V \nvDash X$. By the maximality of $\mathscr{S}$, $V \in \mathscr{S}$. By Proposition 2.1.4, $\xi = \mathscr{O}_X \backslash \mathscr{S}$ is a van der Put point. If ξ has a proper specialization s then by c), $s \subsetneq \xi$ thus, $\mathscr{S} \subsetneq \mathscr{T} = \mathscr{O}_X \backslash s$. Again by the maximality $\mathscr{T} \vDash X$ but $X \notin \mathscr{T}$ a contradiction. □

Definition 2.1.15. Let X be a G_+-space. We say X has sufficiently many van der Put points if for all $U \in \mathscr{O}_X$, and every U-sieve $\mathscr{S}$, $U^* = \bigcup_{V \in \mathscr{S}} V^*$ implies $\mathscr{S} \vDash U$.

Lemma 2.1.1. If X is G_+-quasi-compact and $\mathscr{S}$ is an $\mathscr{O}_X$-sieve such that $X^* = \bigcup_{U \in \mathscr{S}} U^*$ then $\mathscr{S} \vDash X$.

Proof. Assume that $\mathscr{S}$ is not covering X. Set,

$$\mathcal{M} := \{\mathscr{T} \mid \mathscr{S} \subseteq \mathscr{T}, \mathscr{T} \notin \mathcal{T}_X\}.$$

We apply Zorn's lemma to $(\mathcal{M}, \subseteq)$. This is possible as $\mathcal{M} \neq \emptyset$ because $\mathscr{S} \in \mathcal{M}$ and if $\mathcal{L} \subseteq \mathcal{M}$ is non-empty subset linearly ordered by $\subseteq$, then

$$\widetilde{\mathscr{S}} = \bigcup_{\mathscr{T} \in \mathcal{L}} \mathscr{T}$$

is a sieve containing $\mathscr{S}$. If $\widetilde{\mathscr{S}} \in \mathcal{T}_X$, then by the G_+-quasi-compactness of X, there are $U_1, \cdots, U_n \in \widetilde{\mathscr{S}}$ such that $[U_1, \cdots, U_n] \vDash X$. Thus there is $\mathscr{T} \in \mathcal{L}$ such that all $U_1, \cdots, U_n \in \mathscr{T}$ therefore

$$[U_1, \cdots, U_n] \vDash X, [U_1, \cdots, U_n] \subseteq \mathscr{T} \overset{\text{Proposition 2.1.1}}{\Rightarrow} \mathscr{T} \vDash X.$$

This is a contradiction to $\mathscr{T} \in \mathcal{M}$.

It follows that $\mathcal{M}$ has a $\subseteq$-maximal element. By Proposition 2.1.6, $\xi = (\mathscr{O}_X \backslash \mathscr{S}_{\max}) \in X^*$. If $U \in \mathscr{S} \subseteq \mathscr{S}_{\max}$, then $\xi \notin U^*$ (otherwise $U \in \xi$, but by Proposition 2.1.5, $\xi \cap \mathscr{S}_{\max} \neq \emptyset$). A contradiction to our assumption that $X^* = \bigcup_{U \in \mathscr{S}} U^*$. □

Proposition 2.1.7. If X is a G_+-space which has a G_+-base $\mathcal{B}$ where elements are G_+-quasi-compact. Then X has sufficiently many van der Put points. If $\mathcal{B}$ is closed under finite intersection in X (including intersection over an empty set thus $X \in \mathcal{B}$), then X^* is a spectral space.

Proof. Let $U \in \mathscr{O}_X$ and $\mathscr{S}$ any U-sieve such that $U^* = \bigcup_{V \in \mathscr{S}} V^*$. Let $\Omega \in \mathcal{B}_U$ and $\mathscr{S}_\Omega = \mathscr{S}|_\Omega$. Then $\Omega^* \subseteq U^*$ hence

$$\Omega^* = \bigcup_{V \in \mathscr{S}} \Omega^* \cap V^* = \bigcup_{V \in \mathscr{S}} (\Omega \cap V)^*.$$

Since $\Omega \cap V \in \mathscr{S}_\Omega$, it follows that $\Omega^* = \bigcup_{V \in \mathscr{S}_\Omega} V^*$. By Lemma 2.1.1, $\mathscr{S} // \Omega$.

Set $\mathscr{T} = [\mathcal{B}_U]$ which is admissible as $\mathcal{B}$ is a G_+-base. For any $V \to U \in \mathscr{T}$, there is $\Omega \in \mathcal{B}_U$ such that $V \to \Omega \to U$ by definition. Therefore, from $\mathscr{S} // \Omega$ for any $\Omega \in \mathcal{B}_U$ and GT-Trans, we get for any $\nu : V \to U$, $\nu_* \mathscr{S} \vDash V$. By GT-Loc, $\mathscr{S} \vDash U$ hence X has sufficiently many van der Put points.

For the spectrality of X^*, it is easy to see that Ω^* is quasi-compact if Ω is G_+-quasi-compact. Then $\{\Omega^* \mid \Omega \in \mathcal{B}\}$ satisfy the requirement for the quasi-compactness part of the definition of spectrality. If $Z \subseteq X^*$ is an irreducible subset, set,

$$\xi = \{U \in \mathscr{O}_X \mid U^* \cap Z \neq \emptyset\}.$$

Then by definition conditions i), iii) of Proposition 2.1.4 are satisfied. Furthermore, using the Irreducibility of Z, ii) also holds. In particular, ξ is a van der Put point. Moreover, ξ is a generic point of Z as it is contained in all elements $\mathcal{B}^*$ intersecting Z. Finally, if $x, y \in X^*$ are distinct points, then there is $U \in \mathscr{O}_X$ such that $U \in x$ and $U \notin y$. Thus x, y are topologically distinguishable by U^*. □

Remark 2.1.4.

$$\operatorname{glim}_{X^*} \mathcal{U} = \{U \in \mathscr{O}_X \mid U \text{ contains some } \Omega \in \mathcal{B} \text{ such that } \Omega^* \in \mathcal{F}\}$$

All Ω^* with $\Omega \in \mathcal{B}$ are closed under this.

2.2 Sheaves

Definition 2.2.1. A presheaf on X with values in (sets, abelian groups, rings) is a contravariant functor $\mathcal{G}$ from $\mathscr{O}_X$ to one of these categories. Then all $\mathcal{G}(U)$ are objects in these categories and we have

$$V \subseteq U \Rightarrow \mathcal{G}(U) \overset{g \mapsto g|_V}{\to} \mathcal{G}(V),$$

which is a morphism in the target category such that $g \in \mathcal{G}$ then $g|_U = g, (g|_V)_W = g|_W$ if $W \subseteq V \subseteq U$. If this is only given on elements of a certain topology base $\mathcal{B}$, we speak of a presheaf on $\mathcal{B}$. In all cases, if a Grothendieck topology in $\mathcal{B}$ is given, then a sheaf is a presheaf $\mathcal{G}$ satisfying the additional assumption that

$$\mathcal{G}(\Omega) \overset{\sim}{\to} \varprojlim_{\Theta \in \mathscr{S}} \mathcal{G} = \left\{ (g_\Theta)_{\Theta \in \mathscr{S}} \in \prod_{\Theta \in \mathscr{S}} \mathcal{G}(\Theta) \mid g_\Theta|_\Xi = g_\Xi \text{ where } \Theta \in \mathscr{S}, \Xi \in \mathcal{B}_\Theta \right\}.$$

If the map is only injective, then $\mathcal{G}$ is separated.

Definition 2.2.2. Let $\mathcal{G}$ be a sheaf on a G_+-space X.

1. A stalk of $\mathcal{G}$ at $\xi \in X^*$ is such that

$$\mathcal{G}_\xi = \varinjlim_{\Omega \in \xi} \mathcal{G}(\Omega) = \{(\Omega, g) \mid \Omega \in \xi \cap \mathcal{B}, g \in \mathcal{G}(\Omega)\} / \sim$$

   where the equivalent relation is given by $(g, \Omega) \sim (r, \Theta)$ if there is $\Xi \subseteq \Omega \cap \Theta$ such that $\Xi \in \mathcal{B}$ and $g|_\Xi = g|_\Xi$.

2. A G-skyscraper sheaf of $\xi \in X^*$ is

$$([G]_\xi)(U) = \begin{cases} G, & U \in \xi, \\ \{0\} \end{cases}$$

Definition 2.2.3. Let X be a G_+-space with sufficiently many van der Put points and $\mathcal{G}$ be a sheaf on X. For an open subset $U \subseteq X$, pick

$$r \in \prod_{\xi \in U^*} \mathcal{G}_\xi.$$

We denote the sieve $\mathscr{S}_r$ on U generated in $\mathscr{O}_X$ to be such that

$$\mathscr{S}_r := \left[\Omega \in \mathcal{B}_U \,\middle|\, \exists g \in \mathcal{G}(\Omega) \text{ s. t. } \forall \xi \in \Omega^*, r_\xi = g_\xi \right].$$

Then we define,

$$(\mathrm{Gerb}(\mathcal{G}))(U) = \left\{ r \in \prod_{\xi \in U^*} \mathcal{G}_\xi \,\middle|\, \mathscr{S}_r \vDash U \right\}.$$

Proposition 2.2.1. Let X be a G_+-space with sufficiently many van der Put points.

a) The stalk is exact on presheaves.

b) $[G]_\xi$ is a sheaf and $\mathcal{G} \mapsto \mathcal{G}_\xi$ is left adjoint to $G \mapsto [G]_\xi$ on (pre)sheaves.

c) If $\mathcal{G}$ is a presheaf, the morphism $\mathcal{G} \to \mathrm{Sheaf}(\mathcal{G})$ sending $g \in \mathcal{G}(U)$ to (image of g in $\mathcal{G}_\xi)_{\xi \in U^*} \in \prod_{\xi \in U^*} \mathcal{G}_\xi$ defines an isomorphisms on stalks.

d) $(\mathcal{G}_\mathcal{B})_\xi \xrightarrow{\simeq} \mathcal{G}_\xi$ for all $x \in\in X$.

e) A morphism of presheaves defining isomorphism on stalks defines an isomorphism on sheafification.

f) If $\mathcal{G}$ is a sheaf, $\mathcal{G} \to \mathrm{Sheaf}(\mathcal{G})$ is an isomorphism. If $\mathcal{G}$ is only defined on $\mathcal{B}$, $\mathcal{G} \xrightarrow{\simeq} \mathrm{Sheaf}(\mathcal{G})|_\mathcal{B}$.

g) A morphism of sheaves defining isomorphism on all stalks is an isomorphism.

h) Sheafification (Presheaves on $\mathcal{B}$) $\to$ (sheaves on $\mathscr{O}_X$) is left adjoint to the functor of restriction to $\mathcal{B}$.

i) One has an equivalence of category between sheaves on X (for the Grothendieck topology on X) and (ordinary) sheaves on X^*, sending a sheaf $\mathcal{G}$ on X to the sheafification of the (pre)sheaf $U^* \to \mathcal{G}(U)$ on $\{U^* \mid U \in \mathscr{O}_X\}$ (a topology base for X^*) and a sheaf $\mathcal{H}$ on X^* to $\mathcal{G}(U) \to \mathcal{H}(U^*)$.

Proof. a-h) are shown in some ways as in derived sheaf theory. i) follows from these □

3 General non-Archimedean topological ring

3.1 General topological rings and modules

Definition 3.1.1. A topological group is a group G with a topology on the underlying set such that multiplication and inversion are continuous. That is

$$\mu(g,h) = gh, \iota(g) = g^{-1}$$

are continuous.

Definition 3.1.2. A topological ring is a ring R with a topology on the underlying set such that it is a topological group under addition and

$$R \times R \ni (a,b) \mapsto ab \in R$$

is continuous. (We do not assume the continuity of $a \mapsto a^{-1}$ for $a \in R^\times$).

Definition 3.1.3. Let R be a topological ring. A topological R-module M is an R-module with topology on its underlying set such that $(M, +)$ is a topological group and

$$R \times M \ni (a, m) \mapsto am \in M$$

is continuous.

Proposition 3.1.1. For a topological group G, the following conditions are equivalent.

a. G is T_0 (Kolmogorov).

b. G is T_1 (Fréchet).

c. G is T_2 (Hausdorff).

d. $\{1\}$ is closed in G.

Proof.

We have $T_2 \Rightarrow T_1 \Rightarrow T_0$. Therefore we only need to show that a $\Rightarrow$ d) and d) $\Rightarrow$ c). Suppose G is T_0. Recall that two elements are topologically indistinguishable if and only if $x \in \overline{\{y\}}$ and $y \in \overline{\{x\}}$. Suppose $x \in \{1\}$. By the continuity of inversion, $x^{-1} \in \overline{\{x\}}$, hence $1 \in \overline{\{x\}}$ by continuity of $\mu(x, \cdot)$ hence $x, 1$ are topologically indistinguishable thus $x = 1$. Suppose $\{1\}$ is closed. Note that

$$\delta = \mu \circ \mathrm{id} \times \iota[G \times G \ni (x, y) \mapsto xy^{-1} \in G].$$

Therefore such map is continuous. Then,

$$\Delta_G = \delta^{-1}(\{1\}) = \{(x, x) \mid x \in G\}.$$

is closed thus T_2. □

Proposition 3.1.2. Every open subgroup H of a topological group G is also closed, and every group $H \subseteq H' \subseteq G$ is also clopen.

Proof. We have,

$$G \backslash H = \bigcup_{g \in G/H \setminus \{0\}} gH$$

is open therefore H is closed. Every H' subset is a union of right or left H-cosets in G which is also open. □

Definition 3.1.4. Let R be a topological ring and M be a topological R-module. A subset $X \subseteq M$ is bounded or R-bounded if for every neighborhood U of 0 in M, there is a neighborhood V of 0 in R such that

$$V \cdot X = \{vx \mid v \in V, x \in X\} \subseteq U.$$

Example 3.1.1. When R has the discrete topology, every subset is bounded (take $V = \{0\}$).

Example 3.1.2. This defines a boundary on X. If X_1, X_2 are bounded then so is $X_1 \cup X_2$. Every $\{m\}$ is bounded, $M \subseteq N$ and N bounded then M is also bounded.

Example 3.1.3. If $R = K$, equipped with an valuation $|\cdot|$ and M is a (valuative) K-vector space which is not discrete, then a subset X of M is bounded if and only if there is $B \in [0, \infty)_{\mathbb{R}}$ such that

$$\forall x \in X, \|x\| \leq B.$$

For example, take $U = \{x \in X \mid \|x\| \leq 1\}, B = |x|^{-1}$ where $x \in V \backslash \{0\}$ and V is as in the definition.

Definition 3.1.5. For subsets $X_1, X_2 \subseteq R$, we define,

$$X_1 X_2 = \{x_1 x_2 | x_1 \in X_1, x_2 \in X_2\}.$$

Let $X^0 = \{1\}$. We define recursively X^n such that

$$X^{n+1} = X \cdot X^n.$$

We say that. a subset $X \subseteq R$ is powerbounded if

$$\bigcup_{n \in \mathbb{N}} X^n$$

is bounded. It is topologically nilpotent if for every neighborhood U of 0 in R, there is $n \in \mathbb{N}$ such that

$$\bigcup_{m=n}^{\infty} X^m \subseteq U.$$

We call an element $r \in R$ topologically nilpotent / powerbounded if $\{r\}$ satisfies these properties.

Proposition 3.1.3. Let R be a topological ring and M be a topological R-module. We have the following statements.

a. A finite union of bounded subsets of M is bounded.

b. Trivially, every finite subset of M is bounded.

c. If $X_1, X_2 \subseteq M$ are bounded then so is $X_1 + X_2 = \{x_1 + x_2 | x_1 \in X_1, x_2 \in X_2\}$.

d. If $X \subseteq R$ and $Y \subseteq M$ are bounded then so is $XY \subseteq M$.

Proof.

Suppose $X_1, \cdots, X_n \subseteq M$ are bounded. Then for $U \subseteq M$ an open neighborhood of 0, there is $V_1, \cdots, V_n$, such that

$$\forall i = 1, \cdots, n, V_i X_i \subseteq U.$$

Therefore take $V = \bigcap_{i=1}^{n} V_i$. Then,

$$V \bigcup_{i=1}^{n} X_i \subseteq U.$$

To show that a singleton $\{x\} \subseteq M$ is bounded, note that for any open neighborhood U of 0 in M, we have, $0 \cdot x = 0 \in U$. Thus $\mu^{-1}(U)$ contains open neighborhoods V, W of 0 and x, respectively such that $VW \subseteq U$. As $x \in W$ is bounded so is $\{x\}$.

For c), let U be a neighborhood of 0 in M, there is a neighborhood $\tilde{U}$ of 0 in M such that $\tilde{U} + \tilde{U} \subseteq U$ and another neighborhoods V_1, V_2 of 0 in R such that $V_1 X_1, V_2 X_2 \subseteq \tilde{U}$. Take $V = V_1 \cap V_2$ in Definition 3.1.4.

Lastly, as Y is bounded, we have $VY \subseteq U$ for arbitrary open neighborhood U of 0. Then we can find an open neighborhood W of 0 such that $WX \subseteq V$. Therefore $WXY \subseteq U$. □

Proposition 3.1.4. If $X \subseteq M$ is R-bounded and $M \xrightarrow{f} N$ a continuous morphism of R-modules then $f(X)$ is bounded in N.

Proof. If $U \subseteq N$ is a neighborhood of 0, let $\tilde{U} = f^{-1}(U)$ and V a neighborhood of 0 in R such that $VX \subseteq \tilde{U}$. Then $Vf(X) \subseteq U$. □

It is not always true that the image of a bounded subset of a R under. a morphism $R \to S$ of topological ring is bounded. (The image is R-bounded but not necessarily S-bounded) for example take $R = \mathbb{Q}_p, S = \mathbb{Q}_q$. However, the following conditions on R asserts that this does not happen.

R has a topologically nilpotent unit (T)

R has a sequence of units converging to 0 (H)

Every neighborhood of 0 intersects $R^\times$ (E)

Proposition 3.1.5. Let $R \xrightarrow{f} S$ be a continuous morphism of topological ring. If R has (T),(H),(E), then so does S and the image $f(X)$ of a bounded subset of R is a bounded subset of S.

Proof. If U is a neighborhood of 0 in S and R has (E), put $\tilde{U} = f^{-1}(U)$ and $r \in R^\times \cap \tilde{U}$. $f(r) \in S^\times \cap U$. If

$$\lim_{j \to \infty} r_j = 0$$

in R and all $r_j \in R^\times$,

$$\lim_{j \to \infty} f(r_j) = 0$$

in S and all $f(r_j) \in S^\times$. showing that (H). For (T), take a topologically nilpotent unit $r \in R$, then by setting $r_j = r^j$ in the previous argument, we see (T) is inherited.

If R has (E) and $X \subseteq R$ bounded and U is a neighborhood of 0 in S, there is $\tilde{U}$ a neighborhood 0 in S such that $\tilde{U}\tilde{U} \subseteq U$ by the continuity of $S \ni s \mapsto s^2 \in S$. Let $U' = f^{-1}(\tilde{U}) \subseteq R$ a neighborhood of 0. There is a neighborhood V' of 0 in R such that $V'X \subseteq U'$. As R has (E), we can take $\rho \in R^\times \cap f^{-1}(\tilde{U})$. As f is a ring homomorphism, $f(\rho) \in S^\times$. Furthermore, we have,

$$f(\rho)f(X) = f(\rho X) \subseteq f(U') \subseteq \tilde{U}.$$

Therefore,

$$f(X) \subseteq f(\rho)^{-1}\tilde{U}.$$

Now we have,

$$Uf(\rho)f(X) \subseteq f(\rho)f(\rho)^{-1}\tilde{U}^2 \subseteq U.$$

□

Proposition 3.1.6. Let R be a commutative topological ring.

a. Every powerbounded subset is bounded.

b. A finite union of powerbounded subsets is powerbounded.

c. A subset $X \subseteq R$ which is both bounded and topologically nilpotent is powerbounded.

d. Let $(X_i)_{i=1,\cdots,n}$ be such that for each i, X_i is bounded and topologically nilpotent. Then $\bigcup_{i=1}^n X_i$ is also topologically nilpotent.

e. If X is powerbounded or topologically nilpotent and Y is topologically nilpotent, XY is then topologically nilpotent.

f. The image of a topologically nilpotent $X \subseteq R$ under a morphism $R \to S$ of topological rings is topologically nilpotent.

Proof. a follows as every powerbounded subset is a subset of a bounded set. b) holds by the commutativity of R and Proposition 3.1.3 d). For the rest of the proof, we set U to be an arbitrary open neighborhood of 0.

For c), take $\tilde{U}$ open neighborhood of 0 such that $\tilde{U}^2 \subseteq U$. Then there is $m \in \mathbb{N}$ such that $\bigcup_{k \geq m} X^m \subseteq \tilde{U}$. Also there is V open neighborhood of 0 such that $V \bigcup_{k=0}^{m-1} X^k \subseteq U$. Take $V \cap \tilde{U}$, then we have,

$$(V \cap \tilde{U}) \left(\bigcup_{k \geq 0} X^k \right) = (V \cap \tilde{U}) \left(\bigcup_{k \geq 0}^{m-1} X^k \right) \cup (V \cap \tilde{U}) \left(\bigcup_{k \geq m} X^k \right) \subseteq U \cup \tilde{U}^2 \subseteq U.$$

Now to show d), we use induction on n. For $n = 2$, let $0 \in \tilde{U}$ be open such that $\tilde{U}^2 \subseteq U$. Take $0 \in V \subseteq \tilde{U}$ such that

$$V \cdot \bigcup_{k=0}^{\infty} X_i^k \subseteq U$$

for each $i = 1, 2$. Now take m to be such that

$$\bigcup_{k \geq m} X_i^k \subseteq V$$

for $i = 1, 2$. Then for fixed $k \geq 2m$ and any $k_1, k_2 \in \mathbb{Z}_{\geq 0}$ such that $k_1 + k_2 = k$, we have,

$$X_1^{k_1} \cdot X_2^{k_2} \subseteq \begin{cases} \tilde{U}\tilde{U} \subseteq U, & (k_1, k_2 \geq m), \\ V \cdot \bigcup_{k=0}^{\infty} X_1^k \subseteq U, & (k_1 < m, k_2 \geq m), \\ V \cdot \bigcup_{k=0}^{\infty} X_i^k \subseteq U, & (k_2 < m, k_1 \geq m). \end{cases}$$

Therefore,

$$\bigcup_{k \geq 2m} (X_1 \cup X_2)^k \subseteq U.$$

By the induction the statement indeed holds for arbitrary n.

$$\bigcup_{k \geq m_i} X_i^k \subseteq \tilde{U}.$$

Furthermore, as X_1, X_2 are bounded, $X_1, \cdots, X_1^{m_1 - 1}, X_2, \cdots, X_2^{m-2-1}$ are all bounded. Therefore, we can take $0 \in V \subseteq \tilde{U}$ such that

$$V \cup$$

Let $m = m_1 + m_2$,

Let turn our attention to e). If X, Y are both topologically nilpotent, then take $\tilde{U}$ such that $\tilde{U}^2 \subseteq U$ and apply the property to it. If X is powerbounded then there is V open neighborhood of 0 such that

$$V \bigcup_{n=0}^{\infty} X^n \subseteq U.$$

For such V there is $m \in \mathbb{N}$ such that

$$\bigcup_{k \geq m} Y^k \subseteq V.$$

Therefore,

$$\bigcup_{m \geq k} (X^k Y^k) \subseteq \bigcup_{l=0}^{\infty} X^l \bigcup_{m \geq k} Y^k \subseteq V \bigcup_{l=0}^{\infty} X^l \subseteq U.$$

For f), for $U \subseteq S$, as X is topologically nilpotent, we can take $m \in \mathbb{N}$ applying the property to $f^{-1}(U)$. Now such m works for $f(X)$. □

Definition 3.1.6. Let R be a topological ring, we define the following subsets.

$$R^{\circ} = \{r \in R \mid r \text{ is power bounded}\},$$
$$R^{\circ\circ} = \{r \in R \mid r \text{ is topologically nilpotent}\}.$$

Remark 3.1.1. Without further conditions, these may fail to be additionally closed. By Proposition 3.1.3 b), and Proposition 3.1.6 a, $R^{\circ\circ} \subseteq R^{\circ}$ always holds.

Definition 3.1.7. Let $(m_n)_{n\in\mathbb{N}}$ be a sequence in M. It is Cauchy if for every neighborhood U of 0 in M, there is k such that $m_i - m_j \in U$ where $\min\{i,j\} \geq k$. M is sequentially complete if every Cauchy sequence has a limit in M.

Definition 3.1.8. A filter $\mathcal{F}$ of M is Cauchy if every neighborhood U of 0 in M, there is $X \in \mathcal{F}$ such that $x - y \in U$ where $x, y \in X$. M is complete if every Cauchy filter has a limit.

Definition 3.1.9. The sequential completion of M is a set of equivalence classes of Cauchy sequences in M where two Cauchy sequences $(m_i), (n_i)$ are equivalent if for every neighborhood U of M, there is $k \geq 0$ such that $i \geq k$ then $m_i - n_i \in U$.

Remark 3.1.2. If G is a non-commutative topological group, then it typically requires that both $g_i g_j^{-1} \in U$ and $g_i^{-1} g_j \in U$ where $\min\{i,j\} \geq k$. Respectively, $x^{-1}y \in U$ and $xy^{-1} \in U$ for $x, y \in X$ for filters.

Remark 3.1.3. One can complete by taking equivalence classes of Cauchy filters. But how to continue multiplication in that completion of R^2. (Sequential completion does not have this problem).

Remark 3.1.4. It is known that topological vector space over $\mathbb{C}$ or $\mathbb{R}$ is the quotient of a complete topological vector space V over that field such that every bounded subset of V is contained in a finite dimensional subspace.

Proposition 3.1.7. Every quotient of a complete topological group with a countable neighborhood base of 1 by a closed normal subgroup is complete.

3.2 Continuous Valuations

For an ordered group Γ equip $\Gamma \cup \{0\}$ with the topology where U is open if and only if $0 \notin U$ or there is $\gamma \in \Gamma$ such that $[0, \gamma)_\Gamma \subseteq U$.

Consider a valuation $A \xrightarrow{v} \Gamma$ where A is a topological ring. Recall that Γ_v is the subgroup of Γ such that

$$\Gamma_v = \langle \{v(a) \mid a \in A \backslash \operatorname{supp} v\} \rangle_\Gamma.$$

Also recall that for $a \in A$ and $\gamma \in \Gamma$,

$$K_{v<a} = \{\alpha \in A \mid v(\alpha) < v(a)\}, K_{v<\gamma} = \{\alpha \in A \mid v(\alpha) < \gamma\}.$$

Similarly,

$$K_{v\leq a} = \{\alpha \in A \mid v(\alpha) \leq v(a)\}, K_{v\leq\gamma} = \{\alpha \in A \mid v(\alpha) \leq \gamma\}.$$

They come with an exception such that

$$K_{v<a} = \emptyset,$$

when $a \in \operatorname{supp} v$. They are subgroups of $(A, +)$.

Lemma 3.2.1. Let G be a topological group and $H \subseteq G$ be a subgroup. Then H is open if and only if H is a neighborhood of the identity (i.e. H contains an open neighborhood of 0).

Proof. Let $e_G \in U \subseteq H$ be an open neighborhood. Then,

$$H = \bigcup_{h \in H} hU,$$

which is open. For the other direction, simply take $U = H$. □

Proposition 3.2.1. Let A be a topological ring and $v : A \to \Gamma \cup \{0\}$ be a valuation. The following conditions are equivalent.

a. $A \xrightarrow{v} \Gamma_v \cup \{0\}$ is continuous.

b. For all element γ of the convex hull of Γ_v in Γ, $K_{v<\gamma}$ is open.

c. For all $\gamma \in \Gamma_v$, $K_{v<\gamma}$ is open.

d. For all $a \in A \backslash \operatorname{supp}(v)$, $K_{v<a}$ is open.

e. For all $a \in A \backslash \operatorname{supp}(v)$, $K_{v<a}$ is a neighborhood of 0.

Proof. By Lemma 3.2.1 d) and e) are equivalent. d)⇒c) is trivial.

For c)⇒a, consider $\gamma \in \Gamma_v \cup \{0\}$ and a neighborhood U of γ in $\Gamma_v \cup \{0\}$. We must show that $v^{-1}(U)$ is a neighborhood of a in A. If $v(a) \neq 0$, it is sufficient to consider

$$U = \{v(a)\}.$$

Then,

$$v^{-1}(U) = K_{r<a} + a.$$

which is a neighborhood of a. If $v(a) = 0$, we may assume $U = [0, \gamma)_{\Gamma_v \cup \{0\}}$ with $\gamma \in \Gamma_v$. Then $v^{-1}(U) = K_{v<\gamma}$ is open by the assumption c).

a⇒d) is trivial by the definition of the topology on $\Gamma_v \cup \{0\}$, that is for an open set $[0, v(a))_{\Gamma_v \cup \{0\}}$, we have,

$$K_{v<a} = v^{-1}([0, v(a))_{\Gamma_v \cup \{0\}}).$$

b)⇒ c) is trivial. For d)⇒b), every $\gamma \in \Gamma_v$ has the form,

$$\gamma = \frac{v(a)}{v(b)},$$

with $a, b \in A \backslash \operatorname{supp} v$. Then $K_{v<\gamma}$ is the inverse image of $K_{v<a}$ under the endomorphism $A \xrightarrow{\cdot b} A$ of the topological group $(A, +)$. Hence it is also an open subgroup. For general element γ of the convex hull, by definition there is $\delta \in \Gamma_v, \delta \leq 1$ such that

$$\delta \leq \gamma \leq \delta^{-1}. \tag{3.1}$$

Then $K_{v<\delta} \subseteq K_{v<\gamma}$ and in particular, the former is an open subgroup. Therefore, the statement is proven. □

Definition 3.2.1. A valuation of A is called continuous if the equivalent conditions in Proposition 3.2.1 hold.

Remark 3.2.1.

1. This only depends on the equivalent class of v.
2. $\Gamma_v \cup \{0\}$ carries the topology induced from $\Gamma \cup \{0\}$ only if Γ is the convex hull of Γ_v in Γ. Hence Γ can be replaced by Γ_v in c) only if Γ is equal to that convex hull (in which case the c) is b)).

Example 3.2.1.

1. $\mathbb{Q}_p \overset{|\cdot|_p}{\to} [0,\infty)_{\mathbb{R}}$ is a continuous valuation.
2. The trivial valuation for $\mathfrak{p} \in \operatorname{Spec} A$ is continuous if and only if $\mathfrak{p}$ is open.
3. If $\mathfrak{p}$ is an open prime ideal, then any valuation with support $\mathfrak{p}$ is open.

Proposition 3.2.2. If v is a continuous valuation. Then $\operatorname{supp} v$ is closed.

Proof. Indeed $\operatorname{supp} v = v^{-1}(\{0\})$ and $\{0\}$ is closed in $\Gamma_v \cup \{0\}$. □

Definition 3.2.2. We denote the subset of equivalence classes of continuous valuations of A to be $\operatorname{Cont}(A) \subseteq \operatorname{Spv}(A)$. Let $A^+ \subseteq A$ be a subset we then denote $\operatorname{Spa}(A, A^+)$ to be the set of equivalence classes of continuous valuations v of A such that $a \in A^+ \Rightarrow v(a) \leq 1$.

Remark 3.2.2.

1. At this point, we do not yet introduce a topology on these sets.

Definition 3.2.3. A Oka system is a ring A and a set $\mathfrak{M}$ of ideals in A such that for an I in A and $a \in A$, the ideals $I + aA$ and

$$(I : a) := \{\alpha \in A \mid a\alpha \in I\}$$

both belong to $\mathfrak{M}$ then $I \in \mathfrak{M}$.

Example 3.2.2. The set of finitely generated ideals in A form a Oka system. This observation gives Eakin-Nagata or Cohen's theorem which state that a ring is Noetherian if and only if all of its prime ideals are finitely generated.

Proposition 3.2.3. Let $\mathfrak{M}$ be an Oka system and $\mathfrak{p} \neq A$ be a $\subseteq$-maximal element of the set of ideals of A not belonging to $\mathfrak{M}$. Then $\mathfrak{p} \in \operatorname{Spec} A$.

Proof. Assume $ab \in \mathfrak{p}$. Let $a \notin \mathfrak{p}, b \notin \mathfrak{p}$. Then $\mathfrak{p} + aA$ and $(\mathfrak{p} : a)$ contain $\mathfrak{p}$ and not equal to $\mathfrak{p}$ as $b \in (\mathfrak{p} : a)$. By the maximality, both belong to $\mathfrak{M}$ thus $\mathfrak{p} \in \mathfrak{M}$ which is a contradiction. □

Proposition 3.2.4. Let $\mathfrak{M}$ be a set of ideals such that $I \subseteq J$ and $I \in \mathfrak{M}$ implies $J \in \mathfrak{M}$. Suppose further that $\mathfrak{M}$ is closed under products, that is $I, J \in \mathfrak{M}$ then $I \cdot J \in \mathfrak{M}$. Then $\mathfrak{M}$ is an Oka system.

Proof. Indeed $I \supseteq (I + aA)(I : a)$ holds. Therefore by definition, it is Oka. □

Remark 3.2.3. The Oka system from Example 3.2.2 satisfies the assumptions if and only if A is Noetherian.

Corollary 3.2.1. The set $\mathfrak{M}_S$ of ideals not disjoint from a multiplicative subset $S \subseteq A$ is an Oka system. Then every $\subseteq$-maximal non-element of $\mathfrak{M}_S$ is a prime ideal. This could also be seen by that all maximal ideals of A_S are prime. By Zorn's lemma, every ideal not belonging to $\mathfrak{M}_S$ is contained in an ideal which is maximal with respect to this non-containment. By Proposition 3.2.3, such ideal is prime, or even more maximal.

Proposition 3.2.5. Let v be a valuation of a topological ring A satisfying the following conditions.

i. $K_{v\leq 1}$ is a neighborhood of 0 in A.

ii. For each $a \in A \setminus \operatorname{supp} v$, there is $b \in A$ with $v(ab) \geq 1$.

Then v is continuous.

Proof. If $a \in A \setminus \operatorname{supp} v$, and b is as above, then $K_{v<a}$ contains the inverse image under the continuous endomorphism, $A \xrightarrow{\cdot b} A$ of $K_{r\leq 1}$ which is a neighborhood of 0. Then Proposition 3.2.1 e) is satisfied. □

Proposition 3.2.6. Let A be a topological ring, A^+ be an open subring of A which is integrally closed in A and $\alpha \in A \setminus A^+$. Then there is $v \in \operatorname{Spa}(A, A^+)$ such that $v(\alpha) > 1$.

Proof. By assumption, α is not integral over A^+, hence $S = \{p(\alpha) \mid p \in A^+[T] \text{ monic}\}$ is a multiplicative subset of A not containing 0. By Corollary 3.2.1, there is $\mathfrak{p} \in \operatorname{Spec} A$ maximal among all ideals of A disjoint from S. It follows from the maximality of $\mathfrak{p}$, that for every $a \in A \setminus \mathfrak{p}$, there is $b \in A$ and a monic polynomial $P \in A^+[T]$ such that $P(\alpha) - ab \in \mathfrak{p}$. Since $\mathfrak{p}$ is disjoint from S, the image $\alpha \bmod \mathfrak{p}$ of α in $k(\mathfrak{p})$ is not integral over the image $A^\#$ of A^+ in $k(\mathfrak{p})$. By Proposition 1.3.5, there is a valuation ring R of $k(\mathfrak{p})$ such that $A^\# \subseteq R$ and $\alpha \notin R$. Consider a map v which is the composition given by,

$$A \to k(\mathfrak{p}) \overset{v_R \text{ in Prop. } 1.3.3}{\to} \Gamma_R.$$

Since $\alpha \bmod \mathfrak{p} \notin R, v(\alpha) > 1$. By the definition, $K_{v\leq 1}$ contains A^+ as $A^+ \bmod \mathfrak{p} \subseteq A^\# \subseteq R$, hence $K_{v\leq 1}$ is open. If $a \in A$ and $v(a) \neq 0$, then $a \notin \mathfrak{p}$ and we have $b \in A$ and a monic $P \in A^+[T]$ such that $P(\alpha) - ab \in \mathfrak{p}$. Since P is monic and $v(A^+) \subseteq [0, 1]_{\Gamma \cup \{0\}}$, $v(P(\alpha)) > 1$. Since $\operatorname{supp} v = \mathfrak{p}$, $v(ab) = v(P(\alpha)) > 1$. By Proposition 3.2.5, v is continuous. □

Corollary 3.2.2. For a subset A^+ of A, the following conditions are equivalent.

a. $A^+ = \bigcap_{x \in \operatorname{Spa}(A,A^+)} K_{x\leq 1} = \{a \in A \mid |a|_x \leq 1 \text{ for all } x \in Spa(A, A^+)\}$

b. A^+ is the intersection of a set of open and integrally closed subrings of A.

Proof. For b⇒a, note that $\subseteq$ in a is trivial and $\supseteq$ is by Proposition 3.2.6. For a⇒b, note that $K_{x\leq 1}$ for $x \in \mathrm{Spa}(A, A^+)$ are integrally closed by Proposition 1.2.5 and open subrings of A by Proposition 3.2.1.a. □

Proposition 3.2.7. Let $\eta \in \mathrm{Cont}(A)$ and $s \in \mathrm{Spv}(A)$ a specialization of η. Then $s \in \mathrm{Cont}(A)$.

Proof. Let $a \notin \operatorname{supp} s$. Then by Proposition 1.2.4, $\operatorname{supp} \eta \subseteq \operatorname{supp} s$. Hence $|a|_\eta \neq 0$. Thus $K_{\eta<a}$ is a neighborhood of 0 in A by Proposition 3.2.1.e. By Proposition 1.2.6, $K_{\eta<a} \subseteq K_{s<a}$ hence $K_{s<a}$ is a neighborhood of 0 in A and again by Proposition 3.2.1.e, hence for s as well. □

Without further assumptions, Spa and Cont (to be equipped with a topology below) fail to be spectral spaces and there are ultrafilters $\mathcal{U}$ on Spa such that $\mathrm{Cont}(A) \in \mathcal{U}$ but $\mathrm{glim}_{\mathrm{Spa}\, A}\, \mathcal{U}$ fails to have any continuous specializations.

The following condition on A are often useful.

If $I \subseteq A$ is an open ideal, I^2 is also open (oisq)

Every open ideal $I \subseteq A$ contains a finitely generated open ideal of A. (oifin)

Proposition 3.2.8. If I and J are open ideals of a topological ring A satisfying (oisq), then IJ is open. In particular, by induction on n, all I^n are open.

Proof. As $I \cap J$ is open $(I \cap J)^2 \subseteq I \cdot J$ is also open by (oisq). □

Proposition 3.2.9. If $a \in A^{\circ\circ}$ and $x \in \mathrm{Cont}(A)$, then $|a|_x < 1$.

Proof. By assumption, we have,

$$\lim_{n\to\infty} |a|_x^n = 0$$

in $\Gamma_x \cup \{0\}$. By the continuity of x, the above equality does not hold if $|a|_x \geq 1$. □

We further introduce following conditions.

The ideal generated by $A^{\circ\circ}$ of A is open (tn0)

$A^{\circ\circ}$ is a neighborhood of 0 in A. (tn1)

Obviously (tn1)⇒(tn0).

Proposition 3.2.10. If A contains an open subring B satisfying one of (oifin),(oisq), (tn0),(tn1), then A also has that property.

Definition 3.2.4. We say that $x \in \mathrm{Spa}(A)$ can be made continuous if $K_{x<1}$ is a neighborhood of 0 in A.

Remark 3.2.4. The condition in Definition 3.2.4 is already satisfied if x is continuous by Proposition 3.2.1 e).

Proposition 3.2.11.

a. If $x \in \mathrm{Spv}(A)$ can be made continuous then $K_{x<\gamma}$ is a neighborhood of 0 where $\gamma \in c\Gamma_x$.

b. If $K_{x\leq 1}$ is a neighborhood of 0 then $K_{x<\gamma}$ is a neighborhood of 0 where $\gamma \in c\Gamma_x \backslash \{0\}$.

c. If $K_{x\leq 1}$ is a neighborhood of 0, then $c\Gamma_x = \{1\}$ or x can be made continuous.

Proof. For b, If $\gamma > 1$ then $K_{x\leq 1} \subseteq K_{x<\gamma}$ and the claim follows. Otherwise $\gamma < 1$ and there is $a \in A$ such that $\gamma|a|_x \geq 1$. In that case, $\gamma|a|_x^2 > 1$ and $K_{x<\gamma}$ contains the preimage under the continuous map $A \xrightarrow{a^2} A$ of $K_{x\leq 1}$, a neighborhood of 0.

a. For $\gamma = 1$, this follows from Definition 3.2.4 and for $\gamma \neq 1$, we can apply b).

c. Assume $c\Gamma_x \neq \{1\}$ so we can pick $\gamma \in c\Gamma_x \backslash \{1\}$. Without loss of generality, assume $\gamma < 1$. Then $K_{x<1}$ is a neighborhood of 0 as it contains $K_{x<\gamma}$, a neighborhood of 0 by b. □

Proposition 3.2.12. Let A be a topological ring with (oisq) and (tn0), and let $\eta \in \mathrm{Spv}(A)$. Assume $|x|_\eta < 1$ for all $x \in A^{\circ\circ}$. In the following the word "specialization" always means "specialization in Spv".

a. There is a continuous specialization of η if and only if η can be made continuous.

b. If this is the case, then,

I. The following subset of Γ_η is a convex subgroup containing $c\Gamma_\eta$.

$$s\Gamma_\eta := \{\gamma \in \Gamma_\eta \mid K_{\eta<r} \text{ is a neighborhood of 0 in } A\}.$$

II. The horizontal specialization $s = \eta|_{s\Gamma_\eta}$, which is well-defined by I, is continuous.

III. The sets of specializations of s and of continuous specializations of η coincide.

IV. Let $\Omega = \mathcal{R}(f_0|f_1, \cdots, f_m)$ with $(f_i)_{i=0}^{m+1} \in A^{m+1}$ be such that the ideal of A generated by $K_{\eta\leq f_0}$ is open, then $s \in \Omega$ if and only if $\eta \in \Omega$.

Proof. Assume that η has the continuous specialization s. As $K_{s\leq 1} \subseteq K_{\eta\leq 1}$ by Proposition 1.2.7 and $K_{s<1}$ is a neighborhood of 0 by Proposition 3.2.1.e, $K_{\eta\leq 1}$ is a neighborhood of 0. By Proposition 3.2.11, $c\Gamma_\eta = \{1\}$ or η can be made continuous. If $c\Gamma_x = \{1\}$, then $A = K_{\eta\leq 1}$ and $K_{\eta<1}$ is an ideal in A. By assumption about η, $A^{\circ\circ} \subseteq K_{\eta<1}$ and thus $A^{\circ\circ}$ contains an open ideal by (tn0). Hence $K_{\eta<1}$ is an open ideal of A. Therefore η can be made continuous.

Assume that η can be made continuous. To finish a, and to show b, it suffices to show I-IV under this assumption. The inclusion $c\Gamma_\eta \subseteq s\Gamma_\eta$ follows

from Proposition 3.2.11.a. The convexity of $s\Gamma_\eta$ as a neighborhood of Γ_η then follows from Proposition 1.2.8. To show that it is indeed a group, we first show that $\gamma \in s\Gamma_\eta \Rightarrow \gamma^2 \in s\Gamma_\eta$. By the convexity, we can assume $\gamma \leq 1$. If $\gamma \in c\Gamma_\eta$ then it holds as $\gamma^2 \in c\Gamma_\eta$. Otherwise, there is $a \in A$ such that $\gamma \notin [|a|_\eta, |a^{-1}|_\eta]_{\Gamma \cup \{0\}}$ thus $\gamma |a|_\eta < 1$ for all $a \in A$. Let I be the ideal generated by $K_{\eta<\gamma}$ in A. Let $U \subseteq K_{\eta<\gamma}$ be an open neighborhood of 0, then

$$I = \bigcup_{x \in I} (x + U).$$

Therefore I is open. By (oisq) and Proposition 3.2.8, I^3 is open. For $\alpha \in I^3$, we have,

$$\alpha = \sum_{i=1}^{n} a_i b_i c_i d_i,$$

where $a_i \in A$, and $b_i, c_i, d_i \in K_{\eta<\gamma}$ for each i. Hence,

$$|a_i b_i|_\eta = |a_i|_\eta |b_i|_\eta < |a_i|_\eta \gamma < 1.$$

Therefore,

$$|\alpha|_\eta \leq \max_{1 \leq i \leq n} \{|a_i b_i c_i d_i|\} \leq \max_{1 \leq i \leq n} \{|c_i d_i|\} < \max_{1 \leq i \leq n} \gamma \cdot \gamma = \gamma^2$$

Therefore $I^3 \subseteq K_{\eta<\gamma^2}$. In particular $\gamma^2 \in s\Gamma_\eta$. Finally, let $\gamma_1, \gamma_2 \in s\Gamma_\eta$. Then,

$$(\min\{\gamma_1, \gamma_2\})^2 \leq \gamma_1 \gamma_2 \leq (\max\{\gamma_1, \gamma_2\})^2.$$

By the convexity of $s\Gamma, \gamma 1 \gamma_2 \in s\Gamma$.

For II, note that as s is a specialization, we have $K_{\eta<a} \subseteq K_{s<a}$. Let $a \in A$ be such that $|a|_\eta$. Then $K_{s<a}$ is a neighborhood of 0. By Proposition 3.2.1.e, s is continuous.

For IV, by Proposition 1.2.11, it suffices to show that $|f_0|_s \neq 0$ so we can drop the condition. That is $|f_0|_\eta \in s\Gamma_\eta$. if $|f_0|_\eta \in c\Gamma_\eta$, then this follows from I. Otherwise $|a|_\eta |f_0|_\eta < 1$ for all $a \in A$. Moreover, the ideal I of A generated by $K_{\eta \leq f_0}$ is open by our assumption. By (oisq), I^2 is also open. Its elements have the form, $\alpha = \sum_i^n a_i b_i c_i$ with all $a_i \in A, b_i, c_i \in K_{\eta \leq f_0}$. Therefore,

$$|\alpha|_\eta \leq \max_{1 \leq i \leq n} \{|a_i|_\eta |b_i|_\eta |c_i|_\eta\} < \max_{1 \leq i \leq n} |c_i|_\eta \leq |f_0|_\eta.$$

Then $K_{\eta<f_0}$ contains the open ideal I^2, hence is a neighborhood of 0. Thus $|f_0|_\eta \in s\Gamma_\eta$ as claimed.

For III, every specialization of a continuous valuation is continuous by Proposition 3.2.7. In particular, specializations of s are all continuous by II. Let $t \in \mathrm{Spv}(A)$, be any continuous specialization of η. For every rational open subset $\Omega = \mathcal{R}(f_0 | f_1, \cdots, f_n)$, we want to show that $t \in \Omega$ implies $s \in \Omega$. But $t \in \Omega$ implies $|f_0|_t \neq 0$ by definition. Hence $K_{t<f_0}$ is a neighborhood of 0 by Proposition 3.2.1 d. Also $K_{t<f_0} \subseteq K_{t \leq f_0}$ thus $K_{t \leq f_0}$ is a neighborhood of 0.

By Proposition 1.2.7 and $K_{t\leq f_0} \subseteq K_{\eta\leq f_0}$, $K_{\eta\leq f_0}$ is a neighborhood of 0 as well. Apply IV with $\eta \in \Omega$, $s \in \Omega$ follows as $t \in \Omega$ since t is a specialization of η.

We explicitly constructed a continuous specialization of η. Thus a is also shown. □

Definition 3.2.5. The horizontal specialization $s = \eta|_{s\Gamma_\eta}$ in Proposition 3.2.12 is called the continuation of η.

Definition 3.2.6. The strong topology on $\mathrm{Cont}(A)$ or on $\mathrm{Spa}(A, A^+)$ is the one induced from $\mathrm{Spv}(A)$. That is for which the set of all $\mathcal{R}(f_0|f_1, \cdots, f_m)$ for arbitrary $n \in \mathbb{N}$ and $(f_i)_{i=0}^n \in A^{n+1}$ form a topology base.

If A satisfies (oisq), the weak topology on both spaces is the one for which the set $\mathfrak{Rat}_A = \mathfrak{Rat}_A^{(t)}$ of $\mathcal{R}(f_0|f_1, \cdots, f_n)$ for which $\langle f_0, \cdots, f_n\rangle_A$ is an open ideal of A is a topology base.

Remark 3.2.5. Note that $\mathcal{R}(f_0|f_1, \cdots, f_m) \cap \mathcal{R}(g_0|g_1, \cdots, g_m) = \mathcal{R}(f_0g_0|f_ig_j)$ and

$$\langle f_ig_j\rangle_A = \langle f_0, \cdots, f_n\rangle_A\langle g_0, \cdots, g_m\rangle_A$$

is open by (oisq). Hence $\mathfrak{Rat}_A^{(t)}$ is closed under finite intersection. In most case, A will be equipped with a topology and may the notation (t) be dropped. When the ring is equipped with a topology only the elements of $\mathfrak{Rat}_A^{(t)}$ will be called rational open subsets of $\mathrm{Cont}(A)$ or $\mathrm{Spa}(A, A^+)$.

Proposition 3.2.13. If A has (oisq) and (oifin), then these topologies coincide.

Proof. Let $U = \mathcal{R}(f_0|f_1, \cdots, f_n) \subseteq X$ where $X = \mathrm{Cont}(A)$ or $X = \mathrm{Spa}(A, A^+)$. For $x \in U$, we want to show that there is $\Omega \in \mathfrak{Rat}_A^{(t)}$ such that $x \in \Omega \subseteq U$. As x is a continuous valuation, $K_{x<f_0}$ is open and by (oifin), there are finitely many $\varepsilon_1, \cdots, \varepsilon_l \in K_{x<f_0}$ such that $\langle \varepsilon_1, \cdots, \varepsilon_l\rangle_A$ is open. This case

$$\langle f_0, \cdots, f_n, \varepsilon_1, \cdots, \varepsilon_n\rangle_A$$

is open as it contains an open ideal. Then $x \in \Omega = \mathcal{R}(f_0|f_1, \cdots, f_n, \varepsilon_1, \cdots, \varepsilon_l) \subseteq U$ and $\Omega \in \mathfrak{Rat}_A^{(t)}$. □

Proposition 3.2.14. $\mathrm{Cont}(A)$ and $\mathrm{Spa}(A, A^+)$ are T_0 for the strong topology.

Proposition 3.2.15. Let A be a topological ring with (oisq), (oifin) and (tn0). For an ultrafilter $\mathcal{U}$ on $\mathrm{Cont}(A)$, let $\mathcal{V}$ be its direct image on $\mathrm{Spv}(A)$ by the inclusion $\iota : \mathrm{Cont}(A) \to \mathrm{Spv}(A)$. The following conditions are equivalent.

a. $\mathcal{U}$ has a generic limit in $\mathrm{Cont}(A)$.

b. $\mathcal{U}$ has some limit in $\mathrm{Cont}(A)$.

c. The generic limit $g = \mathrm{glim}_{\mathrm{Spv}(A)} \mathcal{V}$ of $\mathcal{V}$ in $\mathrm{Spv}\, A$ can be made continuous.

d. The set

$$\{a \in A \mid \{v \in \mathrm{Cont}(A) \mid |a|_x < 1\} \in \mathcal{U}\} \tag{3.2}$$

is a neighborhood of 0 in A.

In this case, a generic limit γ of $\mathcal{U}$ in $\mathrm{Cont}(A)$ is given by the continuation $g|_{s\Gamma_g}$ of g. If $\mathrm{Spa}(A, A^+) \in \mathcal{U}$, then $\gamma \in \mathrm{Spa}(A, A^+)$ and γ is a generic limit of the intersection of $\mathcal{U}$ with the set of subsets of $\mathrm{Spa}(A, A^+)$.

Proof. a⇒b is obvious. For b⇒c, first observe that due to Proposition 3.2.13, the strong and weak topologies coincide in our case. If λ is any limit of $\mathcal{U}$, it is a limit of $\mathcal{V}$ in $\mathrm{Spv}(A)$. Hence λ is a specialization of g in $\mathrm{Spv}(A)$. By Proposition 3.2.12 a, g can be made continuous.

For c⇒ a, if g can be made continuous and γ is its continuation. γ is in the closure of $g = \mathrm{glim}_{\mathrm{Spv}(A)} \mathcal{V}$, hence is a limit of $\mathcal{V}$. As $\mathrm{Cont}(A)$ carries the induced topology, it is a limit of $\mathcal{U}$ in $\mathrm{Cont}(A)$. Let λ be any limit of $\mathcal{U}$ in $\mathrm{Cont}(A)$. Then λ is a limit of $\mathcal{V}$ in $\mathrm{Spv}(A)$. Hence by Proposition 3.2.12.b.II, λ is a specialization of γ in $\mathrm{Spv}(A)$. Using once again the fact that $\mathrm{Cont}(A)$ carries the induced topology, λ is a specialization of γ in $\mathrm{Cont}(A)$, showing that γ is a $\mathrm{glim}_{\mathrm{Cont}(A)} \mathcal{U}$.

We have shown the equivalence of first three statement and also the

Observe that

$$\{x \in \mathrm{Cont}(A) \mid |a|_x < 1\} = \{x \in \mathrm{Spv}(A) \mid |a|_x < 1\} \cap \mathrm{Cont}(A).$$

Also $a \in A$ belongs to the set in (3.2) if and only if

$$M_a = \{x \in \mathrm{Spv}(A) \mid |a|_x < 1\} \in \mathcal{V}.$$

Therefore, we can rewrite (3.2) as,

$$\{a \in A \mid M_a \in \mathcal{V}\}.$$

Now note that

$$a \in K_{g<1} \Leftrightarrow g \notin \mathcal{R}(a|1) \Leftrightarrow \mathcal{R}(a|1) \notin \mathcal{V} \Leftrightarrow M_a \in \mathcal{V}.$$

Therefore, we see,

$$K_{g<1} = \{a \in A \mid \{v \in \mathrm{Cont}(A) \mid |a|_x < 1\} \in \mathcal{U}\}.$$

By definition, this is a neighborhood of 0 if and only if g can be made continuous. Thus we obtained c⇔d.

Finally let $\mathrm{Spa}(A, A^+) \in \mathcal{U} \subseteq \mathcal{V}$. We have

$$\mathrm{Spa}(A, A^+) = \left[\bigcap_{\alpha \in A^+} \mathcal{R}_{\mathrm{Spv}(A)}(1|\alpha) \right] \cap \mathrm{Cont}(A) \tag{3.3}$$

and since the open sets $\mathcal{R}_{\mathrm{Spv}(A)}(1|\alpha)$ are all closed under generic limits, thus g is contained in all of them, hence $|\alpha|_g \leq 1$ where $\alpha \in A^+$. Since γ is a horizontal specialization of g, this implies $|\alpha|_\gamma \leq 1$. As γ is continuous, by (3.3), $\gamma \in \mathrm{Spa}(A, A^+)$. Since $\mathrm{Spa}(A, A^+)$ carries the topology induced from $\mathrm{Cont}(A)$ and $\mathrm{Spa}(A, A^+) \in \mathcal{U}$, using Proposition 1.1.9

$$\mathcal{U}' = \{U \in \mathcal{U} \mid U \subseteq \mathrm{Spa}(A, A^+)\}$$

is also an ultrafilter and γ is its limit. If λ is any limit of $\mathcal{U}'$ in $\mathrm{Spa}(A, A^+)$, it is a limit of the image $\mathcal{U}$ of $\mathcal{U}'$ under the continuous map $\mathrm{Spa}(A, A^+) \to \mathrm{Cont}(A)$. Hence it is a specialization of γ in $\mathrm{Cont}(A)$. Hence also in $\mathrm{Spa}(A, A^+)$ which has the induced topology. □

Corollary 3.2.3. If A is a topological ring satisfying (oisq),(oifin), and (tn1). For $X = \mathrm{Cont}(A)$ or $\mathrm{Spa}(A, A^+)$ for some subset A^+, X is spectral and the elements of $\mathfrak{Rat}_X^{(t)}$ are quasi-compact open subsets of X.

Proof. Let $\mathcal{U}$ be an ultrafilter on $\mathrm{Cont}(A)$. If $a \in A^{\circ\circ}$, then

$$\{x \in \mathrm{Cont}(A) \mid |a|_x < 1\} = \mathrm{Cont}(A) \in \mathcal{U}$$

By Proposition 3.2.15, $A^{\circ\circ} \subseteq K_{g<1}$. By (tn1), g can be made continuous. Therefore every ultrafilter on $\mathrm{Cont}(A)$ admits a unique generic limit. Hence, $\mathrm{Cont}(A)$ is spectral by Proposition 1.1.20. Furthermore, X is also spectral using the closing claim of Proposition 3.2.15.

We now show that the elements of $\mathfrak{Rat}_X^{(t)}$ are quasi-compact. By Proposition 1.1.20, we need to show that these are closed under taking glim_X. Observe that we have a canonical inclusion $\iota : X \hookrightarrow \mathrm{Cont}(A)$. Let,

$$\Omega = \mathcal{R}_{\mathrm{Spv}(A)}(f_0|f_1, \cdots, f_m),$$
$$\Omega' = \mathcal{R}_X(f_0|f_1, \cdots, f_m) = \Omega \cap X.$$

with $I = \langle f_0, \cdots, f_m \rangle_A$ being open. As Ω is closed under $\mathrm{glim}_{\mathrm{Spv}(A)}$ and again by the closing line of Proposition 3.2.15 and $\Omega' = \Omega \cap X$, Ω' is also closed under taking glim_X. □

Proposition 3.2.16. If $X = \mathrm{Cont}(A)$ or $\mathrm{Spa}(A, A^+)$, and $x \in X$ is closed then $c\Gamma_x = \Gamma_x$.

Proof. Clearly horizontal specializations preserve boundedness on A^+, thus we only need to show that when $X = \mathrm{Cont}(A)$. Let s be a horizontal specialization of x, then by the definition of horizontal specializations, we see that for all $\gamma \in \Gamma_s$,

$$K_{s<\gamma} = K_{x<\gamma}.$$

Therefore, $K_{s<\gamma}$ is open for all $s \in \Gamma_s$. Thus s is continuous. □

Proposition 3.2.17. For a topological ring A, let $X = \mathrm{Cont}(A)$ or $\mathrm{Spa}(A, A^+)$. Suppose $s \preceq_X \eta$ and $x \preceq_{\mathrm{Spv}(A)} \eta$ such that $s \preceq_{\mathrm{Spv}(A)} x$. Show that $x \in X$ and $s \preceq_X x \preceq_X \eta$ also holds.

Proof. By Proposition 1.2.7, we have,

$$A^+ \subseteq K_{s\leq 1} \subseteq K_{x\leq 1}.$$

Therefore we reduce the statement to show when $X = \mathrm{Cont}(A)$. When $a \in A \setminus \mathrm{supp}(x)$ we have,

$$K_{\eta<a} \subseteq K_{x<a}.$$

Therefore x is continuous. □

Proposition 3.2.18. Let A be a topological ring and $\mathcal{M}$ be the set of open ideals in A.

a. If A has (oifin) then every ideal I not in $\mathcal{M}$ is contained in some ideal of A which is $\subseteq$-maximal in the set of ideals of A not in $\mathcal{M}$.

b. If A has (oisq), then $\mathcal{M}$ is an Oka-system. If $\mathfrak{p}$ is any ideal of A which is $\subseteq$-maximal among ones which are not in $\mathcal{M}$. Then $\mathfrak{p}$ is a prime ideal such that either it is closed or it is a dense maximal ideal.

Proof. Let $\mathcal{N}$ be the set of ideals in A which are not open. For a, let $\mathcal{L} \subseteq \mathcal{N}$ be linearly $\subseteq$-ordered non-empty subset. Let $I = \bigcup_{J \in \mathcal{L}} J$. Then I is an ideal in A. If I is open then by (oifin), there are $f_1, \cdots, f_m \in I$ such that

$$K = \langle f_1, \cdots, f_m \rangle_A \subseteq I$$

is open. As $\mathcal{L}$ is linearly ordered, there is $J \in \mathcal{L}$ containing all f_i. Hence $K \subseteq J$ and J is open. This contradicts to that $J \in \mathcal{N}$. Applying Zorn's lemma, we derived the statement.

For b, $\mathcal{M}$ is closed under product by Proposition 3.2.8 and any ideal containing an open ideal is also open, it is an Oka system by Proposition 3.2.4. That $\mathfrak{p}$ must be prime follows from Proposition 3.2.3. If $\mathfrak{p}$ is not closed let $a \in \overline{\mathfrak{p}} \backslash \mathfrak{p}$. As $\overline{\mathfrak{p}}$ is an ideal and $\mathfrak{p}$ is prime, $a^2 \in \overline{\mathfrak{p}} \backslash \mathfrak{p}$. Now, $\mathfrak{p} + a^2 A$ is open by the maximality of $\mathfrak{p}$, as this is an ideal, and by Proposition 3.1.2, this is also closed. By the minimality of closures, $\overline{\mathfrak{p}} = \mathfrak{p} + a^2 A$. Since $a \in \overline{\mathfrak{p}} = \mathfrak{p} + a^2 A$, there is $b \in A$ such that $a - a^2 b \in \mathfrak{p}$. As $a \notin \mathfrak{p}, 1 - ab \in \mathfrak{p}$. Thus $1 \in \overline{\mathfrak{p}}$ as $a \in \overline{\mathfrak{p}}$ and $\overline{\mathfrak{p}}$ is an ideal. Now the previous consideration can be applied to all $a \in A$, and show that $A/\mathfrak{p}$ is a field. Thus $\mathfrak{p}$ is a dense maximal ideal. □

Definition 3.2.7. A continuous valuation v of A and its equivalence class is called analytic if the closed prime ideal $\operatorname{supp} v$ fails to be open. We denote

$$\operatorname{Cont}(A)_a = \{x \in \operatorname{Cont}(A) \mid x \text{ is analytic}\},$$
$$\operatorname{Spa}(A, A^+)_a = \{x \in \operatorname{Cont}(A) \mid x \text{ is analytic}\}.$$

Proposition 3.2.19. For a topological ring A with (tn0), the following conditions are equivalent.

i. The ideal of A generated by $A^{\circ\circ}$ is A.

ii. A has no open maximal ideal.

iii. A has no open prime ideal.

iv. A has no open proper ideal.

v. For every subset $A^+ \subseteq A$, all elements of $\operatorname{Spa}(A, A^+)$ are analytic.

Proof. i⇒iv, as every open ideals contains $A^{\circ\circ}$. iv⇒ii⇒ii are trivial. For ii⇒i, let $I = (A^{\circ\circ})_A$. Consider $B = A/I$, and recall that we have a correspondence of maximal ideals in B and maximal ideals in A containing I. But any ideal containing I is open thus B does not admit any maximal ideal which is possible only when $B = \{0\}$. Therefore $A = I$.

iii⇒v is trivial as $\operatorname{supp} v$ is always prime. For v⇒iii, for every open prime ideal $\mathfrak{p}$, we can consider a trivial valuation with support $\mathfrak{p}$. □

Definition 3.2.8. A ring is called analytic if the ideal of A generated by $A^{\circ\circ}$ is A.

Example 3.2.3. Every ring with (E) ($A^\times$ intersects every neighborhood of 0, see §3.1) is analytic. In particular, every ring with (T) is analytic.

Lemma 3.2.2. Let A be a topological ring with (tn0). A valuation of A is not analytic if and only if its support contains $A^{\circ\circ}$.

Proof. ⇐ is clear. Let x be a non-analytic valuation. As $\operatorname{supp} x$ is an open neighborhood of 0, for each $a \in A^{\circ\circ}$ there is $n \in \mathbb{N}$ such that $a^n \in \operatorname{supp} x$. But the support is prime therefore, $a \in \operatorname{supp} x$. □

Proposition 3.2.20. When A is a topological ring with (oifin) and (tn0) and $X = \operatorname{Cont}(A)$ or $\operatorname{Spa}(A, A^+)$. Then X_a is a finite union of elements of $\mathfrak{Rat}_X$. In particular, it is open.

Proof. By Lemma 3.2.2, $x \in X_a$ if and only if $A^{\circ\circ} \not\subseteq \operatorname{supp} x$. By our assumption, there are finitely many elements $f_1, \cdots, f_m \in A^{\circ\circ}$ generating an open ideal in A. Then $\mathfrak{p} \in \operatorname{Spec} A$ is open if and only if $f_1, \cdots, f_m \in \mathfrak{p}$. Therefore,

$$X_a = \bigcup_{i=1}^{m} \mathcal{R}_x(f_i | f_1, \cdots, f_m).$$

Note that these are only finitely many elements thus each valuation admits i taking maximum among them. □

Remark 3.2.6. If only (tn0) holds then,

$$X_a = \bigcup_{f \in A^{\circ\circ}} \mathcal{R}_X(f|)$$

is still open.

Proposition 3.2.21. Let $A \xrightarrow{f} B$ be a morphism of topological rings which is open. If $M \subseteq A$ is bounded then $f(M)$ is bounded in B.

Proof. Let $U \subseteq B$ be an open neighborhood of 0. Since M is bounded, there is $V \subseteq A$ open neighborhood of 0 such that $V \cdot M \subseteq f^{-1}(U)$. Since f is open, $f(V)$ is a neighborhood of 0 in B. □

Proposition 3.2.22. Let A be a topological ring with (oisq), (oifin), and (tn0). Suppose further that it has no dense maximal ideals. Let A^+ be an open subring of A which is a subset of A°. For $m \in \mathbb{N}$ and $(a_i)_{i=1}^m \in A^m$, the following conditions are equivalent.

a. The ideal $I = \langle a_1, \cdots, a_m \rangle_A$ is open.

b. For all $x \in \mathrm{Spa}(A, A^+)_a$, there is $1 \leq i \leq m$ such that $|a_i(x)| \neq 0$.

Proof. By the definition, $\max_{1 \leq i \leq m} |a_i|_x = 0$ if and only if $I \subseteq \operatorname{supp} x$. If I is open, no such x can be analytic. Thus a⇒b.

Assume that I is not open. By Proposition 3.2.18, there is a closed prime ideal $\mathfrak{p}$ such that $I \subseteq \mathfrak{p}$ which is maximal among non-open ideals. By (tn0), there is $x \in A^{\circ\circ} \backslash \mathfrak{p}$. As A^+ is a neighborhood of 0, without losing generality, (as $s^j \overset{j \to \infty}{\to} 0$), $x \in A^+$. We denote the image of A^+ in $A/\mathfrak{p} \subseteq k(\mathfrak{p})$ by B. By Proposition 3.2.21,

$$B \subseteq (A/\mathfrak{p})^\circ. \tag{3.4}$$

We now show that $(x \mod \mathfrak{p})^{-1} \in k(\mathfrak{p})$ is not integral over B. Suppose not, thus there will be $(p_i)_{i=1}^d \in B$ such that

$$1 = \sum_{i=1}^{d} p_i (x \mod \mathfrak{p})^i = (s \mod \mathfrak{p}) \sum_{i=0}^{d-1} p_{i+1} (x \mod \mathfrak{p})^i \in (x \mod \mathfrak{p}) B.$$

As $(x \mod \mathfrak{p}) \in (A/\mathfrak{p})^{\circ\circ}$ and each p_i is powerbounded, 1 is a sum of topologically nilpotent elements. Thus this implies that 1 is topologically nilpotent in $A/\mathfrak{p}$. This contradicts the fact that $\mathfrak{p}$ is closed in A as $1 \notin \mathfrak{p}$ in A but $1^k \overset{k \to \infty}{\to} 0 \in \mathfrak{p}$. By Proposition 1.3.5, there is a valuation ring R of $k(\mathfrak{p})$ containing B but not $(x \mod \mathfrak{p})^{-1}$. Let v denote the composition of

$$A \to A/\mathfrak{p} \to k(\mathfrak{p}) \overset{\text{valuation defined by } R}{\to} (k(\mathfrak{p})^\times / R^\times) \cup \{0\}.$$

There is a valuation and the support $\mathfrak{p}$ is not open and $v(a) \leq 1$ for $a \in A^+$, as $a \mod \mathfrak{p} \in B \subseteq R$ by construction. Thus the equivalence class of v is in $\mathrm{Spa}(A, A^+)_a$ and all $v(f_i) = 0$ as $I \subseteq \mathfrak{p}$. Hence to show that b cannot hold, it remains to show that v is continuous.

We first show that $K_{v \leq x^m}$ is a neighborhood of 0, for all $m \in \mathbb{N}$. If $m = 0$, this follows from $A^+ \subseteq K_{v \leq 1}$ and the openness of A^+. If there is $a \in A$ such that $v(ax^m) \geq 1$, then $K_{v \leq x^m}$ contains the preimage under $A \overset{a}{\to} A$ of $K_{v \leq 1}$, hence it is a neighborhood of 0. If no such a exists then $K_{v \leq x^n}$ contains $\mathfrak{p} + x^n A$ which is open by the maximality of $\mathfrak{p}$. Let $\alpha \in A \backslash \mathfrak{p}$. Then $\mathfrak{p} + \alpha A$ is an open ideal by the maximality of $\mathfrak{p}$. Since $x \in A^{\circ\circ}$, $x^n \in \mathfrak{p} + \alpha A$ for some positive integer. Thus there is $b \in A$ such that $x^n - b\alpha \in \mathfrak{p}$ and

$$v(b)v(\alpha) = v(x^r)$$

hence $v(b)v(\alpha) > v(x^{n+1})$ as $(s \mod \mathfrak{p})^{-1} \notin R$ and thus $v(s) < 1$. By the continuity and the previous part, $b^{-1} K_{v \leq s^{n+1}}$ is a neighborhood of 0 which is contained in $K_{v < \alpha}$. As $\mathrm{supp}(v) = \mathfrak{p}$, this shows v is continuous. □

Corollary 3.2.4. Under the assumption of Proposition 3.2.22 on A and A^+,

$$\langle a_1, \cdots, a_m \rangle_A = A \iff \forall x \in \mathrm{Spa}(A, A^+), \max_{1 \leq i \leq m} |a_i|_x > 0.$$

Proof. The necessity is obvious from Proposition 3.2.22. If the condition holds, $I = \langle a_1, \cdots, a_m \rangle_A$ is open by Proposition 3.2.22. If it is different from A, then there is an open maximal ideal $\mathfrak{m}$ such that $I \subseteq \mathfrak{m}$. If $x \in \mathrm{Spa}(A, A^+)$ is given by the trivial valuation for $\mathfrak{m}$, then $|a_i|_x = 0$, contradicting to our assumption. □

Remark 3.2.7. In particular, $a \in A^\times$ if and only if $|a|_x \neq 0$ for all $x \in \mathrm{Spa}(A, A^+)$ under the same assumption as in Proposition 3.2.22.

Lemma 3.2.3. Let f, g be continuous homomorphisms of topological groups such that gf is open then g is open.

Proof. Let U be a neighborhood of x. Then

$$g(U) = g(Ux^{-1})g(x).$$

Therefore, g is open if and only if for any neighborhood U of 1, $g(U)$ is a neighborhood of 1. Since gf is open, $gf(f^{-1}(U))$ is a neighborhood of 1. Since $g(U)$ contains $gf(f^{-1}(U))$ is a neighborhood of 1. □

Proposition 3.2.23. Let A be a topological ring and M be a topological A-module. Let $m, n \in \mathbb{N}$, $\vec{a} = (a_i)_{i=1}^m \in A^m$, $I = \langle a_1, \cdots, a_m \rangle$, and $\vec{b} = (b_i)_{i=1}^n \in I^n$. If

$$M^n \xrightarrow{\vec{b}} M, (m_i)_{i=1}^n \mapsto \sum_{i=1}^n b_i m_i$$

is open, then $M^m \xrightarrow{\vec{a}} M$ is also open.

Proof. Set

$$b_i = \sum_{j=1}^m c_{ij} a_j.$$

Then we have a commutative diagram,

$$\begin{array}{ccc} M^n & & \\ {\scriptstyle (c_{ij})_{ij}}\downarrow & \searrow^{\vec{b}} & \\ M^m & \xrightarrow[\vec{a}]{} & M \end{array}$$

Now apply Lemma 3.2.3. □

Remark 3.2.8. In particular, the openness of the map $\vec{a} : M^m \to M$ depends only on $I = \langle a_1, \cdots, a_m \rangle$.

Proposition 3.2.24. Let A be a topological ring such that the set of open ideals form a neighborhood basis of 0 and M be a topological A-module.Then the following conditions are equivalent.

a. For every neighborhood U of 0, there are $m \in \mathbb{N}$ and $\vec{a} \in U^m$ such that $M^m \xrightarrow{\vec{a}} M$ is open.

b. The above statement holds for open ideals.

If A has (oifin), this is equivalent to

c. If $m \in \mathbb{N}$ and $\vec{a} \in A^m$ generates an open ideal, then $M^m \xrightarrow{\vec{a}} M$ is open.

Proof. $\Rightarrow$ is trivial. For the converse, for an arbitrary open neighborhood of 0, take $I \subseteq U$, and apply Lemma 3.2.3. □

Definition 3.2.9. An A-module M is called adic if the equivalent conditions a and b hold in Proposition 3.2.24. An A-algebra is called adic if and only if it is adic as an A-module.

A is called id-adic if itself together with id_A is an A-adic algebra.

Proposition 3.2.25. Every id-adic ring satisfies (oifin) and (oisq).

Proof. If I is an open ideal. Then there are $(a_j)_{j=1}^m \in I^m$ such that $A^m \xrightarrow{\vec{a}} A$ is open. Then $\langle a_1, \cdots, a_m \rangle_A = \vec{a}(A^m)$, hence (oifin). Moreover, the image of I^m under $\vec{a}$ is also open and contained in the ideal square I^2, which is therefore open. □

Proposition 3.2.26. If A is analytic or satisfies (E), then every A-module (hence every A-algebra) is adic. In particular, A is id-adic.

Proof. If A is analytic, there are $(a_i)_{i=1}^m \in (A^{\circ\circ})^m$ such that $\langle a_1, \cdots, a_m \rangle = A$. Then also $\langle a_1^j, \cdots, a_m^j \rangle = A$ for all $j \in \mathbb{N}$. For any open ideal I, I contains all $a_1^j, \cdots, a_m^j$ for big enough j hence $I = A$. As A admits no proper open ideal, Proposition 3.2.24 b can be checked only for A and clearly identity map is open thus $n = 1, a_1 = 1$ works. The latter is trivial as every topological ring A with (E) is analytic. □

Proposition 3.2.27. Let $A \xrightarrow{f} B$ be a morphism of topological rings and M be a topological B-module which is adic as an A-module .Then M is adic as a B-module. In particular, if B is an adic A-algebra, then B is id-adic.

Proof. Let $I \subseteq B$ be an open ideal. By the continuity of f and adicity of A, there are $\vec{a} \in (f^{-1}(I))^n$ such that $M^n \xrightarrow{\vec{a}} M$ is open. Then such open map coincides to $M^n \xrightarrow{f(\vec{a})} M$ thus this is also open and $f(a_1), \cdots, f(a_n) \in I$. □

Proposition 3.2.28. Let $B \subseteq A$ be an open subring, M be a topological A-module, N an open sub-B-module of M. Then M is adic as an A-module if and only if N is adic over B.

Proof. $\Rightarrow$, let $U \subseteq B$ be open neighborhood of 0, then U is open in A, so we find $\vec{a} : M^n \to M$ open. As N is a open submodule of M, and $\vec{a} \in B^n$, $\vec{a} : N^n \to N$ is also open.

For the converse, for any open neighborhood U of A, apply the assumption to $B \cap U$, and use that $N \subseteq M$ is open, we get the statement. □

3.3 Nat-rings, Nat-modules and Huber Ring

Definition 3.3.1. A commutative (additively written) topological group is called nat (for non-Archimedean topological) if it has a neighborhood base of 0 where each element is an open subgroup. A topological ring A is nat if $(A, +)$ is so. A topological A-module M is nat if its underlying group is so.

A morphism of nat rings or nat A-modules is a morphism whose source and target are nat.

Example 3.3.1. $\mathbb{Q}_p$ with topology given by $|\cdot|_p$ or any other field with ultrametric absolute value and any ring with its discrete topology are nat.

Remark 3.3.1. For any topological ring A, A^{nat} is a topological ring A equipped with the topology where the open subgroups of $(A, +)$ are a neighborhood base of 0. Then A^{nat} is a nat-ring. Every morphism $A \to B$ of topological ring with B being nat factors over A^{nat}, and $\mathrm{Cont}(A^{\mathrm{nat}}) \to \mathrm{Cont}(A)$ is bijective.

Proposition 3.3.1. Let R be a topological ring, M be a R-module which is nat, $X \subseteq M$ a subset of M, and $X' \subseteq M$ be the subgroup generated by X. Then X is bounded if and only if X' is.

Proof. As $X \subseteq X'$, $\Leftarrow$ is clear. For $\Rightarrow$, take a neighborhood $U \subseteq M$ of 0. Without loss of generality, U is a subgroup of $(M, +)$. If X is bounded, there is a neighborhood V of 0 in R such that $VX \subseteq U$. Since U is a subgroup, $VX' \subseteq U$. Thus X' is bounded. □

Proposition 3.3.2. Let A be a nat ring, $X \subseteq A$ be a subset, and X' be a subgroup generated by X. Then X is power-bounded if and only if X' is. (Respectively, it is topologically nilpotent and powerbounded if X' is).

Proof. Let $Y = \bigcup_{n=0}^{\infty} X^n$ and $Y' \subseteq A$ be the subgroup generated by Y. Let Y'' be the subgroup generated by $\bigcup_{n=0}^{\infty} {X'}^n$ then $Y' \subseteq Y''$ is clear. $XY' \subseteq Y'$ and $X'Y' \subseteq Y''$ are clear. Thus ${X'}^m \subseteq Y'$ by induction on m. Hence $Y'' \subseteq Y'$. X is powerbounded if and only if Y is bounded by definition. By Proposition 3.3.1,, Y is bounded if and only if $Y' = Y''$ bounded. And Y'' is bounded if and only if X' is powerbounded. The latter case is similar. □

Proposition 3.3.3. Let A be a nat-ring, $X \subseteq A$ be a subset and $X' \subseteq A$ be the subring generated by X. Then the following condition are equivalent.

a. X is powerbounded.

b. X' is bounded.

c. X' is powerbounded.

Proof. Set $X^* = \bigcup_{n=0}^{\infty} X^n$. Then X' is generated by X^*. Therefore a$\Leftrightarrow$ b is clear from Proposition 3.3.1. As X' is a subring, $X'^n \subseteq X'$ for any $n \in \mathbb{Z}_{\geq 0}$. Therefore b$\Leftrightarrow$c. □

Proposition 3.3.4. Let A be a nat ring, $(X_i)_{i=1}^m$ are powerbounded subsets. Then the subring generated by $\bigcup_{i=1}^m X_i$ is powerbounded.

Proof. Follows from Proposition 3.1.6 and Proposition 3.3.3. □

Proposition 3.3.5. For a nat ring A, A° is a subring of A which is integrally closed in A, $A^{\circ\circ}$ is an ideal in A° and $\sqrt{A^{\circ\circ}} = A^{\circ\circ}$. If A is (tn1), then A° is clopen in A.

Proof. The fact that A° is a subring follows from Proposition 3.3.3. That $A^{\circ\circ}$ is a subgroup of $(A^\circ, +)$ follows from Proposition 3.3.2 and $A^\circ A^{\circ\circ} \subseteq A^{\circ\circ}$ follows from Proposition 3.1.6. That $\sqrt{A^{\circ\circ}} = A^{\circ\circ}$ follows from the definition. Indeed, if $r^n \in A^{\circ\circ}$ with $n > 0$, then

$$\lim_{j\to\infty} r^{i+jn} = 0$$

for all $0 \leq i < n$. Hence $\lim_{j\to\infty} r^j = 0$. If $x \in A$ is integral over A°, then there are $n \in \mathbb{N}$ and $(p_j)_{j=0}^{n-1} \in A^\circ$ such that

$$x^n = \sum_{j=0}^{n-1} p_j x^j.$$

Since finite set is bounded, a subgroup S generated by $A^\circ \cup \{p_0, \cdots, p_{n-1}\}$ is bounded by Proposition 3.3.1. Moreover, $x^n \in S$ thus x is powerbounded hence $x \in A^{\circ\circ}$. If (tn1) holds then $A^{\circ\circ}$ is a neighborhood of 0 hence so is A°. Thus A° is open, and in particular it is clopen as it is a subgroup. □

Remark 3.3.2. Note that A° can be unbounded (e.g. $A = \mathbb{Q}_p[T]/T^2\mathbb{Q}_p[T]$, $A^\circ = \mathbb{Z}_p + T \cdot \mathbb{Q}_p$) and $A^{\circ\circ}$ will often happen to be topologically nilpotent. For example, $A = \mathbb{C}_p$, then

$$A^{\circ\circ} = \{x \in \mathbb{C}_p \mid |x| < 1\}$$

such that $(A^{\circ\circ})^2 = A^{\circ\circ}$.

Proposition 3.3.6. Let A be a complete nat ring with (tn1) (hence $A^{\circ\circ}$ is open subgroup). If A is sequentially compact, then the following statements hold.

a. $A^\times$ is an open subset of A.

b. $A^\times \overset{a\mapsto a^{-1}}{\to} A$ is continuous

c. A has no dense proper ideals (hence no dense maximal ideals).

Proof. For a, set

$$s_n = \sum_{k=0}^{n} (-a)^k.$$

Using the sequentially compactness of A, it has a convergent subsequence s_{n_k}. Let the limit be s, then,

$$(1+a)s_{n_k} = 1 + (-1)^{n_k} a^{n_k+1} \overset{k\to\infty}{\to} 1 = (1+a)s.$$

Therefore, $A^\times$ contains $A^{\circ\circ} + 1$ which is an open neighborhood of 1. Using the continuity of multiplication, for each $a \in A^\times$, $A^{\circ\circ} + a$ is open. This also shows b as any preimage of an open subset is a union of $A^{\circ\circ} + a$.

For c, let $I \subseteq A$ be an dense ideal. Then $(A^{\circ\circ} + 1) \cap I \neq \emptyset$. Thus I contains a unit hence it is the whole. □

Proposition 3.3.7. Let A be a nat ring, B be a nat A-algebra which is adic over A, and $X \subseteq A$ bounded. Then the image of X in B is bounded.

Proof. Let U be a neighborhood of 0 in B, we need to find a neighborhood V of 0 in B such that $Vf(X) \subseteq U$, where $A \overset{f}{\to} B$ is the homomorphism making B into an A-algebra. Take a neighborhood U' of 0 in B such that $U'U' \subseteq U$. Without loss of generality, U and U' can be assumed to be subgroups of $(B, +)$. Let $W = f^{-1}(U') \subseteq A$, then take $\Omega \subseteq A$ such that $\Omega X \subseteq W$. By the adicity, there are $(\omega_i)_{i=1}^n := \vec{\omega}$ such that $B^n \overset{\vec{\omega}}{\to} B$ is an open map. Let $V = \vec{\omega}((U')^n)$ then

$$Vf(X) \subseteq Vf(W) \subseteq VU'$$

For $r = f(\omega_i)u$ with $u \in U'$, then $rf(X) \subseteq f(\Omega X)U' \subseteq f(W)U' \subseteq U'U' \subseteq U$, and write? U as a subgroup and the general elements of V is a sum of such r, we indeed have $Vf(X) \subseteq U$. □

Remark 3.3.3. In general, A° may well be unbounded in A for example,

$$A = \mathbb{Q}_p[T]/T^2, A^\circ = \mathbb{Z}_p + \mathbb{Q}_p T.$$

But every finite subset of it is contained in some bounded subring.

Definition 3.3.2. Let A be a ring and $I \subseteq A$ be an ideal. The I-adic topology on A is the topology for which $(A, +)$ is a topological group and $(I^n)_{n=0}^\infty$ is a neighborhood base of 0.

Proposition 3.3.8. For a nat ring A, the following conditions are equivalent.

a. A is id-adic satisfies (tn1) and has an open bounded subring B.

b. A is id-adic, satisfies (tn0) and has an open bounded subring B.

c. A has an open bounded subring B for which there is a finitely generated ideal $I \subseteq B$ such that the topology on B is the I-adic one.

Proof. a⇒b is trivial as (tn1) implies (tn0). For the converse, $A^{\circ\circ}$ generates an open ideal I by (tn0) and using the adicity of A, there are $(a_i)_{i=1}^m \in A^{\circ\circ m}$ such that $A^m \overset{\vec{a}}{\to} A$ is an open map. Note that B is bounded and $B^n \subseteq B$ for any

$n \in \mathbb{N}$ thus it is powerbounded. The image of B^m under this map is the open and topologically nilpotent since all a_i are topologically nilpotent.

b⇒ c, take $\vec{a} = (a_i)_{i=1,\cdots,m}$ as above, then for each $n \in \mathbb{N}$, the ideal generated by the a_i^n is also open since A has (oisq) by Proposition 3.2.25. Without loss of generality, all $a_i \in B$. It is justified as for large enough $n \in \mathbb{N}$, $a_i^n \in B$ by the openness of B thus replace a_i by a_i^n. We claim $I = \langle a_1, \cdots, a_m \rangle_B$ can be taken as such an open ideal. Indeed, $B^m \xrightarrow{\vec{a}} B$ is open and $I^{\cdot(j+1)}$ is the image of $(I^{\cdot j})^{\oplus m}$. Here we used $\cdot$ and $\oplus$ to distinguish ideal powers and Cartesian powers. Thus this is open using (oisq). To show that powers of I forms a neighborhood base of 0, take U is a neighborhood of 0 in A which we can assume to be a subgroup by nat. The boundedness of B implies there is a neighborhood U' of 0 such that $U'B \subseteq U$. As the $\{a_1, \cdots, a_m\}$ is topologically nilpotent, there is N such that for any

$$\forall \alpha \in \mathbb{N}^m, |\alpha| \geq N \Rightarrow \vec{a}^{\alpha} = \prod_{i=1}^{m} a_i \alpha_i \in U'.$$

Then $I^{\cdot N} \subseteq U$, showing $\{I^n\}_{n \in \mathbb{Z}_{\geq 0}}$ is indeed a neighborhood base of 0.

To show c⇒a, note that $I^n \subseteq B$ for any $n \in \mathbb{N}$ thus $I \subseteq A^{\circ\circ}$, in particular A satisfies (tn1). It remains to show id-adicity. If J is an open ideal in A there is N such that $I^{\cdot N} \subseteq J \cap B \subseteq J$ using I-adicity of B. Denote $I^{\cdot N} = \langle b_1, \cdots, b_l \rangle_B$ then $A^l \xrightarrow{\vec{b}} A$ is open. □

Definition 3.3.3. A nat-ring with satisfying the equivalent conditions in Proposition 3.3.8 is called a Huber ring.

Definition 3.3.4. Let A be a Huber ring. A ring of definition of A is an open bounded subring. A pair of definition of A is a pair (B, I) where B is an open bounded subring and $I \subseteq B$ a finitely generated ideal such that B is I-adic.

Corollary 3.3.1. Every ring of definition of a Huber ring can be extended to a pair of definition.

Proof. b⇒c in Proposition 3.3.8. □

Proposition 3.3.9. Every Huber ring satisfies (oisq), (oifin), and (tn1). Moreover A° is clopen and integrally closed subring of A. If A is (sequentially) compact and complete then $A^\times$ is an open subset of A and A has no proper dense ideal. In particular, every maximal ideal is then closed.

Proof. The first part follows from Proposition 3.2.25. The second part is by definition or equivalently by Proposition 3.3.8.b. The integrally closedness and the clopenness follows from (tn1) and Proposition 3.3.5. The last part follows from Proposition 3.3.6. Note that A is compact as it is I-adic and thus every point admits a countable open neighborhood basis. □

Proposition 3.3.10. Let B be an open subring of a nat ring A. Then B is Huber if and only if A is. In this case, B has a pair of definition which is a pair of definition for A.

Proposition 3.3.11. Let (B, I) be a pair of definition for a Huber ring A. Then a neighborhood X of A is bounded if and only if there is $l \in \mathbb{N}$ such that $I^{\cdot l} X \subseteq B$.

Definition 3.3.5. Let $I \subseteq A$ be an ideal of a topological ring. The formal/I-adic completion is the limit

$$\widehat{A} := \varprojlim A/I^n. \tag{3.5}$$

Proposition 3.3.12. Every Huber ring has a countable neighborhood base of 0, the completion of a Huber ring is a Huber ring (If (B, I) is a pair of definition, then $\widehat{B}$ is an open subring of $\widehat{A}$ and the topology of $\widehat{B}$ is $\widehat{I}$-adic with $\widehat{I} = \langle f_1, \cdots, f_n \rangle_{\widehat{B}}$ where $I = \langle f_1, \cdots, f_m \rangle_B$).

Definition 3.3.6. A Huber pair is a pair $A = (A^\lhd, A^+)$ such that $A^\lhd$ is a Huber ring and $A^+ \subseteq (A^\lhd)^\circ$ is an open and integrally closed subring of $A^\lhd$.

A Huber pair is complete if $A^\lhd$ is (sequential) complete.

A morphism $A \to B$ of Huber pairs is a continuous ring homomorphism $A^\lhd \to B^\lhd$ mapping A^+ to B^+. It is adic if $A^\lhd \to B^\lhd$ is adic

Lemma 3.3.1. Let $A = (A^\lhd, A^+)$ be a Huber pair. Then we have $(A^\lhd)^{\circ\circ} \subseteq A^+$.

Proof. Since A^+ is open for each $x \in A^{\circ\circ}$, there is $n \in \mathbb{N}$ such that $x^n \in B$. Using the integrally closedness of A^+, all roots of $T^n - x^n$ is in A^+. Thus $x \in A^+$. □

Lemma 3.3.2. Let $B \subseteq A$ be an open subgroup of a topological group A. Then A is sequentially complete if and only if B is. In our case, a Huber ring $(A^\lhd, A^+)$, $A^\lhd$ is complete if and only if A^+.

Proof. Open subgroups are closed and using that B is an open neighborhood of 0 in A, the statement is proven. □

Theorem 3.3.1. Let $A = (A^\lhd, A^+)$ be a Huber pair and $X = \mathrm{Spa}(A, A^+)$.

a. $A^+ = \{a \in A^\lhd \mid |a|_x \leq 1 \text{ for all } x \in X\}$.

b. X is a spectral space.

If A is complete, we also have,

c. The ideal $\langle a_1, \cdots, a_m \rangle_{A^\lhd}$ is open if and only if $\max_{1 \leq i \leq m} |a_i|_x > 0$ for all $x \in X_a$ This is the case if and only if $A^{\lhd m} \xrightarrow{\vec{a}} A$ is an open map.

d. $A^\lhd = \langle a_1, \cdots, a_m \rangle_{A^\lhd}$ if and only if $\max_{1 \leq i \leq m} |a_i|_x > 0$ for all $x \in X$.

A morphism of Huber pairs $\varphi : (A^\lhd, A^+) \to (B^\lhd, B^+)$ defines a continuous map $Y = \mathrm{Spa}(B^\lhd, B^+) \to \mathrm{Spa}(A^\lhd, A^+) = X$ which is the case of an adic morphisms is a spectral map.

Proof. a follows from Corollary 3.2.2. b is due to Corollary 3.2.3. The first part of c follows from Proposition 3.2.22, and the second part is from id-adicity with Proposition 3.2.24.c. d follows from Corollary 3.2.4. The morphism $\mathrm{Spa}(\varphi) : Y \to X$ is well-defined as for any $y \in Y$,

$$a \in A^+ \Rightarrow \varphi(a) \in B^+, y \circ \varphi(a) \leq 1.$$

The continuity is easily observed as the topology is induced from Spv and proven for Spv. Combining Theorem 3.3.1 and Proposition 3.2.13, the strong and the weak topologies on X, Y coincide. Suppose $B^\triangleleft/A^\triangleleft$ is adic and take $\Omega = \mathcal{R}(a_0|a_1, \cdots, a_m) \in \mathfrak{Rat}_X$ with $\langle a_0, \cdots, a_m\rangle_A$ open. By the adicity, $B^{m+1} \xrightarrow{\vec{a}} B$ is open hence $\langle \varphi(a_0), \cdots, \varphi(a_m)\rangle_B$ is open hence the preimage

$$\mathrm{Spa}(\varphi)^{-1}(\Omega) = \mathcal{R}_Y(\varphi(a_0)|\varphi(a_1), \cdots, \varphi(a_m)),$$

is rational open by the definition of the weak topology. It follows from Proposition 1.1.6 such maps are spectral. □

Remark 3.3.4. Note that $D(T) = \mathcal{R}(T|)$ is quasi-compact in $\mathrm{Spv}\,\mathbb{T}_{1,\mathbb{Q}_p}$ but not in $\mathrm{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ)$ as

$$D(T) = \bigcup_{l=0}^{\infty} \mathcal{R}(T|p^l).$$

Therefore $\mathrm{Spa}(\mathbb{T}_1, \mathbb{T}_1^\circ) \to \mathrm{Spv}(\mathbb{T}_1)$ is not spectral.

Proposition 3.3.13. Let A be a (sequentially) complete Huber ring. Let $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in A^{m+1}$ be such that

$$I = (f_0, \cdots, f_m)_A$$

is open in A. Let $X = \mathrm{Cont}(A)$. Put $f_i = 0$ when $i > m$. Then there is a neighborhood U of 0 in A such that for $n \geq m$ and $(g_i)_{i=0}^n \in A^{n+1}$ such that $g_i - f_i \in U$ for $0 \leq i \leq n$ with the following additional properties.

1. $I = (g_0, \cdots, g_n)_A$,
2. $\mathcal{R}_X(f_0|f_1, \cdots, f_m) = \mathcal{R}_X(g_0|g_1, \cdots, g_n)$.

Proof. Let $\vec{f} = (f_i)_{i=0}^m$ and $\vec{g} = (g_i)_{i=0}^m$. By the assumption, $A^{m+1} \xrightarrow{\vec{f}\cdot} A$ is open, hence the image U of $(A^{\circ\circ})^{m+1}$ under $\vec{f}\cdot$ is a neighborhood of 0. By our assumption about $\vec{g}$, all $g_i - f_i \in U \subseteq I$, hence all $g_i \in I$. We have $\vec{g} = (I_{m+1} + N)\vec{f}$ where I_{m+1} is an identity $(m+1)$-square matrix and N is an $(m+1)$-matrix with coefficients from $A^{\circ\circ}$. Then $det(I_{m+1} + N) \in 1 + A^{\circ\circ} \subseteq A^\times$ as the completeness of A is assumed. Hence $1_{m+1} + N$ is invertible and,

$$\langle f_0, \cdots, f_m\rangle = \langle g_0, \cdots, g_m\rangle.$$

For 2)., we distinguish the case $x \notin X_a$, ($\mathfrak{p} = \mathrm{supp}\, x$ open) and $x \in X_a$, ($\mathfrak{p} \in \mathrm{supp}\, x$ not open). In the first case, $A^{\circ\circ} \subseteq \mathfrak{p}$ ($a \in A^{\circ\circ} \Rightarrow a^n \in \mathfrak{p}$ when n is

large $\Rightarrow a \in \mathfrak{p}$) hence $U \subseteq \mathfrak{p}$ hence $g_i - f_i \in \mathfrak{p}$ and $|f_i|_x = |g_i|_x$ and x is in the right hand side of 2).

In the second case, $M = \max_{0 \leq i \leq m} |f_i|_x$ is positive (otherwise $I \subseteq \mathfrak{p}$ hence $\mathfrak{p}$ is open ad I is). For $a \in A^{\circ\circ}$, $|a|_x < 1$ as,

$$\lim_{n \to \infty} |a|_x^n = \lim_{n \to \infty} |a_i^n|_x = \left| \lim_{n \to \infty} a^n \right|_x = 0$$

by the continuity. Let $\rho = \max_{\substack{0 \leq i \leq n \\ 0 \leq j \leq m}} |n_{ij}|_x$ where

$$g_i - f_i = \sum_{j=1}^{m} n_{ij} f_j.$$

Then $|f_i - g_i|_x \leq \rho M$ hence,

$$\max(|f_i|_x, \rho M) = \max(|g_i|_x, \rho M).$$

Thus, $|g_i|_x = M$ when $M = |f_i|_x$. Hence M □

Remark 3.3.5. By Proposition 3.3.13.2, the same about $\mathrm{Spa}(A, A^+) \subseteq X$ follows.

Definition 3.3.7. A Tate ring is a nat ring A satisfying,

i. $A^{\circ\circ} \cap A^\times \neq \emptyset$,

ii. A contains a bounded open subring.

A ring of definition of A is an open bounded subring of A. A pair of definition of a Tate ring A is a pair (B, s) where B is a bounded open subring of A and $s \in B^{\circ\circ} \cap A^\times$.

Proposition 3.3.14. Let A be a Tate ring and B be a ring of definition.

a. There is $s \in B$ such that (B, s) is a ring of definition in Tate sense.

b. If (B, s) is a Tate pair of definition for A, $(s^j B)_{j \in \mathbb{Z}}$ or $(s^j B)_{j=n}^{\infty}$ for any $n \in \mathbb{N}$ are neighborhood bases of 0, and moreover $X \subseteq A$ is bounded if and only if $X \subseteq s^j B$ for some $j \in \mathbb{Z}$.

c. If (B, s) is a Tate pair of definition for A then (B, sB) is a pair of definition in the sense of Huber rings. Hence any Tate ring is Huber.

Proof. For a, by assumption, there is $s \in A^{\circ\circ} \cap A^\times$. By the openness of B, there is $j \in \mathbb{N}$ such that $s^j \in B$. Thus s^j can be taken.

For b, the first part is clear from that $s \in B^{\circ\circ}$. For the second statement, we must show that $VX \subseteq U$ for any U. But since $\{s^j B\}_{j \in \mathbb{Z}}$ is a topology basis $V = s^j B$ for some $j \in \mathbb{Z}$. And by taking $U = s^k B$, we obtain the statement. c follows immediately from b. □

Remark 3.3.6. Even in the Tate case, A° can fail to be bounded. For example,

$$A = \mathbb{Q}_p[T]/(T^2), A^\circ = \mathbb{Z}_p + T\mathbb{Q}_p.$$

Remark 3.3.7. If A is a Tate ring then the image of a bounded or power-bounded subset of A under the morphism $A \to B$ of topological rings is bounded or powerbounded. Thus the universal property of $A\langle X_1, \cdots, X_0\rangle$ takes the exact same one for $A[X_1, \cdots, X_n]$.

3.4 The Structure Presheaves on $\mathrm{Spa}(A^\triangleleft, A^+)$

We start with the special case of a topological localization of $A^\triangleleft$ with respect to $f \in A^\triangleleft$ such that $A^\triangleleft \xrightarrow{f} A^\triangleleft$ is open, giving $\mathscr{O}_X(\mathcal{R}_X(f|))$.

Lemma 3.4.1. If $C \xrightarrow{r} C$ is open then $(C/I) \xrightarrow{r \bmod I} (C/I)$ is always open.

Proof. Let U be a neighborhood of 0 in C/I then $V = \pi^{-1}(U)$ is also a neighborhood of 0 in C where $C \xrightarrow{\pi} C/I$. To show that $r \bmod I$ is open, we need to show that rU is open which is equivalent to $\pi^{-1}(rU)$ being open by definition. Since π is C-equivariant, $\pi^{-1}(rU) \supseteq rV$ and rV is also a neighborhood of 0 in C thus the statement follows. □

Proposition 3.4.1. Let A be a complete nat ring with a countable neighborhood base of 0 (e.g. a complete Huber ring). Let $f \in A$ such that $A \xrightarrow{f} A$ is an open map and $I = \bigcup_{j=0}^{\infty} \mathrm{Ker}(A \xrightarrow{f^j} A)$ and $\overline{I}$ the closure of I in A. Then we have the following statements,

1. $B = A/\overline{I}$ is a (sequentially) complete nat ring.
2. $B \xrightarrow{f \bmod \overline{I}} B$ open.
3. $A \to B$ has the universal property for morphisms $A \xrightarrow{\tau} T$ such that $T \xrightarrow{\tau(f)} T$ is injective and T is a complete nat ring. That is such τ must factor through $\pi : A \to B$.

Proof. Take a countable neighborhood basis $A = U_0 \supseteq U_1 \supseteq \cdots$ of A. Let $(b_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in B, then we can consider a lift $(a_n)_{n\in\mathbb{N}}$ in A. By the construction, for any $k \in \mathbb{N}$, there is N_k such that

$$m, n \geq N_k \Rightarrow a_m - a_n \in U_k + \overline{I}.$$

Therefore we modify $(a_n)_{n\in\mathbb{N}}$ inductively so that

$$m, n \geq N_k \Rightarrow a_m - a_n \in U_k.$$

Now $(a_n)_{n\in\mathbb{N}}$ is Cauchy and admits a limit a in A. Since $\overline{I}$ is closed, B is Hausdorff thus $\pi(a)$ is indeed a limit of $(b_n)_{n\in\mathbb{N}}$. Therefore B is complete.

The second assertion follows from Lemma 3.4.1.

For each $x \in \overline{I}$, there is $n \in \mathbb{N}$ such that $f^n x = 0$. Thus

$$0 = \tau(0) = \tau(f^n x) = \tau(f)^n \tau(x).$$

But $\tau(f)$ is injective thus $\tau(x) = 0$. More generally, $\tau(\overline{I}) = 0$.

It remains to show that π itself satisfies the condition in 3. Let $a \in A$ be such that $fa \in \overline{I}$. We may pick elements $(i_n)_{n\in\mathbb{N}}$ from I so that $fa - i_n \in f \cdot U_n$ with $i_0 = 0$. Since $i_{n-1} - i_n \in fU_{n-1}$, we may pick d_n from U_{n-1} such that

$$i_{n-1} - i_n = fd_n.$$

Therefore, $fd_n \in I$ and by construction $d_n \in I$. Now set,

$$j = \sum_{n=0}^{\infty} d_n.$$

Then $j \in \overline{I}$. Furthermore $f(a - j) = 0 \in I \subseteq \overline{I}$. Again by construction $a_j \in I$, therefore $a - j \in \overline{I}$. In particular, $a \in \overline{I}$. □

Corollary 3.4.1. Inheriting the notions from Proposition 3.4.1, if A is Huber (Tate, Analytic), then so is B (as these are identified by A/I).

Corollary 3.4.2. Let C be a complete nat ring with a countable neighborhood base of 0, $J_0 \subseteq C$ be a closed ideal, and $f \in C$ such that $C \xrightarrow{f} C$ is an open map. Let $J = \bigcup_{j=0}^{\infty}(J_0 : f^j)$ and $\overline{J}$ be the closure of J in C. The following statements hold.

a. $B = C/\overline{J}$ is a complete nat ring and $B \overset{f \bmod \overline{J}}{\to} B$ is an injective open map.

b. $C \to B$ has the universal property for morphisms $C \xrightarrow{\tau} T$ to complete topological rings such that $\tau(J_0) = 0$ and $T \overset{\tau(f)}{\to} T$ is injective. (i.e. such τ factors through $C \to B$)

c. C is Huber, Tate, or analytic then so is B.

Proof. We would like to apply Proposition 3.4.1 to $A = C/J_0$. As J_0 and $C \xrightarrow{\cdot f} C$ is open, we can apply Lemma 3.4.1 and Proposition 3.4.1. That is A is a complete nat ring and $\cdot\overline{f} = \cdot f \bmod J_0$ is open. Set

$$I := \bigcup_{j=0}^{\infty}(0 : (f \bmod J_0)^j)_A = \bigcup_{j=0}^{\infty} J_j/J_0 = J/J_0.$$

Using this we will show that $B \cong A/\overline{I}$. To do so we need to show that $\pi : C \to A$, $\pi(\overline{J}) = \overline{I}$. The above equation shows $\pi(J) = I$ and also clearly $\pi(\overline{J}) \subseteq \overline{I}$. As π is a quotient map and $\pi^{-1}(\overline{J}/J_0) = \overline{J}$, $\pi(\overline{J})$ is closed hence $\pi(\overline{J}) = \overline{I}$.

We now apply Proposition 3.4.1 to $A/\overline{I}$. Observe that $B \cong A/\overline{I}$. First τ factors through A, using the universality from Proposition 3.4.1, it factors through B. □

Proposition 3.4.2. Let A be a nat ring and $f \in A$ such that $A \xrightarrow{f} A$ is an injective open map. Consider A as a subring of the localization $A_f = A[T]/(1 - T \cdot f)_{A[T]} = A[f^{-1}]$. Equip $(A_f, +)$ with the topology for which U is a neighborhood of 0 in A_f if and only if $U \cap A$ is a neighborhood of 0 in A. We have the following statements.

a. A_f is a nat ring and has the universal property for morphisms $A \xrightarrow{\tau} T$ of topological rings such that $\tau(f) \in T^\times$.

b. A is an open subring of A_f.

c. A is Huber (respectively complete, sequentially complete) if and only if A_f is.

Proof. We need to check that A_f is indeed a topological ring. The addition follows easily from the fact that A is open in A_f as $A \cap A = A$ and the additive structure inherits directly from the one of A. More concretely, the continuity of addition can be checked at the origin and it translates to other points.

Now let us show the continuity of the multiplication of $A_f \times A_f \to A_f$. Certainly $A \times A \to A \subseteq A_f$ is continuous. Also observe that $A \xrightarrow{\cdot f} A$ is open, thus for an open set $U \subseteq A_f$, $(fU \cap A) = f(U \cap A)$ which is open. Hence $A_f \xrightarrow{\cdot f} A_f$ is a bijective open map, therefore a homeomorphism. Now take $m \in \mathbb{Z}$, $(f^{-m}A, +)$ is an open subgroup of $(A, +)$ and the continuity of

$$(f^{-m}A) \times (f^{-m}A) \to A_f$$

follows from the diagram below in which the vertical arrows are homeomorphisms and the bottom horizontal arrow is continuous by our initial observation. As $(A_f, +)$ is the ascending union of its open subgroups $f^{-m}A$, this shows the continuity of $A_f \times A_f \to A_f$.

$$\begin{array}{ccc} (f^{-m}A) \times (f^{-m}A) & \xrightarrow{\cdot} & A_f \\ {\scriptstyle (f^m\cdot)\times(f^m\cdot)}\downarrow & & \downarrow{\scriptstyle f^{2m}} \\ A \times A & \longrightarrow & A_f \end{array}$$

If $A \xrightarrow{\tau} T$ is a morphism of topological rings and $\tau(f) \in T^\times$, then we have a ring homomorphism $A_f \xrightarrow{\tilde{\tau}} T$ by the universal property of localization. This is continuous as for any $U \subseteq T$ open, $\tilde{\tau}^{-1}(U) \cap A = \tau^{-1}(U) \cap A$. □

Corollary 3.4.3. In the situation of Corollary 3.4.2, $B_{f \bmod \overline{J}}$ is a complete nat-ring and $C \to B_{f \bmod \overline{J}}$ has the universal property for morphisms $C \xrightarrow{\tau} T$ such that $\tau(J_0) = 0$ and $\tau(f) \in T^\times$. If C is Huber, (Tate, analytic), then so is $B_{f \bmod \overline{J}}$.

Proof. Follows from Corollary 3.4.2 and Proposition 3.4.2. □

Definition 3.4.1. Let A be a nat-ring. We equip $A[X_1, \cdots, X_n]$ with the topology such that $(A[X_1, \cdots, X_n], +)$ is a topological group and a neighborhood base of 0 is given by

$$\{U[X_1, \cdots, X_n] \mid U \text{ a neighborhood of 0 in } A\},$$

where

$$U[X_1, \cdots, X_n] = \left\{ f = \sum_{\alpha \in \mathbb{N}^n} f_\alpha X^\alpha \in A[X_1, \cdots, X_n] \mid \forall \alpha \in \mathbb{N}^n, f_\alpha \in U \right\}.$$

We define $A\langle X_1, \cdots, X_n \rangle$ as the completion of $A[X_1, \cdots, X_n]$. If M is an A-module, then $M[X_1, \cdots, X_n]$ a module over $A[X_1, \cdots, X_n]$, is the set of formal sum $\sum_{\alpha \in \mathbb{N}^n} m_\alpha X^\alpha$ where $m_\alpha \in M$, all but finitely many $m_\alpha = 0$. The neighborhood base of 0 is given by $U[X_1, \cdots, X_n]$, where U is a neighborhood of 0 in M as above. Similarly, $M\langle X_1, \cdots, X_n \rangle$ is defined as the completion of $M[X_1, \cdots, X_n]$.

Lemma 3.4.2. Let A be a nat-ring, B be a nat-A-algebra.

a. $A[X_1, \cdots, X_n]$ is a nat ring in which $\{X_1, \cdots, X_n\}$ are powerbounded.

b. If $C \subseteq A$ is an open subring, $C[X_1, \cdots, X_n]$ is an open subring of $A[X_1, \cdots, X_n]$.

c. If A is Huber(Tate, analytic), then so is $A[X_1, \cdots, X_n]$.

d. If (C, I) is a pair of definition for A, $(C[X_1, \cdots, X_n], I[X_1, \cdots, X_n])$ is also a pair of definition for $A[X_1, \cdots, X_n]$.

e. If $(s_1, \cdots, s_n) \in (A^{\circ\circ})^n$ generates A as an ideal (i.e. $\langle s_1, \cdots, s_n \rangle_A = A$) then they generate $A[X_1, \cdots, X_n]$ in $A[X_1, \cdots, X_n]$.

f. If $(b_i)_{i=1}^n$ are powerbounded in B, then the homomorphism,

$$A[X_1, \cdots, X_n] \ni p \mapsto p(b_1, \cdots, b_n) \in B,$$

is continuous.

g. If B is an adic A-algebra (e.g. as A is Tate or analytic), then we have a bijection,

$$\mathrm{Hom}_{(A-\mathrm{alg})^{\mathrm{top}}}(A[X_1, \cdots, X_n], B) \ni \varphi \overset{\cong}{\mapsto} (\varphi(X_i))_{i=1}^n \in (B^\circ)^n.$$

h. Suppose $\langle f_0, \cdots, f_n \rangle_A$ is open. Set $I = \langle f_0 X_1 - f_1, \cdots, f_0 X_n - f_n \rangle_{A[X_1, \cdots, X_n]}$ then $A[X_1, \cdots, X_n]/I \overset{f_0}{\to} A[X_1, \cdots, X_n]/I$ is an open map.

i. If M is an adic A-module, then so is $M[X_1, \cdots, X_n]$. In particular, A being id-adic implies $A \to A[X_1, \cdots, X_n]$ is adic.

Proof. For **b**, clearly $(C[X_1, \cdots, X_n], +)$ is a topological group. Being open follows from the definition and C itself is open. That $(A, +)$ is indeed a nat-group follows from the fact that the U in Definition 3.4.1 can be taken to be open subgroups of $(A, +)$. Therefore we obtain an open subgroup $(U[X_1, \cdots, X_n], +)$ of $A[X_1, \cdots, X_n]$.

If $P, Q \in A[X_1, \cdots, X_n]$ and U is an open subgroup of $(A, +)$, then by the continuity of multiplication in A, we have a neighborhood V of 0, such that

$$\forall \alpha \in \mathbb{N}^n, P_\alpha \cdot V, Q_\alpha \cdot V, V \cdot V \subseteq U.$$

(Note that only finitely many $\alpha \in \mathbb{N}^n$, $P_\alpha, Q_\alpha \neq 0$). Then

$$(P + V[X_1, \cdots, X_n])(Q + V[X_1, \cdots, X_n]) \subseteq (PQ) + U[X_1, \cdots, X_n].$$

Thus, multiplication is continuous.

For **a**, we need to show that

$$\Theta = \{X^\alpha \mid \alpha \in \mathbb{N}^n\}$$

is bounded. By Definition 3.4.1, it suffices to consider $U' = U[X_1, \cdots, X_n]$ in which case, obviously $\Theta U' \subseteq U'$.

f. Consider $A[X_1, \cdots, X_n] \xrightarrow{\varphi} B$ such that $\varphi(P) = P(b_1, \cdots, b_n)$. This is a ring homomorphism. To show its continuity, consider a neighborhood Ω of 0 in B. We must find a neighborhood V' of 0 in $A[X_1, \cdots, X_n]$ such that $\varphi(U') \subseteq \Omega$. Without loss of generality, we may take Ω to be a subgroup of $(B, +)$. By the powerboundedness of $\{b_1, \cdots, b_n\}$, there is a neighborhood V of 0 in B such that $b^\alpha V \subseteq \Omega$. Let U be the preimage of V in A. If $P \in U[X_1, \cdots, X_n]$, then all $b^\alpha P_\alpha \in b^\alpha V \subseteq \Omega$ hence $\varphi(U[X_1, \cdots, X_n]) \subseteq \Omega$ as Ω is a subgroup of $(B, +)$.

g. The surjectivity follows from **d** and the well-definedness follows from Remark 3.3.4.

h. Let U be an open subgroup of $(A, +)$ and $V \subseteq A$ the image of U^{n+1} under $A^{n+1} \xrightarrow{\vec{f}} A$. Observe that by Proposition 3.2.23, $\vec{f}$ is an open map hence V is open. Since $U[X_1, \cdots, X_n]$ is a basis element of the topology of $A[X_1, \cdots, X_n]$. Need to show that $f_0 U[X_1, \cdots, X_n]/I$ is open. Recall that for any $v \in V$, there are $u_0, \cdots, u_n \in U$ such that

$$v = \sum_{i=0}^{n} f_i u_i.$$

Observe that

$$v + I = f_0(u_0 + u_1 X_1 + \cdots + u_n X_n) + I.$$

Hence $V[X_0, \cdots, X_n] \subseteq U[X_0, \cdots, X_n]$ and this proves that $f_0 \bmod I$ is open as U, V were taken topological groups.

i. If U is a neighborhood of 0 in A, then by the assumption, there are $(\omega_i)_{i=1}^m \in U$ such that $M^m \xrightarrow{\vec{\omega}} M$ is an open map. Taking U to be a subgroup and $V = \vec{\omega}(U^m)$, we see that $M[X_1, \cdots, X_n]^m \xrightarrow{\vec{\omega}} M[X_1, \cdots, X_n]$ is also open. $\square$

Proposition 3.4.3. Let $X \xrightarrow{f} Y$ be a continuous homomorphism between abelian nat-groups which have countable neighborhood bases of 0. Let $\widehat{X}$ and $\widehat{Y}$ be their completions. If f is an open map then so is $\widehat{X} \xrightarrow{\widehat{f}} \widehat{Y}$.

Proof. There are open subgroups $(X_i)_{i=1}^{\infty}$ of X forming a neigh base of 0, with

$$X_1 \supseteq X_2 \supseteq \cdots$$

As $f(X_i)$ is open, there is a countable neighborhood base,

$$Y_1 \supseteq Y_2 \supseteq \cdots$$

of Y satisfying $Y_i \subseteq f(X_i)$ where Y_i is an open subgroup of Y for each i. Thus we aim to prove the statement by showing that $\widehat{Y_i} \subseteq \widehat{f}(\widehat{X_i})$ for each i. An element y of $\widehat{Y_i}$ is represented as an equivalence class of Cauchy sequence $(y_j)_{j=1}^{\infty}$ with elements $y_i \in Y_i$. Without loss of generality by taking subsequences, we assume $y_{j+1} - y_j \in Y_j$ for all $j \geq 1$. By assumption, there is $x_1 \in X_1$ with $f(x_1) = y_1$ and there are $(\xi_j)_{j=1}^{\infty}$ where $\xi_j \in X_j$ such that

$$f(\xi_j) = y_{j+1} - y_j.$$

Let

$$x_j = x_1 + \sum_{l=1}^{j-1} \xi_l.$$

Then

$$x_{j+1} - x_j = \xi_j \in X_j.$$

Therefore $(x_j)_{j=1}^{\infty}$ is Cauchy and $f(x_j) = y_j$. Thus $\widehat{Y_1} \subseteq f(\widehat{X_1})$. The argument generalizes directly to $\widehat{X_i}$ and proves our statement. □

Proposition 3.4.4. Let A be a nat-ring with a countable neighborhood base of 0 and B be a complete nat-A-algebra.

a. $A\langle X_1, \cdots, X_n\rangle$ is a nat ring in which $\{X_1, \cdots, X_n\}$ is powerbounded.

b. If C is an open subring of A then $C\langle X_1, \cdots, X_n\rangle$ is an open subring of $A\langle X_1, \cdots, X_n\rangle$ in which $C[X_1, \cdots, X_n]$ is dense.

c. If A is Huber (Tate, analytic) then so is $A\langle X_1, \cdots, X_n\rangle$. For example, in **b**, if C is Huber and $\langle c_1, \cdots, c_n\rangle_C$-adic, then $C\langle X_1, \cdots, X_n\rangle$ is $\langle c_1, \cdots, c_n\rangle$-adic.

d. For each $(b_i)_{i=1}^n \in (B^\circ)^n$, there is a unique continuous $A\langle X_1, \cdots, X_n\rangle \to B$ sending X_i to b_i.

e. If B is adic over A (e.g. A is Tate, analytic or 3.1(E) holds), then we have the following bijection,

$$\mathrm{Hom}_{\mathrm{nat}-A-\mathrm{alg}}(A\langle X_1, \cdots, X_n\rangle, B) \ni \varphi \overset{\sim}{\mapsto} (\varphi(X_i))_{i=1}^n \in (B^\circ)^n.$$

f. If $\langle f_1, \cdots, f_n \rangle_A$ is open and $I = A\langle X_1, \cdots, X_n \rangle$ is an arbitrary closed ideal containing all $f_0 X_i - f_i$. Set $B = A\langle X_1, \cdots, X_n \rangle / I$. We have $B \overset{f_0}{\to} B$ is open.

g. If M is an adic A-module with a countable neighborhood base of 0 (e.g. $M = A$ if A is id-adic). Then $M\langle X_1, \cdots, X_n \rangle$ is an adic A-module.

Proof. **a.** By definition a subset $S \subseteq A\langle X_1, \cdots, X_n \rangle$ is powerbounded if for any neighborhood U of 0 there is another neighborhood V of 0 such that $VS \subseteq U$. The topology base of $A\langle X_1, \cdots, X_n \rangle$ is given by

$$\{U\langle X_1, \cdots, X_n \rangle \mid U \subseteq A \text{ an open additive subgroup}\}.$$

Note that $X_i U\langle X_1, \cdots, X_n \rangle \subseteq U\langle X_1, \cdots, X_n \rangle$, therefore $\{X^\alpha \mid \alpha \in \mathbb{N}^n\}$ is powerbounded.

b. The openness follows from that for any open subgroup $U \subseteq C$,

$$U\langle X_1, \cdots, X_n \rangle \subseteq C\langle X_1, \cdots, X_n \rangle.$$

The density follows from the definition.

c. Lemma 3.4.2 and Proposition 3.3.12.

d. Since B is complete and morphisms of topological rings are continuous, we have,

$$\mathrm{Hom}_{\mathrm{nat}-A-\mathrm{alg}}(A\langle X_1, \cdots, X_n \rangle, B) \overset{\text{Continuity}}{\cong} \mathrm{Hom}_{\mathrm{nat}-A-\mathrm{alg}}(A[X_1, \cdots, X_n], B) \overset{\text{Lemma 3.4.2.f}}{\cong} (B^\circ)^n.$$

e. By Proposition 3.2.27, all such φ-s are adic and use that $X_1, \cdots, X_n$ are powerbounded from Lemma 3.4.2.a.

f. Use Lemma 3.4.2 and Proposition 3.4.3.

g. Follows from Proposition 3.4.3. □

Definition 3.4.2. Let $\mathscr{C}$ be a category. A functor $F : \mathscr{C} \to (\mathrm{Sets})$ is said to be represented by $B \in \mathrm{Ob}(\mathscr{C})$ if there is a natural isomorphism,

$$F \cong \mathrm{Mor}_{\mathscr{C}}(B, \cdot).$$

Equivalently, B represents F if for any $C \in \mathrm{Ob}(\mathscr{C})$, there is $b \in F(B)$ such that evaluation at b gives a bijection

$$\mathrm{Mor}_{\mathscr{C}}(B, C) \ni \varphi \overset{\mathrm{ev}_b \circ F}{\mapsto} F(\varphi)(b) \in (\mathrm{Sets}).$$

Remark 3.4.1. If $F : \mathscr{C} \to (\mathrm{Sets})$ is represented by $B \in \mathrm{Ob}(\mathscr{C})$, then such B is unique up to isomorphisms.

Notation 3.4.1. $\mathcal{P}$ denotes the category of complete Huber pairs.

Definition 3.4.3. Let $A = (A^\triangleleft, A^+)$ and $X = \mathrm{Spa}(A^\triangleleft, A^+)$. For a rational open subset $\Omega = \mathcal{R}_X(f_0 | f_1, \cdots, f_m)$, we define a functor $\mathcal{F}_\Omega : \mathcal{P}^{\mathrm{op}} \to (\mathrm{Sets})$ to be

$$\mathcal{P} \ni C \overset{\mathcal{F}_\Omega}{\mapsto} \mathcal{F}_\Omega(C) := \{\varphi \in \mathrm{Hom}_{\mathrm{Huber}}(A, C) \mid \mathrm{Spa}(\varphi)(\mathrm{Spa}(C)) \subseteq \Omega\}.$$

If $\mathcal{F}_\Omega$ is representable, we will denote the representing object to be $(\mathcal{O}_X(\Omega), \mathcal{O}_X^+(\Omega))$.

Theorem 3.4.1. Let $A = (A^\triangleleft, A^+)$, $X = \mathrm{Spa}(A^\triangleleft, A^+)$, and $\Omega = \mathcal{R}_X(f_0|f_1, \cdots, f_m)$ be rational open. Then,

$$\mathcal{O}_X(\Omega) = (A^\triangleleft\langle X_1, \cdots, X_m\rangle/J)_{f_0}$$

for some ideal J and $\mathcal{O}_X^+(\Omega)$ to be the integral closure of $A^+\langle X_1, \cdots, X_m\rangle + J$ in the localization. More concretely, let

$$I := \overline{\langle f_i - f_0 X_i\rangle_{i=1,\cdots,m}}.$$

Then we can take

$$J = \overline{\bigcup_{l=0}^{\infty}(I : f_0^l)}.$$

Proof. Using J in the statement, set

$$B^\triangleleft := (A^\triangleleft\langle X_1, \cdots, X_m\rangle/J)\,[f_0^{-1}],$$

and B^+ to be the integral closure of $A^+\langle X_1, \cdots, X_m\rangle + J$ in the localization.

$A^\triangleleft\langle X_1, \cdots, X_m\rangle$ is complete by definition and Huber by Proposition 3.4.4.c. By Corollary 3.4.2, $A^\triangleleft\langle X_1, \cdots, X_m\rangle/J$ is again complete and Huber. Using Proposition 3.4.2, $B^\triangleleft$ is complete and Huber. Using Proposition 3.4.4.b and Proposition 3.4.2, we see that $A^+\langle X_1, \cdots, X_m\rangle + J$ is open. Since $A^+\langle X_1, \cdots, X_m\rangle + J \subseteq B^+$ (i.e. it contains an open subring), B^+ is also open. Furthermore,

$$A^\triangleleft\langle X_1, \cdots, X_m\rangle \to A^\triangleleft\langle X_1, \cdots, X_m\rangle/J \to (A^\triangleleft\langle X_1, \cdots, X_m\rangle)_{f_0} = B^\triangleleft \quad (3.6)$$

is a composition of open maps, hence open. By Proposition 3.4.4.a, each indeterminate is powerbounded and by definition $A^+ \subseteq (A^\triangleleft)^\circ$, hence

$$A^+\langle X_1, \cdots, X_m\rangle \subseteq (A^\triangleleft\langle X_1, \cdots, X_m\rangle)^\circ.$$

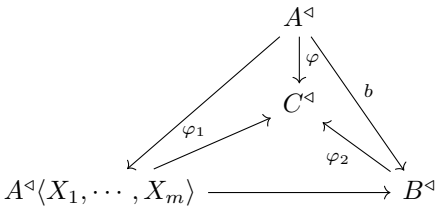
By Proposition 3.2.21 and using the openness of the morphism in Equation (3.6), the image of $A^+\langle X_1, \cdots, X_m\rangle$ is open. By Proposition 3.3.5, $B^+ \subseteq (B^\triangleleft)^\circ$. Precomposing the canonical inclusion $A^\triangleleft \to A^\triangleleft\langle X_1, \cdots, X_m\rangle$ to the morphism in Equation (3.6), we obtain a morphism of complete Huber pairs $b : (A^\triangleleft, A^+) \to (B^\triangleleft, B^+)$. We now show that $b \in \mathcal{F}_\Omega(B)$, that is $\mathrm{Spa}(b)(\mathrm{Spa}(B)) \subseteq \Omega$. Let $q_i := b(X_i)$ for each $X_i \in A^\triangleleft\langle X_1, \cdots, X_m\rangle$. Then by Theorem 3.3.1.a and the construction of I,

$$\forall y \in \mathrm{Spa}(B), y(b(f_i)) = y(b(f_0X_i)) = y(q_i)y(b(f_i)) \leq y(b(f_i)).$$

As f_0 is a unit in $B^\triangleleft$ by construction, we have $y(b(f_0)) \neq 0$. Hence $\mathrm{Spa}(b)(\mathrm{Spa}(B)) \subseteq \Omega$ showing $b \in \mathcal{F}_\Omega(B)$.

It remains to show that evaluation at b gives a desired bijection. Let $C \in \mathcal{P}$ and $\varphi \in \mathcal{F}_\Omega(C)$. By definition and Theorem 3.3.1, $\varphi(f_0) \in (C^\triangleleft)^\times$. Set $q_i = \frac{\varphi(f_i)}{\varphi(f_0)}$, then for any $z \in \mathrm{Spa}(C)$, $z(q_i) \leq 1$. Hence by Theorem 3.3.1.a, $q_i \in C^+$ for each $i = 0, \cdots, m$. Hence by Proposition 3.4.4.d, there is a unique morphism

of nat-rings $\varphi_1 : A\langle X_1, \cdots, X_m\rangle \to C$ sending X_i to q_i and the composition with $A^\triangleleft \to A^\triangleleft\langle X_1, \cdots, X_m\rangle$ gives φ. By construction and the continuity of φ_1, we obtain $\varphi_1(I) = 0$. Recalling that $\varphi(f_0)$ is invertible, we see φ_1 annihilates $(I : f_0^l)$ for any $l \geq 0$. Again by the continuity $\varphi_1(J) = 0$. Using Corollary 3.4.3, φ_1 factors through $B^\triangleleft$. We denote the lifted morphism $\varphi_2 : B^\triangleleft \to C^\triangleleft$. It remains to show that φ_2 is a morphism of Huber pairs. At the moment, we have the following diagram,



Using that $\varphi(A^+) \subseteq C^+$ and $q_i \in C^+$ for each $i = 0, \cdots, m$, we obtain

$$\varphi_1(A^+[X_0, \cdots, X_m]) \subseteq C^+.$$

Using the density of $A^+[X_0, \cdots, X_m]$ in $A^+\langle X_1, \cdots, X_m\rangle$ by Proposition 3.4.4.b, the continuity of φ_1, and C^+ is an open subgroup hence also closed, we get

$$\varphi_1(A^+\langle X_1, \cdots, X_m\rangle) \subseteq C^+.$$

Using the commutative diagram,

$$\varphi_2(A^+\langle X_1, \cdots, X_m\rangle) \subseteq C^+.$$

Finally $\varphi_2(B^+)$ is integral over $\varphi_2(A^+\langle X_1, \cdots, X_m\rangle)$ by construction and C^+ is integrally closed by definition, we obtain,

$$\varphi_2(B^+) \subseteq C^+.$$

We have shown the surjectivity. Furthermore, the construction of morphisms has been all unique thus the injectivity also follows. □

Corollary 3.4.4. In the situation of Theorem 3.4.1, if $\langle f_0, \cdots, f_m\rangle_A = A$ holds (e.g. A is Tate or analytic), then we can take,

$$B^\triangleleft := A^\triangleleft\langle X_1, \cdots, X_m\rangle/I.$$

Proof. By the assumption, we have,

$$\begin{aligned} A^\triangleleft\langle X_1, \cdots, X_m\rangle/I &= (f_0, \cdots, f_m)(A^\triangleleft\langle X_1, \cdots, X_m\rangle/I), \\ &= (f_0, f_0X_1 \cdots, f_0X_m)(A^\triangleleft\langle X_1, \cdots, X_m\rangle/I), \\ &= f_0(A^\triangleleft\langle X_1, \cdots, X_m\rangle/I). \end{aligned}$$

Hence f_0 is already a unit in $(A^\triangleleft\langle X_1, \cdots, X_m\rangle/I)$. In particular,

$$(I : f_0^l) = I \text{ and } (A^\triangleleft\langle X_1, \cdots, X_m\rangle/I)_{f_0} = A^\triangleleft\langle X_1, \cdots, X_m\rangle/I.$$

□

Corollary 3.4.5. Let $A = (A^\triangleleft, A^+), X = \mathrm{Spa}(A^\triangleleft, A^+), \Omega = \mathcal{R}_X(f_0|f_1, \cdots, f_m)$ a rational open subset. We denote $\mathcal{P}$ to be the category of complete Huber pairs $(B^\triangleleft, B^+)$.

a. $\mathcal{F}_\Omega$ is representable.

b. The image of f_0 in $\mathcal{O}_X(\Omega)$ is a unit and the image of the resulting morphism $A^\triangleleft_{f_0} \to \mathcal{O}_X(\Omega)$ is a dense subring of $\mathcal{O}_X(\Omega)$.

c. The morphism $A^\triangleleft \to \mathcal{O}_X(\Omega)$ is adic. Furthermore, if $A^\triangleleft$ is Tate or analytic then so is $\mathcal{O}_X(\Omega)$.

d. The subring $\mathcal{O}_X^+(\Omega)$ is integral over the closure in $\mathcal{O}_X(\Omega)$ of the subring generated by $b(A^+)$ and the $\frac{b(f_i)}{b(f_0)}$.

Proof. **a** follows immediately from Theorem 3.4.1. For the sake of simplicity, we denote $(B^\triangleleft, B^+) = (\mathcal{O}_X(\Omega), \mathcal{O}_X^+(\Omega))$.

d and **e** are already shown in the proof of Theorem 3.4.1. For **b.** we have seen that by Proposition 3.4.2.b and Proposition 3.4.4.b that

$$\overline{A^\triangleleft[X_0, \cdots, X_m]} \subseteq A^\triangleleft\langle X_1, \cdots, X_m\rangle \subseteq A_{f_0}\langle X_1, \cdots, X_m\rangle.$$

By construction,

$$B^\triangleleft = \bigcup_{j\geq 0} f^{-j}(A^\triangleleft\langle X_1, \cdots, X_m\rangle/J).$$

However, $f^{-j}(A[X_0, \cdots, X_m] + J)$ is dense in $f^{-j}(A^\triangleleft\langle X_1, \cdots, X_m\rangle/J)$ and $X_i = \frac{f_i}{f_0}$ in $B^\triangleleft$, the image of $A^\triangleleft_{f_0} \to B^\triangleleft$ does contain every elements of $f^{-j}(A[X_0, \cdots, X_m]+J)$, hence it is dense.

c. Use Proposition 3.4.4.c, Corollary 3.4.2.c, and Corollary 3.4.3 for the latter statement. For the first part, note that A is Huber hence id-adic by definition. Using 3.4.4.g, $A^\triangleleft\langle X_1, \cdots, X_m\rangle$ is adic. Hence for any open ideal $I \subseteq A$, there are finitely many $\{a_1, \cdots, a_r\} \subseteq I$ such that

$$(A^\triangleleft\langle X_1, \cdots, X_m\rangle)^r \ni (\alpha_i)_{i=1,\cdots,r} \overset{\vec{a}}{\mapsto} \sum_{i=1}^r a_i\alpha_i \in A^\triangleleft\langle X_1, \cdots, X_m\rangle$$

is open. Consider the following diagram,

$$\begin{array}{ccc} (A^\triangleleft\langle X_1, \cdots, X_m\rangle)^r & \xrightarrow{\vec{a}} & A^\triangleleft\langle X_1, \cdots, X_m\rangle \\ \downarrow & & \downarrow \\ (B)^r & \xrightarrow[\vec{a}]{} & B \end{array}$$

Since $\vec{a}$ on the top and the canonical morphism $A^\triangleleft\langle X_1, \cdots, X_m\rangle \to B$ is open, using Lemma 3.2.3, we see $\vec{a}$ on the bottom is also open. Hence B is an adic A-module. □

It remains to prove the last two assertions of Corollary 3.4.5. Bijectivity of $Y \overset{\beta}{\to} \Omega$, plus identification of topologies of $\mathfrak{Rat}$ and $(\mathscr{O}(\cdots), \mathscr{O}^+(\cdots))$. That $\beta(Y) \subseteq \Omega$ follows from $b \in \mathcal{F}_\Omega(B)$ which was checked before. For the bijectivity of β, we need the following proposition.

Proposition 3.4.5. Let A be the (completion or) sequential completion of a topological ring $\mathcal{A}$. Then $\mathrm{Cont}(A) \to \mathrm{Cont}(\mathcal{A})$ is bijective.

Proof. We will first show that for a valuation $v \in \mathrm{Cont}(\hat{A})$ and the restriction v_0 of v by precomposing $A \to \hat{A}$, we have $\Gamma_v = \Gamma_{v_0}$. First the inclusion $\Gamma_{v_0} \subseteq \Gamma_v$ is immediate. For any $\alpha \in \hat{A}$ and $\gamma = v(a) \neq 0$, $W = K_{v<\gamma}$ is open by assumption. As A is dense in $\hat{A}$, there is $a \in A$ such that $a \in \alpha + W$. By definition, $v(\alpha + W) = \{\gamma\}$. Hence $v_0(a) = \gamma$, therefore $\Gamma_v \subseteq \Gamma_{v_0}$.

Let $u, v \in \mathrm{Cont}(A)$ such that $u_0 \sim v_0$. Hence $\Gamma_{u_0} \cong \Gamma_{v_0}$. By the previous claim, we have

$$\Gamma_u = \Gamma_{u_0} \cong \Gamma_{v_0} = \Gamma_v.$$

Thus we may assume $u_0 = v_0$. Then u, v coincide on A which is dense in $\hat{A}$. By the continuity and the topology of the codomain is Hausdorff, $u = v$.

For the surjectivity, let $v \in \mathrm{Cont}(A)$ and we need to show that this lifts to $\hat{v} \in \mathrm{Cont}(\hat{A})$. Let $(a_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in A. If $a_n \overset{n\to\infty}{\to} 0$, then we put $\hat{v}((a_n)_{n\in\mathbb{N}}) = 0$. Otherwise, there is $\delta > 0$ such that $n \gg 0 \Rightarrow a_n \notin [0, \delta) \subseteq \Gamma \cup \{0\}$. As $(a_n)_{n\in\mathbb{N}}$ is Cauchy, there is $N \in \mathbb{N}$ such that

$$i, j \geq N \Rightarrow v(a_j - a_i) < \delta.$$

Suppose $N \gg 0$ so that $i \geq N \Rightarrow v(a_i) \geq \delta$. By definition,

$$v(a_j) = v(a_i + (a_j - a_i)) = v(a_i).$$

Hence $(v(a_n))_{n\in\mathbb{N}}$ stabilizes eventually to some γ. Hence we put $\hat{v}((a_n)_{n\in\mathbb{N}}) = \gamma$. Since we have defined the value for the Cauchy sequence converging to 0, this gives us the independence of the choice of Cauchy sequence. Furthermore, all axioms of valuations are closed conditions, hence by the continuity such $\hat{v}$ is indeed a valuation. □

Theorem 3.4.2. In the situation of Corollary 3.4.5, the map $Y := \mathrm{Spa}(B) \overset{\mathrm{Spa}(b)}{\to} \Omega$ where b is the universal element is a homeomorphism sends following $\mathfrak{Rat}_Y$ with $(\mathfrak{Rat}_Y)_\Omega := \{\Theta \in \mathfrak{Rat}_X \mid \Theta \subseteq \Omega\}$.

Proof. We denote $\beta = \mathrm{Spa}(b)$. By the assumption $\beta(Y) \subseteq \Omega$ hence the question is well-defined. Let $v \in \mathrm{Spa}(A^\triangleleft, A^+)$ be a valuation since $v \in \Omega$, under the homeomorphism in Remark 1.2.9, we lift $v_{f_0} \in \mathrm{Cont}(A_{f_0})$. Then by precomposing the evaluation map,

$$A^\triangleleft[X_1, \cdots, X_m] \ni P \overset{\mathrm{ev}}{\mapsto} P\left(\frac{f_1}{f_0}, \cdots \frac{f_m}{f_0}\right) \in A^\triangleleft_{f_0},$$

we obtain a lift $v_1 \in \mathrm{Spa}(A^\triangleleft[X_1, \cdots, X_m], A^+[X_1, \cdots, X_m])$.

$$\begin{aligned} v_1(P) &= v_{f_0}\left(\sum_{\alpha\in\mathbb{N}^m} p_\alpha \prod_{i=1}^m \left(\frac{f_i}{f_0}\right)^{\alpha_i}\right), \\ &\leq \max_{\alpha\in\mathbb{N}^m} v_0\left(p_\alpha \prod_{i=1}^m \left(\frac{f_i}{f_0}\right)^{\alpha_i}\right), \\ &\leq \max_{\alpha\in\mathbb{N}^m} v(p_\alpha) \prod_{i=1}^m \left(\frac{v(f_i)}{v(f_0)}\right)^{\alpha_i}, \\ &\leq \max_{\alpha\in\mathbb{N}^m} v(p_\alpha) \end{aligned} \tag{b}$$

Hence for any $[0,\gamma) \subseteq \Gamma\cup\{0\}$ open, $K_{v\leq\gamma}[X_1, \cdots, X_n]$ is open and the image under v_1 is contained in $[0,\gamma)$, hence $v_1 \in \mathrm{Spa}(A^\triangleleft[X_1, \cdots, X_n], A^+[X_1, \cdots, X_n])$.

By Proposition 3.4.5, there is a unique continuous extension,

$$\widehat{v_1} : A^\triangleleft\langle X_1, \cdots, X_m\rangle \xrightarrow{r} \Gamma \cup \{0\}.$$

By construction $\widehat{v_1}$ annihilates I since $\widehat{v_1}(f_0X_i - f_i) = 0$ and $\widehat{v_1}$ is continuous. Furthermore, as $\widehat{v_1} = v(f_0) \neq 0$, hence this also kills J. Again using the identification in Remark 1.2.9, we obtain $w : B^\triangleleft = (A^\triangleleft\langle X_1, \cdots, X_m\rangle/J)_{f_0} \to \Gamma \cup \{0\}$. As $v_1 \in \mathrm{Spa}(A^\triangleleft[X_1, \cdots, X_m], A^+[X_1, \cdots, X_m])$, using the continuity,

$$A^+\langle X_1, \cdots, X_m\rangle \subseteq K_{w\leq 1}.$$

By Proposition 1.2.5, $K_{w\leq 1}$ is integrally closed. Since B^+ is the integral closure of $A^+\langle X_1, \cdots, X_m\rangle$, $B^+ \subseteq K_{w\leq 1}$. Hence $\beta(w) = v$.

Suppose we obtained another valuation $\widehat{v}_1'$ of $A^\triangleleft\langle X_1, \cdots, X_m\rangle$ from v_1. Let $P \in A^\triangleleft[X_1, \cdots, X_m]$ be of degree d, then set

$$Q(X_0, \cdots, X_m) := X_0^d P\left(\frac{X_1}{X_0}, \cdots, \frac{X_m}{X_0}\right).$$

Set $q := (f_0, \cdots, f_m)$, then $f_0^d P - q \in I$ since $Q \equiv q \mod I$. Since $\widehat{v}_1'$ kills I, hence

$$\widehat{v}_1'(f_0)^d \widehat{v}_1'(P) = \widehat{v}_1'(q).$$

However, $\widehat{v}_1'(f_0) = v_{f_0}(f_0) \neq 0$ and $\widehat{v}_1'(q) = v_{f_0}(q)$. Hence, $\widehat{v}_1'(P)$ depends solely on v_{f_0} therefore it must coincide with $\widehat{v}_1$ on a dense subset. Using the continuity, we have shown the uniqueness of the extension. All the other extensions were obtained from homeomorphisms, hence showing the injectivity.

It remains to show that β is a homeomorphism. To do so, it is enough to show that β is an open map. Consider a rational open $\Theta = \mathcal{R}_Y(g_0|g_1, \cdots, g_k)$ where $\langle g_0, \cdots, g_k\rangle_B$ generates an open ideal. Using Proposition 3.3.13 together with the density of $A_{f_0}^\triangleleft$ in $B^\triangleleft$ by Corollary 3.4.5.b, there is an open subset $U \subseteq B^\triangleleft$ such that $g_i - b(r_i) \in U$ for each $i = 0, \cdots, k$ and

$$\langle g_0, \cdots, g_k\rangle = \langle b(r_0), \cdots, b(r_k)\rangle,$$

hence open. Furthermore, as f_0 is a unit in $B^\triangleleft$, we may assume that $g_i = b(\alpha_i)$ for some $\alpha_i = f_0^d r_i \in A^\triangleleft$ for each $i = 0, \cdots, k$ and fixed $d \geq 0$. As $f_0^d U$ is an open subset, $V = b^{-1}(f_0^d U)$ is also open. Since $A^\triangleleft$ is Huber, by Proposition 3.3.12, there is an finitely generated open ideal whose generators are topologically nilpotent elements $\{s_1, \cdots, s_l\}$. We may choose these elements small enough so that,

$$\langle s_1, \cdots, s_l \rangle_{A^\triangleleft} \subseteq V.$$

We extend α_i by those in order to obtain an open ideal $\langle \alpha_0, \cdots, \alpha_r \rangle_{A^\triangleleft}$.

Let $y \in Y$ and $x = \beta(y)$, then

$$y(b(\alpha_i)) = x(\alpha_i).$$

As we can take the extending elements small enough, we conclude,

$$y \in \Theta \iff x \in \mathcal{R}_X(\alpha_0 | \alpha_1, \cdots, \alpha_r).$$

Hence,

$$\beta(\Theta) = \Omega \cap \mathcal{R}_X(\alpha_0 | \alpha_1, \cdots, \alpha_r)$$

which is open. Hence we have shown β is a continuous open bijection hence a homeomorphism. □

Theorem 3.4.3. In the setting of Corollary 3.4.5, let $\Theta \in \mathfrak{Rat}_Y, C = (\mathscr{O}_Y(\Theta), \mathscr{O}_Y^+(\Theta))$. Set $\Theta_X := \beta(\Theta)$ and take $c \in \mathcal{F}_\Theta(C)$ the universal element. The composition $A \xrightarrow{b} B \xrightarrow{c} C$ is an element of $\mathcal{F}_{\Theta_X}$ which is a universal element induces an isomorphism,

$$(\mathscr{O}_X(\Theta_X), \mathscr{O}_X^+(\Theta_X)) \to (\mathscr{O}_Y(\Theta), \mathscr{O}_Y^+(\Theta)) = C.$$

Proof. By Theorem 3.4.2, we have

$$\Theta_X = \beta(\Theta) = \Omega \cap \mathcal{R}_X(\varphi_0 | \varphi_1, \cdots, \varphi_m) = \mathcal{R}_X(f_0\varphi_0 | \text{ all other } f_i\varphi_j)$$

The topological claim then follows. If $\Theta_X = \beta(\Theta)$, one checks that

$$\mathcal{F}_\Theta(C) \ni (B \xrightarrow{c} C) \to (A \xrightarrow{b} B \xrightarrow{c} C) \in \mathcal{F}_{\Theta_X}(C),$$

is bijective using the definition and the bijectivity of β. Thus the composition of structure presheaves also holds. □

Corollary 3.4.6.

$$\mathrm{Spa}(A) \cong \mathrm{Spa}(\widehat{A})$$

Proof. Note that $\mathrm{Spa}(A) = \mathcal{R}_{\mathrm{Spa}(A)}(1|)$. Hence the corresponding ring B is just the completion of A. □

3.5 A Version of The Weierstraß Preparation Theorem

Definition 3.5.1. Let A be a complete Tate ring, we call $P \in A\langle T\rangle$, (A,T)-distinguished of degree d if $P_d \in A^\times$ and $\frac{P_i}{P_d}$ belongs to A° if $0 \leq i \leq d$, if othewise $i > d$, then it belongs to $A^{\circ\circ}$.

Lemma 3.5.1. Let R be a (discrete topology) ring, $I \subseteq R$ be a nilpotent ideal, M be a R-module, $P = \sum_{i=0}^N P_i T^i \in R[T]$, and $d \in \mathbb{N}$ such tha $P_d \in R^\times$ and $P_j \in I$ for any $j > d$. Then

$$M^d \ni (m_0, \cdots, m_{d-1}) \mapsto \left(\sum_{i=0}^{d-1} m_i T^i\right) \mod PM[T] \in M[T]/P \cdot M[T].$$

is bijective.

Proof. The proof proceeds induction on k such that $I^k = 0$. If this holds with $k = 0$, then $I = 0$¡ hence $P_i = 0$ for $i > d$ and since P_d is a unit in R, the assertion is a straitforward polynomial division with remainders.

Let $I^k = 0$ with $k >$ and suppose the assertion is shown for R' and I' with $I^{k-1} = 0$. In particular, it holds for $R' = R/I^{k-1}$, $I' = I/I^{k-1}$ and the image of P in R'. Let $M' = IM, M'' = M/M'$. These are R'-modules and the claim holds for them. Now consider the diagram,

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M'^d & \longrightarrow & M^d & \longrightarrow & M''^d & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M'^d[T]/P \cdot M'^d & \longrightarrow & M^d[T]/P \cdot M^d & \longrightarrow & M''^d[T]/P \cdot M''^d & \longrightarrow & 0 \end{array}$$

and the claim for M follows from the five lemma. □

Lemma 3.5.2. Let A be complete with respect to the i-adic topology, where $I \subseteq A$ is a finitely generated ideal (hence $I \subseteq A^{\circ\circ}$). Let $P = \sum_{k=0}^\infty P_k T^k \in A\langle T\rangle$ and $d \in \mathbb{N}$ be such that $P_d \in A^\times$ and $P_k \in A^{\circ\circ}$ for $k > d$. Then $P \cdot A\langle T\rangle$ is closed in $A\langle T\rangle$.

Proof. By assumption, $A \xrightarrow{\simeq} \varprojlim A/I^j$ and $A\langle T\rangle \xrightarrow{\simeq} \varprojlim (A/I^j)[T]$. By Lemma 3.5.1,

$$\begin{array}{ccc} A^d \oplus A\langle T\rangle & \xrightarrow{((a_i)_{i=0,\cdots,d-n}, f) \mapsto f \cdot P + \sum_{i=0}^{d-1} a_i T^i} & A\langle T\rangle \\ \simeq \downarrow & & \downarrow \simeq \\ (\varprojlim (A/I^j))^d \oplus \varprojlim (A/I^j)\langle T\rangle & \xrightarrow[((a_i)_{i=0,\cdots,d-n}, f) \mapsto f \cdot P + \sum_{i=0}^{d-1} a_i T^i]{\text{bijective by Lemma 3.5.1}} & \varprojlim (A/I^j)\langle T\rangle \end{array}$$

where the vertial arrows are homeormophisms, and so is the bottom horizontal arrow by Lemma 3.5.1 from which the claim follows as in the proof of the proposition. □

Proposition 3.5.1 (Weierstraß Division)**.** Let A be a complete Tate ring and $P \in A\langle T\rangle$ be (A,T)-distinguished of degree d. Then $I = P \cdot A\langle T\rangle$ is closed and $A\langle T\rangle/I$ is a free A-module with generators $(1, \cdots, T^{d-1})$. Moreover,

$$A^d \ni (a_0, \cdots, a_{d-1}) \mapsto \sum_{i=0}^{d-1} a_i T^i \mod I$$

is a homermorphism when A/I has the quotient topology.

Proposition 3.5.2. Without loss of generality, we assume $P_d = 1$. Then $P_i \in A^\circ$ for all i, $\lim_{i\to\infty} P_i = 0$, and $P_i \in A^{\circ\circ}$ for $i > d$. There is an open bounded subring $A^\#$ and $N \in \mathbb{N}$ such that $P_i \in A^\#$ for $i > N$. There is an open bounded subrig $A^b \supseteq A^\#$ such that $P_i \in A^b$ for $0 \le i < N$. Then all $P_i \in A^b$. If there is $s \in (A^b)^\circ \cap A^\times$, then Lemma 3.5.2 can be applied to A^b and the ideal $s \cdot A^b$.

If

$$\sum_{i=0}^{d-1} a_i T^i = P \cdot f \tag{$\dagger$}$$

where $f = \sum_{k=0}^{\infty} f_k T^k$, we would like for all $a_i = 0$, where all but finitely many $f_i \in A^b$ as $\lim_{i\to\infty} f_i = 0$. Therefore by multiplying with an appropriate power of s all $f_i \in A^b$ and all $a_i \in A^b$. Then ($\dagger$) holds in A^b, contradicting to Lemma 3.5.2.

Let $g \in A\langle T\rangle$ which will like to express in $g = \sum_{j=0}^{d-1} a_i T^i + P \cdot f$ with $f \in A\langle T\rangle$. It is possible to replace g by $s^i g$ (afterwards $f* = s^{-j} f_i, a_j* = s^j a_i$). Since $\lim_{i\to\infty} g_i = 0$, all but finitely many $g_i \in A^b$ clearly. Choosign i, (in s^i) large enough, all $s^i g j \in A^b$, adn the same application of Lemma 3.5.2 shows our claim. This is

$$A^d \oplus A\langle T\rangle \ni (a_0, \cdots, a_{d-1}, f) \mapsto \left(\sum_{i=0}^{d-1} a_i T^i\right) + P \cdot f \in A\langle T\rangle,$$
$$s^j (A^b)^d \oplus sj A^b\langle T\rangle \overset{\simeq}{\to} s^j A^b\langle T\rangle$$

is topological and these form neighborhood basis of 0 on both sides. Sice A^d is closed in the left hand side and mapped onto $P \cdot A\langle T\rangle, P \cdot A\langle T\rangle$ is closed in the right hand side.

Corollary 3.5.1 (Weierstrass Preparation Lemma)**.** Let A be a complete Tate ring.

1. Eery (A,T)-distinguished $f \in A\langle T\rangle$ is a multiplicatively equivalent to a unique irreducible $P \in A^\circ[T]$ which is called a Weierstrass poynomial, where the polynomial degree of P equals to the degree of distinguished one of f.

2. If $P \in A^\circ[T]$ is a Weierstrass polynomialand $f \in A[T]$ then P divides f i n$A[T]$ if and only if P divides f in $A^\langle T\rangle$.

Proof. **1.** Let f be (A,T)-distinguished of degree d. After dividing by f_d, without loss of generality that all $f_i \in A^\circ$ and $f_d = 1$. Choosing a ring of definition $A^\#$ containing all coefficients of f (as in the proof of Proposition 3.5.1),

$$T^d = -\sum_{i=0}^{d-1} P_i T^i + q(T)f(T).$$

Hence $P(T) = T^d + \sum_{i=0}^{d-1} P_i T^i \in A^\circ[T]$ is a Weierstrass polynomial ofdegree d divisible by f in $A\langle T\rangle$. Hence there is a morphism,

$$A\langle T\rangle/P \cdot A\langle T\rangle \to A\langle T\rangle/f \cdot A\langle T\rangle$$

sending $g \mod P$ to $g \mod f$. Thisis an isomorphism since using Proposition 3.5.1, both modules are free with generatorss $(T^i)_{i=0,\cdots,d-1}$. Then, f is also divisible by P, hence P and f are multiplicatively equivalent. If f is multiplicatively equivalent to the Weierstrass polynomial Q of the degree e, then $e = d$ as $A\langle T\rangle/Q = A\langle T\rangle/f$ as A-modules and these have ranks d and e, respectively. In **1**, without loss of generality, $A^\#$ contains all coefficient of Q. Then $P = Q$ by the unique remark made after **1**, f being divisible by Q.

2. Trivial as we have unique division by P with remainders of degree less than d in $A\langle T\rangle$ and $A[T]$, by Proposition 3.5.1, and elementary algebra. □

3.6 Topological Noetheriannes

Definition 3.6.1. We say that $(m_i)_{i=1}^k$ generates the A-module M topologically if $\vec{m} : A^k \to M$ is open and surjective. We say that M is topologically finitely generated if such k and $(m_i)_{i=1,\cdots,k} \in M^k$ can be taken.

Proposition 3.6.1. We have the following fact about Definition 3.6.1.

1. Every topological finitely generated M is also finitely generated in the algebraic sense. However, the converse is not true in general.

2. In the notation of Definition 3.6.1, if $l > k$, and $(m_i)_{i=k+1,\cdots,l} \in M^{l-k}$ are given then $(m_i)_{i=1,\cdots,l}$ also generate M topologically.

3. If M is topologically finitely generated then the topology of M is the finite nat-A-module topology. In particular, it does not depend the chosen generator.

Proof. **3.** Indeed, if $U \subseteq M$ is a neighborhood of 0 in M for some nat-A-module topology, then by the continuity of the module operation, there is a neighborhood V of 0 in A such that $\vec{m}(V^k) \subseteq U$. By the assumption $\vec{m}(V^k)$ is also open hence U is a neighborhood of 0 in M in the original topology. □

Definition 3.6.2. Let M be a nat-A-module, we say that M is topologically Noetherian if every submodule $N \subseteq M$ is topologically finitely generated when equipped with the induced topology.

Proposition 3.6.2. We have following facts about Definition 3.6.2.

1. Every topologically Noetherian A-module is Noetherian.

2. Every submodule of a topologically Noetherian module is topologically Noetherian when equipped with the induced topology.

Proposition 3.6.3. Let M be a nat-A-module ad $M' \subseteq M$ a submodule equipped with the induced topology. We equip $M'' = M/M'$ the quotient topology. If M', M'' are both topologically Noetherian/topologically finitely generated, then so is M. If $N \subseteq M$ is a submodule such that $N \cap M'$ is topologically finitely generated for the topology induced from M and $N'' = N/N'$ is topologically finitely generated for the topology induced from M'', then N is also topologically finitely generated.

Proof. For the finitely generated case, let $N = M, N' = M', N'' = M''$ other wise let $N \subseteq M$ be any submodule and $N' = M' \cap N$, both equipped with the topology induced from M, and $N'' = N/N' \subseteq M''$ equipeed with the topology induced from M''. In each case, N' and N'' are topologically finitely generated. By the assertion there are topological generators $(\eta_i \mod N')_{i=1}^k$ of N'' and $(\nu_i)_{i=0}^l$ of N'. THen $(\eta_1, \cdots, \eta_k, \nu_1, \cdots, \nu_l)$ generates N and we would like to show that they do as topologically.

We demand the open surjective maps $\vec{\eta} : A^k \to N'', \vec{\nu} : A^l \to N'$. Let $U \subseteq A$ be an open subgroup. We must find an open subgroup $V \subseteq M$ such that $\vec{m}(U^{k+l}) \supseteq V$ where $\vec{m}$ corresonds to $(\eta_1, \cdots, \eta_k, \nu_1, \cdots, \nu_l)$. By assumption about $\nu_1, \cdots, \nu_l$, there is an open subgroup $v_1 \subseteq U$ such that $\vec{\nu}(U^l) \supseteq V_1$. Similarly there is $U_1 \subseteq U$ such that $\vec{\eta}(U^k) \subseteq V_1 \cap N'$. By the continuity of the module operation there is an open subgroup $U_1 \subseteq U$ such that $\vec{\eta}(U_1^k) \subseteq V_1$. by the assumption on $\vec{\eta}$, there is an open subgroup $W \subseteq M''$ such that $\vec{\eta}(U_1^k) \supseteq W \cap N''$. Then $V = \pi^{-1}(W) \cap V_1$ will do where $M \xrightarrow{\pi} M''$ denotes the projection. Indeed, if $n \in N \cap V$ then $\pi(n) \in W$, hence there are $(a_i)_{i=1}^k \in U_1^k \subseteq U^k$ such that

$$n - \sum_{i=1}^{k} a_i \eta_i \in V_1 \cap N$$

(it is in N' and subtrahend is in V_1 by selection of V_1). Thus there are $(\alpha)_{i=1}^l \in U^l$ such that

$$n = \vec{\eta}(\vec{a}) + \vec{\nu}(\vec{\alpha})$$

as claimed for V. The latter statement is shown in the proof. □

Proposition 3.6.4. If A is complete and has a countable neighborhood basis of 0 and M is a complete nat-A-module with a countable neighborhood base of 0, then every topologically finitely generated (with induced topology) submodule $N \subseteq M$ is closed in M.

Proof. Let $U_1 \supseteq U_2 \supseteq \cdots$ be a sequence of open subgruops of A which forms a neighborhood basis of 0 in A. Let $m = \lim_{i \to \infty} n_i$ in M and all $n_i \in N$, let

$\vec{\nu} = (\nu_i)_{i=1,\cdots,k}$ be topological generator of N hence $\vec{\nu} : A^k \to N$ is open and surjective. Without loss of generality (taking subsequence), $n_{i+1} - n_i$ is in the image of U_i^l under the above map. Let $\alpha_1^i, \cdots, \alpha_l^i$ be such that

$$n_{i+1} - n_i = \vec{\nu}(\vec{\alpha^i}).$$

let $n_0 = \vec{\nu}(\vec{\alpha^0})$ and set

$$\alpha_j = \sum_{i=0}^{\infty} \alpha_j^i.$$

THese converge in A as they are Cauchy and $\vec{\nu}(\vec{\alpha}) = m$ hence $m \in N$. □

Proposition 3.6.5. A and M be complete with countable neighborhood bases of 0, and $N \subseteq M$ be a topologically finitely generated with topology induced from M, then N is closed in M.

Proposition 3.6.6. Let A be a complete, topologically Noetherian ring with a countable neighborhood basis of 0. The following statements hold.

1. All A^n equipped with the product topologies are topologically Noetherian.
2. Every finitely generated A-module M has a unique nat-A-module topology from which it is topologically Noetherian. This is the final nat-A-topology on M. (follows from Fact 1). If $M = A^n$ this coincides with the product topology.
3. Let M, N be finitely generated A-module with the topology in **2**. Then every surjective morphism $M \xrightarrow{f} N$ is continuous and open.
4. Every topologically Noetherian A-module is Hausdorff and every submodule is closed.

Proof.

1. Induction on n using Proposition 3.6.3.

2. Let M be finitely generated. Choose $\vec{m} = (m_i)_{i=1}^n \in M^n$ such that $A^n \xrightarrow{\vec{m}} M$ is surjective and M is equipped with the quotient topology. Then M is a topologically Noetherian (need to check?). Automatically the continuity is clear as any $M \to N$ can be lifted as

$$\begin{array}{ccc} A^k & \longrightarrow & A^l \\ {\scriptstyle \vec{m}}\downarrow & & \downarrow{\scriptstyle \vec{n}} \\ M & \xrightarrow[f]{} & N \end{array}$$

The upper horizontal arrow is continuous. If the map is surjective, the nthe set of generators of M consisting of $\vec{m}$ can be chosen such that it induces preimage under f of the generator and for N, and $A^k \to A^l$ to be surjective, hence split surjective hence open.

4. Since M is topologically Noetherian it is topologically finitely generated and carries the quotient topology from $A^k \xrightarrow{\vec{m}} M$ for some set of k generators of M. Let $N \subseteq M$ be a submoule. Since A^k is topologically Noetherian by **1**, $\vec{m}^{-1}(M) \subseteq A^k$ is topologically finitely generated for the induced topology hence closed in A^k by Proposition 3.6.5. As M carries the quotient topoloy, N is closed. If $N = \{0\}$, this shows that M is Hausdorff. □

3.7 Ring Norms

Definition 3.7.1. Let R be a ring. A map $\|\cdot\| : R \to \mathbb{R}$ is calle a ring norm if it satisfies the following conditions.

i. $\|x + y\| \leq \max\{\|x\|, \|y\|\}$.

ii. $\|1\| = 1$.

iii. $\|ab\| \leq \|a\|\|b\|$.

iv. $\|x\| = 0 \iff x = 0$.

Remark 3.7.1. A ring norm induces a metric on R and hence a topology on R as well.

$$\{x \in R \mid \|x\| \leq 1\} \subseteq R^\circ,$$
$$\{x \in R \mid \|x\| < 1\} \subseteq R^{\circ\circ}.$$

Definition 3.7.2. A ring norm is power-multiplicative if for any $x \in R$ and $r \in \mathbb{N}$, we have $\|x^r\| = \|x\|^r$.

Remark 3.7.2. If R is power-multiplicative, we have

$$\{x \in R \mid \|x\| < 1\} = R^{\circ\circ}.$$

Definition 3.7.3. A ring norm is multiplicative if the third condition in Definition 3.7.1 is an equality.

Remark 3.7.3. If $\|\cdot\| : R \to \mathbb{R}$ is multiplicative, then any subset X of R is bounded if $\|X\|$ is bounded with respect to the Euclidean norm on $\mathbb{R}$. Furthermore, we have

$$\{x \in R \mid \|x\| \leq 1\} = R^\circ.$$

Proposition 3.7.1.

1. If M is a nat-A-module, $M' \subseteq N \subseteq M$ submodules with the induced topology, and M/M' is equipped with the quotient topology, then the topology on N/M' is induced from M/M' conincides with the quotient topology.

2. The quotient of a topologically Noetherian module by a submodule is topologically Noetherian when equipped with the quotient topology.

Proof. **1.** It is clear that any open set for the induced topology is open for the quotient topology. Indeed, let $M \xrightarrow{\pi} M/M'$ denotes the projection. If $\Omega \subseteq M/M'$ is open then $\pi^{-1}(\Omega \cap (N/M')) = N \cap \pi^{-1}(\Omega)$ is open in N. Let $\Theta \subseteq N/M'$ be such that $\pi^{-1}(\Theta)$ is an open subgroup of N. Then there is an open subgropu $V \subseteq M$ such that $\pi^{-1}(\Theta) \supseteq N \cap V$. Without loss of generality, we assume $V \supseteq M'$, otherwise $V \leftarrow V + M'$, as $\pi^{-1}(\Omega)$ is invariant under translation by elements of M' and $M' \subseteq N$. Let $\Omega = \pi(V)$, then $\Omega \cap (N/M') \subseteq \Theta$. If $\pi(n) \in \Theta$, then $n \in \Theta + M' = \Theta$ hence $n \in N \cap V \subseteq \pi^{-1}(\Omega)$.

2. Follows from **1**. If $N'' \subseteq M/M''$ then $\pi^{-1}(N'')$ is topologically finitely generated with the induced topology from M. Hence the same holds for its quotient by M'' which by **1**, coincides with the topology induced from M/M'. □

End of proof of Proposition 2

Since A^N is topologically Noetherian by Proposition 2.1, so is any finitely generated A-module with the quotient topology from A^N which is the topologically finitely generated, hence the finite nat-topology by Fact 1, hence is independent of the choice of generators.

Remark 3.7.4. Without $M' \subseteq N$, Fact 4.1 will be false.

Let V be a $\mathbb{Q}_p$-Banach space and $M = V \oplus V$, $M' = V \oplus \{0\}$, $N \subseteq V \oplus V$ the graph of a bounded linear on-surjective operator with a dense image by $V = l_\infty$, $L(X_i)_{i=0}^\infty = (p^i X_i)_{i=0}^\infty$, $N =$ (Graph of L).

4 Acyclicity proofs

The Acyclicity proofs usually consider a special kinds of Čech cohomology to show that $\mathcal{O}_X$ extends from $\mathfrak{Rat}_X$ to $\mathscr{O}_X$. Acyclicity will typically follow using the following statements.

Proposition (Grothendieck's acyclicity lemma)**.** Let $\mathscr{A} \xrightarrow{F} \mathscr{B}$ be a left exact functor between abelian categories where $\mathscr{A}, \mathscr{B}$ contain sufficiently many injective objects. Let $\mathscr{X}$ be a class of objects of $\mathscr{A}$ such that

a. Every direct summand of (in particular, every object isomorphic to) an element of $\mathscr{X}$ is in $\mathscr{X}$.

b. Every objet A of $\mathscr{A}$ has a monomorphism $A \to X$ with $X \in \mathscr{X}$.

c. If
$$0 \to X' \to X \to X'' \to 0,$$
is a short exact sequence in $\mathscr{A}$ such that $X', X \in \mathscr{X}$, then $X' \in \mathscr{X}$ and
$$0 \to FX' \to FX \to FX'' \to 0.$$

Proof. See Grothendieck Tohoku Paper Lemma 3.3.1. □

Corollary. All injective objects of $\mathscr{A}$ belongs to $\mathscr{X}$ and furthermore $X \in \mathscr{X}$ and $p > 0$, we have $R^p FX = 0$.

4.1 Acyclicity Criteria for General Sheaves on $\operatorname{Spa} A$

In this subsection, we denote $X = \operatorname{Spa} A$ where A is a Huber pair.

Proposition 4.1.1. Let $x \in X$ be such that $c\Gamma_x = \Gamma_x$ and U be an open subset of X such that $x \in U$. Then there is $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in (A^\triangleleft)^{m+1}$ such that

$$\langle f_0, \cdots, f_m \rangle_{A^\triangleleft} = A^\triangleleft$$

and

$$x \in \mathcal{R}_X(f_0 | f_1, \cdots, f_m) \subseteq U.$$

In particular, Proposition 3.2.16, this holds when x is closed.

Proof. Since rational open subsets form a topology base, there are $(\varphi_i)_{i=0}^m$ such that

$$x \in \mathcal{R}_X(\varphi_0 | \varphi_1, \cdots, \varphi_m) \subseteq U.$$

If $a \in A$ is such that $|a|_x \neq 0$, then this implies

$$x \in \mathcal{R}(a\varphi_0 | a\varphi_1, \cdots, a\varphi_m) \subseteq \mathcal{R}(\varphi_0 | \varphi_1, \cdots, \varphi_m) \subseteq U.$$

As $|\varphi_0|_x \in \Gamma_x = c\Gamma_x$, such a can be chosen so that

$$|\varphi_0 a|_x \geq 1.$$

Then

$$x \in \mathcal{R}(a\varphi_0 | a\varphi_1, \cdots, a\varphi_m, 1) \subseteq \mathcal{R}(a\varphi_0 | a\varphi_1, \cdots, a\varphi_m) \subseteq U.$$

Obviously,

$$\langle a\varphi_0, \cdots, a\varphi_m, 1 \rangle_{A^\triangleleft} = A^\triangleleft.$$

□

Remark 4.1.1.

1. A being Tate or analytic is also sufficient to the above assertion for arbitrary x.
2. Thus $\mathcal{O}_X(\mathcal{R}(f_0 | f_1, \cdots, f_m))$ can be bounded by the ? Then 3b) (where ? f_0 is not necessary).

Definition 4.1.1.

a. A rational covering of X is a covering $X = \bigcup_{i=1}^m \mathcal{R}_X(f_i | f_1, \cdots, f_m)$ such that

$$\langle f_1, \cdots, f_m \rangle_{A^\triangleleft} = A^\triangleleft.$$

b. The rational sieve defined by such $(f_i)_{i=1}^m$ is

$$[\mathcal{R}(f_i | f_1, \cdots, f_m) \mid i = 1, \cdots, m].$$

Proposition 4.1.2.

1. Every open covering $\mathcal{U}$ of X has a refinement which is a rational covering.

2. Every covering sieve (i.e. $X = \bigcup_{U \in \mathscr{S}} U$) of X contains a rational sieve.

Proof. Let $\mathscr{S}$ be a covering sieve of X. By Proposition 4.1.1, for $x \in X$, we have $U_x \in \mathscr{S}$ such that $x \in U_x$ and $(f_{x,i})_{i=0,\cdots,m_x}$ with

$$\langle f_{x,0}, \cdots, f_{x,m_x} \rangle_{A^\triangleleft} = A^\triangleleft$$

such that $x \in \mathcal{R}(f_{x,0} | f_{x,1}, \cdots, f_{x,m_x}) \subseteq U_x$. Then

$$V = \bigcup_{x \in X_{\mathrm{cl}}} \mathcal{R}_X(f_{x,0} | f_{x,1}, \cdots, f_{x,m_x}).$$

Looking at the right hand side V contains all closed points. If $V \subsetneqq X$, then $X \backslash V$ would be a non-empty closed subset without closed points, a contradiction by Proposition 1.1.7.

As X is spectral, it is quasi-compact, hence there are finitely many closed points $(x_i)_{i=1}^n$ such that

$$X = \bigcup_{i=1}^n \mathcal{R}(f_{x_i,0} | f_{x_i,1}, \cdots, f_{x_i,m_i}).$$

We denote $f_{i_j} = f_{x_i,j}, n_i = m_{x_i}, U_i = \mathcal{R}(f_{i,0} | f_{i,1}, \cdots, f_{i,m_i})$. Let J be the set of functions defined on $\{1, \cdots, n\}$ such that

$$\forall \iota \in J, 0 \leq \iota(j) \leq m_j,$$

and let $J_0 \subseteq J$ be the subset set of all $\iota \in J$ such that there is $j \in \{1, \cdots, n\}$ with $\iota(j) = 0$. For all $\iota \in J$, let

$$f_\iota = \prod_{j=1}^n f_{j,\iota(j)}.$$

Clearly,

$$\langle f_\iota \mid \iota \in J \rangle_{A^\triangleleft} = \prod_{j=1}^n \langle f_{j,0}, \cdots, f_{j,m} \rangle_{A^\triangleleft} = A^\triangleleft$$

by the assumption. Note that

i. $\langle f_{j,0}, \cdots, f_{j,m_j} \rangle_{A^\triangleleft} = A^\triangleleft$.

ii. Since for every $x \in X$, there is j such that $x \in U_j$, we can improve any $f_{j,\iota(j)}$ by replacing it to $f_{j,0}$ hence $\max_{\iota \in J} |f_\iota|_x = \max_{\iota \in J_0} |f_\iota|_x$.

iii. Using Theorem 3.3.1 twice, to $\langle f_\iota \mid \iota \in J \rangle$ and $\langle f_\iota \mid \iota \in J_0 \rangle_{A^\triangleleft}$. We see that the latter also generates $A^\triangleleft$.

Then the $(f_\iota)_{\iota\in J_0}$ defines a rational covering. We will show that the rational sieve generated by this is contained in $\mathscr{S}$. To prove this claim, it suffices to consider,

$$\Omega_\lambda = \mathcal{R}(f_\lambda| \text{ all } f_\iota \text{ with } \iota \in J_0).$$

with $\lambda \in J_0$ and to show that $\Omega_\lambda \in \mathscr{S}$, it suffices to find $j \in \{1, \cdots, n\}$ such that

$$\Omega_\lambda \subseteq U_j.$$

We claim that this holds when $\lambda(j) = 0$. Indeed, let $x \in \Omega_\lambda$ then

$$|f_\lambda|_x = \prod_{k=1}^{n} |f_{k,\lambda(k)}| \neq 0.$$

Hence $0 \neq |f_{j,\lambda(j)}|_x = |f_{j,0}|_x$. If $x \notin U_j$, there would be $i \in \{1, \cdots, m_j\}$ such that

$$|f_{j,i}|_x > |f_{j,0}|.$$

Let

$$\lambda'(k) = \begin{cases} \lambda(k), & k \neq j, \\ i, & k = j. \end{cases}$$

Then

$$|f_{\lambda'}(x)| = |f_{j,i}(x)| \prod_{k\neq j} |f_{k,\lambda(k)}|_x > |f_{j,0}|_x \prod_{k\neq j} |f_{k,\lambda(k)}|_x = |f_\lambda(x)|$$

Hence there is $\theta \in J_0$ such that

$$|f_\theta(x)| \geq |f_{\lambda'}|_x > |f_\lambda(x)|,$$

contradicting $x \in \mathcal{R}(f_\lambda| \text{ all } f_\iota \text{ for } \iota \in J_0) \subseteq \mathcal{R}(f_\lambda|f_\theta)$. □

Proposition 4.1.3. Let A be a Huber pair and $X = \operatorname{Spa} A$ and $\mathcal{G}$ be a presheaf of abelian groups on $\mathfrak{Rat}_X$ such that for every $\Omega \in \mathfrak{Rat}_X$ and every rational covering $\mathcal{U}$ of Ω, the map,

$$\mathcal{G}(\Omega) \to \check{H}^0(\mathcal{U}, \mathcal{G})$$

is bijective. Then

$$\mathcal{G} \to (\underline{\text{Sheaf}}(\mathcal{G}))_{\mathfrak{Rat}_X}$$

is an isomorphism in the category of presheaves on $\mathfrak{Rat}_X$.

Proposition 4.1.4. Let $X = \operatorname{Spa} A$ where A is a Huber pair. Set $\mathscr{X}$ to be a class of sheaves of abelian groups such that

$$\mathscr{X} = \left\{ \mathcal{F} \in \operatorname{Sh}_{\mathrm{Ab}}(X) \;\middle|\; \substack{\forall p>0, \forall \Omega \in \mathfrak{Rat}_X, \\ \text{and } \mathcal{U} \text{ rational covering of } \Omega \check{H}^p(\mathcal{U},\mathcal{F})=0} \right\},$$

together with a left exact functor

$$\mathcal{F} \mapsto \mathcal{F}(\Omega)$$

where $\Omega \in \mathfrak{Rat}_X$ satisfies the assumption of Grothendieck's acyclicity lemma. Hence due to corollary of it, all injective sheaves are in $\mathscr{X}$. In particular, $\Omega \in \mathfrak{Rat}_X, \mathcal{F} \in \mathscr{X}$ and $p > 0$ then $H^p(\Omega, \mathcal{F}) = 0$.

Under the assumption of A being Tate, we have more convenient acyclicity criterion.

Definition 4.1.2. Let $X = \operatorname{Spa} A$ where $(A^\triangleleft, A^+)$ is a Huber pair. We inductively define the property of a sieve $\mathscr{S}$ in $\Omega \in \mathfrak{Rat}_X$ to have a Laurent order at most o as follows.

1. A sieve has Laurent order 0 if it is the all sieve.
2. If $o > 0$ and Laurent order $\leq o - 1$ has already been defined then $\mathscr{S}$ has Laurent order $\leq o$ if there is $g \in \mathcal{O}_X(\Omega)$ such that $\mathscr{S}|_{\mathcal{R}_\Omega(g|1)}$ and $\mathscr{S}|_{\mathcal{R}_\Omega(1|g)}$ both have Laurent order $\leq o - 1$.

The Laurent order of $\mathscr{S}$ is the smallest $o \in N$ such that $\mathscr{S}$ has Laurent order $\leq o$ if no such o exists then its Laurent order is ∞. We denote $o_L(\mathscr{S})$ to be the Laurent order of $\mathscr{S}$.

Proposition 4.1.5. Let $\mathscr{S}$ be a sieve on Ω. Then the following statements hold.

a. $\mathscr{S} \subseteq \mathcal{T} \Rightarrow o_L(\mathcal{T}) \leq o_L(\mathscr{S})$.

b. $\Theta \subseteq \Omega \Rightarrow o_{L,\Theta}(\mathscr{S}|_\Theta) \leq o_{L,\Omega}(\mathscr{S})$.

c. $o_L(\mathscr{S}) < \infty \Rightarrow \mathscr{S}$ is a covering sieve.

Lemma 4.1.1. Let $G = \{g_i \mid 1 \leq i \leq n\}$ be a non-empty finite subset of $\mathcal{O}_X(\Omega)^\times$. Then the rational sieve defined by G has finite Laurent order.

Proof. We write $g_i \preceq g_j$ if $\Omega = \mathcal{R}(g_j|g_i)$. Obviously, this is a partial order with allowed equivalence. We call (i, j) in order if at least one of $g_i \preceq g_j$ or $g_j \preceq g_i$ holds. We prove the claim by induction on the number of pairs not in order.

If all pairs are in order then without loss of generality by renumbering G, we can assume $g_1 \preceq \cdots \preceq g_m$. Thus, all $g_i \preceq g_m$. In this case, observe that

$$\Omega = \mathcal{R}_\Omega(g_m|g_0, \cdots, g_{m-1}).$$

Hence the rational sieve generated by the rational cover of G contains Ω hence is the all sieve. Therefore by definition, it has Laurent order 0.

Suppose (i, j) is not in order and the assertion shown for a smaller number of pairs not in order. Let $q = \frac{g_i}{g_j}$ and $\Theta_1 = \mathcal{R}_\Omega(q|1)$ and $\Theta_2 = \mathcal{R}_\Omega(1|q)$. Then the pairs in order stay in order after restricting to Θ_i as they are subsets of Ω. Furthermore, $g_j|_{\Theta_1} \preceq g_i|_{\Theta_1}$ and $g_i|_{\Theta_2} \preceq g_j|_{\Theta 2}$. Thus (i, j) becomes in order for $G_k = \{g_i|_{\Theta_k} \mid 1 \leq i \leq n\}$ where $k = 1, 2$. By the induction assumption and Definition 4.1.2, the rational sieve defined by G has finite Laurent order. □

Lemma 4.1.2. Let $A = (A^{\triangleleft}, A^{+})$ be a Huber pair such that $A^{\triangleleft}$ is Tate and there is $(f_i)_{i=0}^{m} \in A^{m+1}$ with $\langle f_0, \cdots, f_m \rangle_{A^{\triangleleft}} = A^{\triangleleft}$. Then we can take $\varepsilon \in (A^{\triangleleft})^{\times} \cap (A^{\triangleleft})^{\circ\circ}$ satisfying

$$\forall x \in X, |\varepsilon|_x \leq \max_{0 \leq i \leq n} |f_i|_x.$$

Proof. Let $s \in (A^{\triangleleft})^{\circ\circ} \cap (A^{\triangleleft})^{\times}$. By assumption, there are $(\lambda_i)_{i=0,\cdots,m} \in (A^{\triangleleft})^{m+1}$ giving

$$\sum_{i=0}^{m} \lambda_i f_i = 1.$$

Since A^{+} is an open subring and $\lim_{j \to \infty} s^j \lambda_i = 0$, there is $j \in \mathbb{N}$ such that for each $i = 0, \cdots, m$, $\lambda_i s^j \in A^{+}$. Then one can set $\varepsilon = s^j$. Indeed,

$$|\varepsilon|_x = \left| \sum_{i=0}^{m} (s^j \lambda_i) f_i \right|_x \leq \max_{0 \leq i \leq m} |\lambda_i s^j|_x |f_i|_x \leq \max_{0 \leq i \leq m} |f_i|_x.$$

□

Proposition 4.1.6. In the situation of Definition 4.1.2, assume A (i.e $A^{\triangleleft}$) to be Tate. Then a sieve $\mathscr{S}$ on $\Omega \in \mathfrak{Rat}_X$ covers Ω if and only if $o_L(\mathscr{S}) < \infty$.

Proof. The proof will work by showing that every rational sieve has finite o_L which proves the claim by Proposition 4.1.2 and 4.1.5 a.

It is sufficient to show that every rational sieve has finite Laurent order. Let $\mathscr{S}$ be the rational sieve defined by the finite subset G of $\mathcal{O}_X(\Omega)$. We use induction on the number of elements of G which are not in $\mathcal{O}_X^{\times}(\Omega)$. If this number is 0, Lemma 4.1.1 can be applied. Suppose $g_1 \notin \mathcal{O}_X^{\times}(\Omega)$ and the assertion shown for a smaller number of non-elements. Since A is Tate, there is $\delta \in \mathcal{O}_X^{\times}(\Omega)$ such that $|\delta|_\omega < \max_{1 \leq i \leq m} |g_i|_\omega$ holds for all $\omega \in \Omega$. Indeed, from the proof of Lemma 4.1.2 shows the strict inequality can be attained. Consider

$$\Theta_1 = \mathcal{R}_{\Omega}(g_1|\delta) = \mathcal{R}\left(\frac{g_1}{\delta}\middle|1\right), \Theta_2 = \mathcal{R}_{\Omega}(\delta|g_1) = \mathcal{R}\left(1\middle|\frac{g_1}{\delta}\right).$$

On Θ_1, g_1 becomes a unit as $|g_1|_\theta \geq |\delta|_\theta \neq 0$ when $\theta \in \Theta_1$ hence the induction assumption applies to $G|_{\Theta_1}$. On the other hand for $\theta_2 \in \Theta_2$,

$$|g_1|_{\theta_2} \leq |\delta|_{\theta_2} < \max_{1 \leq i \leq m} |g_i|_{\theta_2}.$$

Hence, we obtain,

$$\mathcal{R}_{\Theta_2}(g_1|g_2, \cdots, g_m) = \emptyset.$$

Furthermore, the inequality shows,

$$\forall 2 \leq i \leq m, \mathcal{R}_{\Theta_2}(g_i|g_1, \cdots, g_m) = \mathcal{R}_{\Theta_2}(g_i|g_2, \cdots, g_m).$$

Applying Theorem 3.3.1.d, and that $g_2|_{\Theta_2}, \cdots g_m|_{\Theta_2}$ have no common zero on Θ_2, these also define a rational covering, hence the induction hypothesis can be applied. □

Proposition 4.1.7. Let $X = \operatorname{Spa} A$ where A is Tate, $\mathcal{G}$ be a presheaf of abelian groups on $\mathfrak{Rat}_X$ such that for all $\Omega \in \mathfrak{Rat}_X$ and all $g \in \mathcal{O}_X(\Omega)$, the sequence,

$$0 \to \mathcal{G}(\Omega) \xrightarrow{r \mapsto (r|_{\Omega_1}, r|_{\Omega_2})} \mathcal{G}(\Omega_1) \oplus \mathcal{G}(\Omega_2) \xrightarrow{(r_1,r_2) \mapsto (r_1|_{\Omega_{12}} - r_2|_{\Omega_{12}})} \mathcal{G}(\Omega_{12}) \to 0.$$

is exact where $\Omega_1 = \mathcal{R}_\Omega(g|1), \Omega_2 = \mathcal{R}_\Omega(1|g)$ and $\Omega_{12} = \Omega_1 \cap \Omega_2$.

Then $\mathcal{G} \to (\underline{\text{Sheaf}}(\mathcal{G}))|_{\mathfrak{Rat}_X}$ is an isomorphism in the category of presheaves of abelian groups on X.

Proposition 4.1.8. Let $X = \operatorname{Spa} A$ where the Huber pair A is Tate. Let $\mathscr{X}$ be a subclass of $\operatorname{Sh}_{\mathrm{Ab}}(X)$ containing any $\mathcal{F} \in \operatorname{Sh}_{\mathrm{Ab}}(X)$ such that

$$0 \to \mathcal{F}(\Omega) \xrightarrow{f \mapsto (f|_{\Omega_1}, f|_{\Omega_2})} \mathcal{F}(\Omega_1) \oplus \mathcal{F}(\Omega_2) \xrightarrow{(f_1,f_2) \mapsto (f_1|_{\Omega_1 2} - f_2|_{\Omega_{12}})} \mathcal{F}(\Omega_{12}) \to 0$$

is exact for all $\Omega \in \mathfrak{Rat}_X$ and all $g \in \mathcal{O}_X(\Omega)$ where Ω_1, Ω_2 and Ω_{12} are as in Proposition 4.1.7. Then the assumption in Grothendieck's acyclicity lemma holds for $\mathscr{X}$ and $\mathscr{A} = \operatorname{Sh}_{\mathrm{Ab}}(X)$ with the functor $\mathcal{F} \mapsto \mathcal{F}(\Omega)$ for any $\Omega \in \mathfrak{Rat}_X$.

4.2 Acyclicity Criterion for Tate Maps

Definition 4.2.1. Let $A = (A^\lhd, A^+)$ be a Huber pair such that $A^\lhd$ is Tate. Set $X = \operatorname{Spa}(A)$ then for $\Omega \in \mathfrak{Rat}_X$ and $g \in \mathcal{O}_X(\Omega)$, $\Omega_1 = \mathcal{R}_\Omega(g|1), \Omega_2 = \mathcal{R}_\Omega(1|g)$ are called Laurent covering of Ω.

Remark 4.2.1.

$$\Omega_{12} = \Omega_1 \cap \Omega_2 = \mathcal{R}_\Omega(g|g^2, 1).$$

Note that $|g|_x^2 \leq |g|_x$ gives that for all $x \in \Omega_{12}, |g|_x \leq 1$.

We know that for $\Theta = \mathcal{R}_X(f_0|f_1, \cdots, f_m)$, the structure presheaf $\mathcal{O}_X(\Theta) = A^\lhd\langle X_1, \cdots, X_m\rangle/\overline{I}$ where

$$I = \langle f_i - f_0 X_i \mid i = 1, \cdots, m\rangle_{A^\lhd\langle X_1, \cdots, X_m\rangle}$$

provided that

$$\langle f_0, f_1, \cdots, f_m\rangle_{A^\lhd} = A^\lhd,$$

which is automatic as $A^\lhd$ is Tate and Θ is rational open.

Definition 4.2.2. Let A be a sequentially complete nat-ring. We denote

$$A\langle T, T^{-1}\rangle = \left\{ f = \sum_{k=-\infty}^{\infty} f_k T^k \;\middle|\; \forall k \in \mathbb{Z}, f_k \in A, \lim_{k \to \pm\infty} f_k = 0 \right\}.$$

The addition is defined coefficient wise and the multiplication is define by

$$fg = \sum_{k \in \mathbb{Z}} \left(\sum_{i+j=k} f_i g_j \right) T^k.$$

The neighborhood of 0 in this space is given by the subsets of. the form,

$$U\langle T, T^{-1}\rangle = \left\{ f = \sum_{k=-\infty}^{\infty} f_k T^k \;\middle|\; \forall k \in \mathbb{Z}, f_k \in U, \lim_{k\to\pm\infty} f_k = 0 \right\},$$

where U running over all open subgroups of A.

Proposition 4.2.1. Such $A\langle T, T^{-1}\rangle$ in Definition 4.2.2 has the following properties.

a. $A\langle T, T^{-1}\rangle$ is a complete nat ring.

b. If A is Tate then so is $A\langle T, T^{-1}\rangle$.

c. The ideal $I = (XY - 1)A\langle X, Y\rangle$ is closed in $A\langle X, Y\rangle$ and there is an isomorphism,
$$A\langle X, Y\rangle/(1 - XY)A\langle X, Y\rangle \overset{X\mapsto T, Y\mapsto T^{-1}}{\to} A\langle T, T^{-1}\rangle.$$

Proof. Exercise in Sheet 1 beside Tateness.

Let $s \in A^{\circ\circ} \cap A^\times$, then s is a topologically nilpotent unit of $A\langle X, Y\rangle$ and also of $A\langle T, T^{-1}\rangle$.

Let $A^\# \subseteq A$ be an open bounded subring. Then $A^\#\langle T, T^{-1}\rangle$ is an open subring. If $W \subseteq A\langle T, T^{-1}\rangle$ is a neighborhood of 0, there are open subgropus $U \subseteq A$ such that $U\langle T, T^{-1}\rangle \subseteq W$ and as $A^\#$ is bouned, an open subgroup $V \subseteq A$ such that $A^\# V \subseteq U$. In this case $\Omega = V\langle T, T^{-1}\rangle$ is a neighborhood of 0 in $A\langle T, T^{-1}\rangle$ and $\Omega A^\#\langle T, T^{-1}\rangle \subseteq U\langle T, T^{-1}\rangle \subseteq W$ sharing boundedness of $A^\#\langle T, T^{-1}\rangle$. □

The following observation does not assume Tateness yet.

Let $Z = \mathrm{Spa}(A^\lhd, A^+)$ with $A^\lhd$ complete and $\Omega_1 = \mathcal{R}_Z(g|1), \Omega_2 = \mathcal{R}_Z(1|g)$ be a Laurent covering for some $g \in A^\lhd$. Due to Theorem 3.4.1, we have

$$\begin{aligned}
\mathcal{O}_Z(\Omega_1) &\cong A^\lhd\langle T\rangle/\overline{(1 - gT)A^\lhd\langle T\rangle}, \\
\mathcal{O}_Z(\Omega_2) &\cong A^\lhd\langle T\rangle/\overline{(g - T)A^\lhd\langle T\rangle}, \\
\mathcal{O}_Z(\Omega_{12}) &\cong A^\lhd\langle X, Y\rangle/\overline{(1 - gX, g^2 - gY)A^\lhd\langle X, Y\rangle}.
\end{aligned}$$

In the last ring, g is a unit and $g^2 = gY$, hence $Y = g$ and $X = g^{-1}$. Hence showing $XY = 1$ and by Proposition 4.2.1.3, the presheaf ring is isomorphic to

$$\begin{aligned}
\mathcal{O}_Z(\Omega_{12}) &\cong A^\lhd\langle T, T^{-1}\rangle/\overline{(1 - gT)A^\lhd\langle T, T^{-1}\rangle}, \\
&\cong A^\lhd\langle T, T^{-1}\rangle/\overline{(T^{-1} - g)A^\lhd\langle T, T^{-1}\rangle}.
\end{aligned}$$

Definition 4.2.3. For a Huber pair $A = (A^\lhd, A^+)$, We put

$$\begin{aligned}
A^\lhd\langle g\rangle &:= A\langle T\rangle/\overline{(T - g)A^\lhd\langle T\rangle}, \\
A^\lhd\langle g^{-1}\rangle &:= A\langle T\rangle/\overline{(1 - gT)A^\lhd\langle T\rangle}, \\
A^\lhd\langle g, g^{-1}\rangle &:= A^\lhd\langle T, T^{-1}\rangle/\overline{(1 - gT)A^\lhd\langle T, T^{-1}\rangle}.
\end{aligned}$$

Definition 4.2.4. Let $B^\triangleleft$ be a complete Tate ring. A Huber pair $A = (A^\triangleleft, A^+)$ is called sheafy if the following holds for $X = \mathrm{Spa}(A)$.

i. The morphism $\mathcal{O}_X \to \underline{\mathrm{Sheaf}}(\mathcal{O}_X)|_{\mathfrak{Rat}_X}$ is an isomorphism.

ii. $H^p(X, \mathcal{O}_X) = 0$ for all $p > 0$.

Corollary 4.2.1. For a complete Tate ring $B^\triangleleft$. Suppose that for all $A^\triangleleft$ which are quotients of $B^\triangleleft\langle X_1, \cdots, X_n\rangle$ by the closure of a finitely generated ideal and for all $g \in A^\triangleleft$, the sequence

$$0 \to A^\triangleleft \to A^\triangleleft\langle g\rangle \oplus A^\triangleleft\langle g^{-1}\rangle \to A^\triangleleft\langle g, g^{-1}\rangle \to 0 \tag{4.1}$$

is exact. Then for all such $A^\triangleleft$, all Huber pairs whose first component is $A^\triangleleft$ are sheafy.

Proposition 4.2.2. Let A be a sequentially complete nat ring. Assume that all principal ideals in $A\langle T\rangle$ and $A\langle T, T^{-1}\rangle$ are closed. Then Equation (4.1) is exact for all $g \in A^\triangleleft$.

Proof. Since no Huber pair is being considered, we simply put $A = A^\triangleleft$. Set

$$\begin{aligned} I &:= \overline{(1 - gX)A\langle X\rangle} = (1 - gX)A\langle X\rangle, \\ J &:= \overline{(g - Y)A\langle Y\rangle} = (g - Y)A\langle Y\rangle, \\ K &:= \overline{(1 - gT)A\langle T, T^{-1}\rangle} = (1 - gT)A\langle T, T^{-1}\rangle. \end{aligned}$$

These equalities hold as principal ideals are already closed by the assumption. For

$$\begin{aligned} a &:= \sum_{k=0}^{\infty} a_k X^k \in A\langle X\rangle, \\ b &:= \sum_{k=0}^{\infty} b_k Y^k \in A\langle Y\rangle, \\ f &:= \sum_{k=-\infty}^{\infty} f_k T^k \in A\langle T, T^{-1}\rangle, \end{aligned}$$

we define

$$\begin{aligned} p(a, b) &:= a(T) - b(T^{-1}) = \sum_{k=0}^{\infty} \left(a_k T^k - b_k T^{-k}\right), \\ q(a, b) &:= (a(g^{-1}), b(g)) = \left(\sum_{k=0}^{\infty} a_k g^{-k}, \sum_{k=0}^{\infty} b_k g^k\right), \\ r(f) &:= f(g^{-1}) = \sum_{k=-\infty}^{\infty} f_k g^{-k}. \end{aligned}$$

Then we obtain the following commutative diagram,

$$\begin{array}{ccccccccc}
 & & & & 0 & & 0 & & \\
 & & & & \downarrow & & \downarrow & & \\
 & & 0 & \longrightarrow & I \oplus J & \xrightarrow{p} & K & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A & \longrightarrow & A\langle X\rangle \oplus A\langle Y\rangle & \xrightarrow{p} & A\langle T, T^{-1}\rangle & \longrightarrow & 0 \\
 & & \Vert & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A & \longrightarrow & A\langle g^{-1}\rangle \oplus A\langle g\rangle & \longrightarrow & A\langle g, g^{-1}\rangle & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & &
\end{array}$$

The second row is clearly exact. Moreover, the first row p maps I onto $(1 - gT)A\langle T\rangle$ and and J onto $(g - T^{-1})A\langle T^{-1}\rangle = T^{-1}(1 - gT)A\langle T^{-1}\rangle$. Using Laurent expansion, we have

$$\forall f \in A\langle T, T^{-1}\rangle, f = f_+(T) + T^{-1}f_-(T^{-1})$$

where $f_+, f_- \in A\langle T\rangle$, thus $A\langle T, T^{-1}\rangle = A\langle T\rangle \oplus T^{-1}A\langle T^{-1}\rangle$. Hence showing the injectivity and the first row is exact. Also the colums are exact by constructions. Hence by Snake lemma, the third row is also exact. □

Proposition 4.2.3. Under the assumption of Proposition 4.2.2 of A and g, pick $a \in A$.

1. The image of a in $A\langle g\rangle$ is 0 if and only if there is a sequence $(b_i)_{i=0,\cdots}$ of elements of A such that $a = gb_0$ and $b_i = gb_{i-1}$ for all $i \in \mathbb{N}$ such that $\lim_{i\to\infty} b_i = 0$ in A.

2. The image of a in $A\langle g^{-1}\rangle$ is 0 if and only if $\lim_{i\to\infty} g^i a = 0$ in A.

Proof. **1.** If $a = (g - T)b$ with $b = \sum_{i=0}^{\infty} b_i T^i$ then $a = gb_0$, $b_0 = gb_1$ vice versa.

2. If $a = (1 - gT)b$, then $a = b_0, b_1 = gb_0, \cdots$, hence $b_i = g^i a$ and some $b \in A\langle T\rangle$, it follows that $\lim_{i\to\infty} g^i a = 0$ and vice versa. □

Proposition 4.2.4. Let A be as in Proposition 4.2.2. Assume also that every ideal $I \subseteq A$ there are finitely many generators $(f_1, \cdots, f_N)$ of it such that

$$A^N \ni (a_1, \cdots, a_N) \overset{\vec{f}}{\mapsto} \sum_{i=1}^{N} a_i f_I \in I,$$

is open when I is equipped with the topology induced from A, then Equation (4.1) is exact for every $g \in A$.

Proof. Due to Proposition 4.2.2, the injectivity of $A \to A\langle g\rangle \oplus A\langle g^{-1}\rangle$ remains to be shown. Let $a \in A$ be in the kernel of this morphism. Then $\lim_{i\to\infty} g^i a = 0$ and there are $(b_i)_{i\in\mathbb{Z}_{\geq 0}}$ as in Proposition 4.2.3. Let $U \subseteq A$ be any open sub

group, we will show $a \in U$. This implies $a = 0$ as A is complete, hence Hausdorf. As A is a nat-ring, there is an open subgroup V such that $VV \subseteq U$. Consider

$$I = \{\alpha \in A \mid \lim_{i\to\infty} g^i\alpha = 0\}.$$

Since $g^i b_j = g^{i-j-1}a$ when $i > j$, all $b_i \in I$. Let $(f_i)_{i=1,\cdots,N}$ be generators of I such that $\vec{f}$ is an open map. Then

$$W = \left\{ \sum_{i=1}^{N} v_i f_i \middle| v_i \in V \right\}$$

is an open subgroup of I. Since $\lim_{i\to\infty} b_i = 0$ in A and I has the topology induced from A, there is a natural number j_0 such that $j \geq j_0$ then $b_j \in W$. Also as $f_i \in I$, there is $k_0 \in \mathbb{N}$ with $k \geq k_0 \Rightarrow g^k f_i \in V$. When $j \geq \max\{j_0, k_0\}$, we have

$$a = g^1 b_j a \in \sum_{i=1}^{N} g^{j+1} f_i \cdot V \subseteq U$$

as $g^{j+1} f_i \cdot V \subseteq VV \subseteq U$. □

Proposition 4.2.5. For a complete nat ring with a countable neighborhood basis of 0 and a topologicaly nilpotent unit (e.g., any complete Tate ring), the following conditions are equivalent.

i. A is Noetherian.

ii. Every ideal of A is closed.

iii. Every ideal $I \subseteq A$ has generator $(f_1, \cdots, f_N)$ such that $\vec{f}$ is open.

In this case, if $A^\triangleleft$ is a complete Tate ring and all rings of the form $A^\triangleleft\langle X_1, \cdots, X_m\rangle$ for every $m \in \mathbb{N}$ are Noetherian, then all Huber pairs B fo which $B^\triangleleft$ is the quotient of some $A^\triangleleft\langle X_1, \cdots, X_m\rangle$ by an ideal I are sheafy.

Proof. Follows from Proposition 4.2.4 and Corollary 4.2.1. □

Remark 4.2.2. Note that **iii**⇒**ii** for trivial reasons. Also in the situation of **iii**, if $(r_j)_{j\in\mathbb{Z}_{\geq 0}}$ is a Cauchy sequence in I there are Cauchy sequences $(a_{j,k})_{j\in\mathbb{N}}$ in A for $1 \leq k \leq N$ such that

$$i_j = \sum_{k=1}^{N} a_{j,k} f_k,$$

then $\lim_{j\to\infty} i_j = \sum_{k=1}^{N} (\lim_{j\to\infty} a_{j,k})\, f_k \in I$. Showing that I is closed hence **iii**⇒**i**. It will then be sufficient to verify **iii**.

5 Ultrametric Fields and Ultrametric Functional Analysis

5.1 Ultrametric Fields and Their Algebraic Extensions

Throuout the subsection, K will be a field equipped with some ultrametric absolute value $|\cdot|_K$.

Definition 5.1.1. A K-norm on a K-vectorspace V is a map satisfying the following conditions.

i. $\forall x, y \in V \|x+y\| \leq \max\{\|x\|, \|y\|\}$.

ii. $\forall \lambda \in K, x \in V, \|\lambda x\| = |\lambda|_K \|x\|$.

iii. $\forall x \in V, \|x\| = 0 \iff x = 0$.

Remark 5.1.1. If the last conditions fails whiel others hold in Definition 5.1.1, we say $\|\cdot\|$ is a seminorm.

We will always assume that K is non-discrete.

Proposition 5.1.1.

$$K^\circ = \{x \in K \mid |x|_K \leq 1\},$$
$$K^{\circ\circ} = \{x \in K \mid |x|_K < 1\}.$$

Proposition 5.1.2. If K is complete then two norm $\|\cdot\|$ and N on a finite dimensional Kvector space are equivalent.

Moreover for these norms, we have the following statements.

1. Every linear map between finite dimensional K-vectorspace is continuous is open onto its image.
2. Every subspace of V is closed.
3. Every dense subspace o V is equal to V.
4. V is complete (as $V \cong K^n$ with maximum norm).

Lemma 5.1.1. If K is any field equipped with any multiplicative absolute value, for a natural number d, $(\alpha_i)_{i=1,\cdots,d} \in K^d$ and

$$\prod_{i=1}^{d}(T-\alpha_i) = \sum_{i=0}^{d} p_i T^i \in K[T]$$

where $p_d = 1$, we have

$$\max\{|\alpha_1|, \cdots, |\alpha_d|\} = \max\{\sqrt[d-i]{|p_i|} \mid 0 \leq i < d\}.$$

Remark 5.1.2. Lemma 5.1.1 is a special case of the Newton polygon.

Proposition 5.1.3. Assume that K is complete with respect to $|\cdot|_K$. If L is an algebraic extension of K, then there is a unique extension of $|\cdot|_K$. to an absolute value $|\cdot|_R$ on L. If $\|\cdot\|_L$ is any K-norm on L, we have

$$|l|_L = \lim_{n\to\infty} \sqrt[n]{\|l^n\|_L}.$$

If L/K is finite this is complete.

Remark 5.1.3. The right hand side of the equality in Proposition 5.1.3 is called a spectral norm.

Corollary 5.1.1. In the situation of Proposition 5.1.3, suppose L/K is Galois and $\alpha, \beta \in L$ be such that $\alpha \neq \beta$ and

$$\forall \sigma \in \mathrm{Gal}(L/K)\backslash\{1\}, |\alpha - \beta| < |\sigma(\alpha) - \beta|_L$$

where $\tilde{\alpha}$ is a conjugate of α. Then $\alpha \in K(\beta)$.

Corollary 5.1.2. If $L/\mathbb{K}$ is an algeraic extension of $\mathbb{K}$ which is algebraically closed (separably closed, perfect), then the completion $\mathbb{L}$ of L with respect to $|\cdot|_L$ has these properties.

Example 5.1.1. $\mathbb{C}_p$, the completion of the algebraic closure of $\mathbb{Q}_p$.

We will assume that $\mathbb{K}$ is complete from now on.

Definition 5.1.2. A normed $\mathbb{K}$-vectorspace is a $\mathbb{K}$-vectorspace equipped with a $\mathbb{K}$-norm.

Definition 5.1.3. For normed $\mathbb{K}$-vectorspaces V, W, we will write

$$\mathcal{L}(V, W) := \{f : V \to W \mid f \text{ is continuous and } \mathbb{K}\text{-linear}\}.$$

For $f : V \to W$, $\mathbb{K}$-linear map, we define,

$$\|f|\mathcal{L}(V, W)\| := \inf\{C \in [0, a)_{\mathbb{R}} \mid \forall v \in V \|f(v)|W\| \leq C\|x|V\|\}.$$

With this notion, we have,

$$\mathcal{L}(V, W) := \{f : V \to W \mid f \text{ is } \mathbb{K}\text{-linear and } \|f|\mathcal{L}(V, W)\| \text{ is finite}\}.$$

Proposition 5.1.4. The open maping theorem holds for $\mathbb{K}$-Banach spaces (i.e., a complete $\mathbb{K}$-normed spaces). With a bit more effort, this can also be seen for closed graph theorem. It also holds for Banach-Steinhaus by Baire's principle.

Remark 5.1.4. If $|\mathbb{K}^\times|$ is not discrete in $(0, \infty)_{\mathbb{R}}$ and $\mathbb{K}$ has a dense countable subset then $\mathbb{K}$ is not spherically complete hence Hahn-Banach fials for normed $\mathbb{K}$-vectorspaces.

6 Affnoid $\mathbb{K}$-Algebra and Their Tate Spectrum

6.1 Basic Properties of Affnoid $\mathbb{K}$-Algebras

Let $\mathbb{K}$ be a complete field with respect to a non-discrete ultrametric absolute value $|\cdot|$. The extension o $|\cdot|$ to any algebraic field extensio nof K will also be denoted by $|\cdot|$.

Definition 6.1.1.

$$\mathbb{T}_n := \left\{ \sum_{\alpha\in\mathbb{N}^n} f_\alpha X^\alpha \;\middle|\; \forall \varepsilon \in (0,\infty)_{\mathbb{R}}, \exists N \in \mathbb{N} \text{ such that } |\alpha| \geq N \Rightarrow |f_\alpha| < \varepsilon \right\}.$$

We define a norm on this space by

$$\|f|\mathbb{T}_n\| = \max\{|f_\alpha| \mid \alpha \in \mathbb{N}^n\}.$$

By the condition on formal sum, this is finite and is a max? ? ? ?.

Obviously this is $\mathbb{K}\langle X_1, \cdots, X_n\rangle$ as a nat ring.

Proposition 6.1.1. With $\|\cdot|\mathbb{T}_n\|$, $\mathbb{T}_n$ becomes a $\mathbb{K}$-Banach algebra and $\|\cdot|\mathbb{T}_n\|$ is multiplicative.

Proof. Exercise sheet 1 from previous term. □

Proposition 6.1.2.

$$\mathbb{T}_n^\circ = \{f \in \mathbb{T}_n \mid \|f|\mathbb{T}_n\| \leq 1\}, \mathbb{T}_n^{\circ\circ} = \{f \in \mathbb{T}_n \mid \|f|\mathbb{T}_n\| < 1\}.$$

Furthermore any subset $B \subseteq \mathbb{T}_n$ is bounded if and only if $\{\|f|\mathbb{T}_n\| \mid f \in B\}$ is bounded in $\mathbb{R}$.

Proof. Follows from the multiplicativity of the norm. □

Proposition 6.1.3. Let $k = K^\circ/K^{\circ\circ}$ be the residue field of K°. Then we have

$$\mathbb{T}_n \circ /\mathbb{T}_n^{\circ\circ} \ni f \overset{\cong}{\mapsto} \sum_{\alpha\in\mathbb{N}^n} (f_\alpha \bmod K^{\circ\circ}) X^\alpha \in k[X_1, \cdots, X_n]. \tag{6.1}$$

We then obtain a decomposition of $\mathbb{T}_n^\times$.

Proposition 6.1.4.

$$\mathbb{T}_n^\times = \mathbb{K}^\times(1 + \mathbb{T}_n^{\circ\circ}) = \{f \in \mathbb{T}_n^\times \mid |f_\alpha| < |f_0|, \alpha \neq 0\}.$$

Proof. The inclusion $\supseteq$ follows from $1 + \mathbb{T}_n^{\circ\circ} \subseteq \mathbb{T}_n^\times$ (geometric series). Let $f \in \mathbb{T}_n^\times$, without loss of generality we assume $\|f|\mathbb{T}_n\| = 1$. This can be achieved by inversion by any $x \in K^\times$ such that $|x| = \|f|\mathbb{T}_n\|$. Then $\|f^{-1}|\mathbb{T}_n\| = 1$ by Fact 1. Then $f^{-1} \in \mathbb{T}_n^\circ$, and $f \in \mathbb{T}_n^{\circ\times}$. Then the image of f under Equation (6.1) is a unit in $k[X_1, \cdots, X_n]$ hence constant and $|f_\alpha| < 1$ when $\alpha \neq 0$. □

Remark 6.1.1. For any (discrete) ring R,

$$\mathfrak{N}(R) = \{r \in R \mid \exists n \in \mathbb{N}, r^n = 0\} = \bigcap_{\mathfrak{p} \in \operatorname{Spec} R} \mathfrak{p},$$

$$\mathfrak{J}(R) = \{r \in R \mid 1 + rx \in R^\times \forall x \in R\} = \bigcap_{\mathfrak{m} \in \operatorname{MaxSpec} R} \mathfrak{m}.$$

Proposition 6.1.5.

$$\mathfrak{J}(\mathbb{T}_n) = \{0\}.$$

Proof. Let $f \in \mathbb{T}_n \backslash \{0\}$. If f_0 and $x \in K^\times$ is such that $\|f|\mathbb{T}_n\| = |x|$. Then $1 + \frac{f}{x} \notin \mathbb{T}_n^\times$ by Proposition 6.1.4. if $f_0 \neq 0$, then the same holds for $1 - \frac{f}{f_0}$ which has 0-th coefficient equal to 0 by Proposition 6.1.4. □

Proposition 6.1.6.

1. Every $f \in \mathbb{T}_1 \backslash \{0\}$ is (K, T)-distinguished of degree d, for a unique $d = \underline{d}(f) \in \mathbb{N}$ is the same as in Defintion 3.5.1. By $\mathbb{T}_1 \backslash \{0\} \xrightarrow{\underline{d}} \mathbb{N}$, $\mathbb{T}_1$ becomes a Euclidean ring.
2. We have $\mathbb{T}_1^\times = K^\times(1 \oplus T\mathbb{T}_1^{\circ\circ}) = \{f \in \mathbb{T}_1 \backslash \{0\} \mid \underline{d}(f) = 0\}$.
3. A set of representatives for the multiplicative equivalence classes of norm zero elements of $\mathbb{T}_1$ is given by the Weierstrass polynomials ($P \in \mathbb{K}^\circ[T]$ with leading term 1) or, equivalently $\{f \in K[T] | f$ is monic and all roots z of f, $|z| \leq 1\}$.

Proof. Set

$$\underline{d}(f) = \max\{k \in \mathbb{N} \mid |f_k| = \|f|\mathbb{T}_1\|\}.$$

Then **1** follows from Definition in 3.5.

2. The equality with the set of Weierstrass polynomials is given in Corollary 3.5.1.

$$\{f \in K[T] \mid f \text{ is monic and all roots } z \text{ of } f, |z| \leq 1\}.$$

by Lemma 5.1.1.

Addition to 1. That we get an Euclidean ring follows from Proposition 3.5.1. □

Lemma 6.1.1. Let $F \subseteq \mathbb{N}^n$ be a finite subset. For $\alpha \in \mathbb{N}^n$ and $\vec{k} = (k_i)_{i=1}^{n-1}$ as in the Lemma, let

$$w(\vec{k}, \alpha) = \sum_{i=1}^{n} k_1 \alpha_i,$$

with $k_n = 1$. Then it is possible to choose $\vec{k}$ in such a way that $w(\vec{k}, \alpha) \neq w(\vec{k}, \beta)$ when $\alpha, \beta \in F$ and $\alpha \neq \beta$.

Proof. CHoose k_i by downward induction on i such that $k_n = 1$ and

$$k_{i,n} > \max\left\{ \sum_{j=1}^{n} k_j \alpha_j \;\middle|\; \alpha \in F \right\}.$$

Then if $\alpha \neq \beta$ are elements of F such that $\alpha_l = \alpha_l$ for $1 \leq l < j$ and $\alpha_j \neq \beta_j$, the large difference difference,

$$k_j |\alpha_j - \beta_j| \geq k_j,$$

can not be compensated by

$$\left| \sum_{l=j+1}^{n} k_l(\alpha_l - \beta_l) \right| = \left| \left(\sum_{l=j+1}^{n} k_l \alpha_l \right) - \left(\sum_{l=j+1}^{n} k_k \beta_l \right) \right|.$$

□

Lemma 6.1.2. Let $(f_i)_{i=1}^m \in (\mathbb{T}_n \backslash \{0\})^m$ where $m, n \in \mathbb{N}$. Then there are positive integers $(k_i)_{i=1}^{n-1}$ such that the continuous ring homomorphism $\varphi : \mathbb{T}_n \to \mathbb{T}_n$ is given by

$$\varphi|_{\mathbb{K}} = \mathrm{id}_{\mathbb{K}}, \varphi(X_i) = \begin{cases} X_i, \ i = n, \\ X_i + X_n^{k_i}, \ 1 \leq i < n. \end{cases}$$

is an isometric $\mathbb{K}$-isomorphism of $\mathbb{K}$-algebraof $\mathbb{T}_n$ onto itself, and such that all $\varphi(f_i)$ are $(\mathbb{T}_n, X_n)$-distinguished in $\mathbb{T}_n \cong \mathbb{T}_{n-1}\langle X_n \rangle$.

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